

Classical and Quantum Random Walks

Rupanshu Shah

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Chapter 1

Classical Random Walks

1.1 Random walks on integer lattices

1.1.1 The one-dimensional simple random walk

Definition 1.1 (One-dimensional simple symmetric random walk). Let $(\xi_i)_{i \geq 1}$ be independent random variables with

$$\mathbb{P}(\xi_i = 1) = \mathbb{P}(\xi_i = -1) = \frac{1}{2}.$$

The *one-dimensional simple symmetric random walk* on $\mathbb{Z}$ is the stochastic process

$$X_0 = 0, \quad X_n = \sum_{i=1}^n \xi_i.$$

Equivalently, at every step the walker moves one unit to the right with probability $1/2$ and one unit to the left with probability $1/2$.

The random variable X_n denotes the signed displacement from the origin after n steps. Since $\mathbb{E}[\xi_i] = 0$, linearity of expectation gives

$$\mathbb{E}[X_n] = \sum_{i=1}^n \mathbb{E}[\xi_i] = 0.$$

Thus the expected signed displacement after n steps is zero.

This does not mean that the walker remains close to the origin. The more meaningful quantity is the average distance from the origin.

Proposition 1.2 (Average distance after n steps). *For the one-dimensional simple symmetric random walk,*

$$\mathbb{E}[|X_n|] = \Theta(\sqrt{n}).$$

Proof in Appendix.

Remark 1.3. A sharper estimate is

$$\mathbb{E}[|X_n|] \sim \sqrt{\frac{2n}{\pi}},$$

which matches the normal approximation $X_n \approx N(0, n)$.

1.1.2 Product and cardinal walks on $\mathbb{Z}^d$

There are two common versions of a simple random walk on $\mathbb{Z}^d$.

Definition 1.4 (Product walk on $\mathbb{Z}^d$). The *product walk* on $\mathbb{Z}^d$ is obtained by running d independent one-dimensional simple symmetric random walks, one in each coordinate. Equivalently, at every step the walker moves by a vector

$$(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_d) \in \{-1, +1\}^d,$$

where each ε_j is independently equal to $+1$ or -1 with probability $1/2$. Thus the step set is

$$\mathcal{S}_{\text{prod}} = \{-1, +1\}^d,$$

and each step has probability 2^{-d} .

Definition 1.5 (Cardinal walk on $\mathbb{Z}^d$). The *cardinal walk* on $\mathbb{Z}^d$ is the nearest-neighbour walk along the coordinate axes. At every step, the walker chooses one of the d coordinate directions uniformly and then chooses one of the two signs uniformly. Thus the step set is

$$\mathcal{S}_{\text{card}} = \{\pm \mathbf{e}_1, \pm \mathbf{e}_2, \dots, \pm \mathbf{e}_d\},$$

where $\mathbf{e}_j$ is the j th standard basis vector. Each of the $2d$ possible steps has probability $1/(2d)$.

Observation 1.6. *In one dimension, the product walk and the cardinal walk are identical. In two dimensions, the product walk moves along diagonal directions*

$$(1, 1), (1, -1), (-1, 1), (-1, -1),$$

whereas the cardinal walk moves along coordinate directions

$$(1, 0), (-1, 0), (0, 1), (0, -1).$$

The two pictures differ only by a 45° rotation and a scaling. Hence, for recurrence questions in two dimensions, they behave the same.

Details in Appendix.

1.1.3 Recurrence and transience

Definition 1.7 (Recurrence and transience). Let $(X_n)_{n \geq 0}$ be a random walk starting at the origin $\mathbf{0}$. Define the first positive return time to the origin by

$$T_{\mathbf{0}}^+ = \inf\{n \geq 1 : X_n = \mathbf{0}\}.$$

The walk is called *recurrent* if

$$\mathbb{P}_{\mathbf{0}}(T_{\mathbf{0}}^+ < \infty) = 1.$$

It is called *transient* if

$$\mathbb{P}_{\mathbf{0}}(T_{\mathbf{0}}^+ < \infty) < 1.$$

Here $\mathbb{P}_{\mathbf{0}}$ means that the walk starts from the origin.

A useful way to test recurrence is to count the expected number of visits to the origin.

Lemma 1.8 (Expected number of visits criterion). *Let*

$$N_{\mathbf{0}} = \sum_{n=0}^{\infty} \mathbf{1}_{\{X_n = \mathbf{0}\}}$$

be the total number of visits to the origin, including the visit at time 0. Then

$$\mathbb{E}_{\mathbf{0}}[N_{\mathbf{0}}] = \sum_{n=0}^{\infty} \mathbb{P}_{\mathbf{0}}(X_n = \mathbf{0}).$$

Moreover, the walk is recurrent if and only if

$$\sum_{n=0}^{\infty} \mathbb{P}_{\mathbf{0}}(X_n = \mathbf{0}) = \infty.$$

Proof in Appendix.

1.1.4 Pólya's recurrence theorem

Theorem 1.9 (Pólya's theorem for lattice random walks). *For both the product walk and the cardinal walk on $\mathbb{Z}^d$,*

$$\begin{cases} d = 1, 2 & \text{the walk is recurrent,} \\ d \geq 3 & \text{the walk is transient.} \end{cases}$$

Proof in Appendix.

The theorem is governed by the decay of the return probability. In all the lattice walks above, the key estimate has the form

$$\mathbb{P}(X_{2k} = \mathbf{0}) \asymp k^{-d/2}.$$

Therefore

$$\sum_{k \geq 1} \mathbb{P}(X_{2k} = \mathbf{0})$$

behaves like the series $\sum_{k \geq 1} k^{-d/2}$, which diverges for $d = 1, 2$ and converges for $d \geq 3$.

For the product walk, this estimate follows directly from independence of the coordinates:

$$\mathbb{P}(X_{2k} = \mathbf{0}) = \left[\binom{2k}{k} 2^{-2k} \right]^d \sim \frac{1}{(\pi k)^{d/2}}.$$

For the cardinal walk, the same order is obtained from the local central limit estimate

$$\mathbb{P}(X_{2k} = \mathbf{0}) = \Theta(k^{-d/2}).$$

The product-walk argument is elementary; the cardinal-walk argument in dimensions $d \geq 3$ needs a separate Fourier/local-central-limit estimate because the simple 45° rotation trick only works in two dimensions.

Takeaway. Pólya's theorem is essentially a comparison between return probability and the p -series $\sum_k k^{-d/2}$. Low dimensions return often enough to force recurrence; dimensions three and higher do not.

Chapter 2

Random Walks on Graphs

2.1 Simple random walk on an undirected graph

Let $G = (V, E)$ be a finite undirected graph with $|V| = n$. Unless explicitly stated otherwise, assume that there are no isolated vertices. For a vertex v , write d_v or $\deg(v)$ for its degree.

Definition 2.1 (Simple random walk on an undirected graph). The *simple random walk* on G is the Markov chain whose transition matrix P is given by

$$P_{v,w} = \begin{cases} \frac{1}{\deg(v)}, & (v, w) \in E, \\ 0, & (v, w) \notin E. \end{cases}$$

Thus, from the current vertex v , the walker chooses a uniformly random neighbour of v .

Let A be the adjacency matrix of G . If G is d -regular, then every vertex has degree d , and

$$P = \frac{1}{d}A.$$

For a non-regular graph, let D be the degree matrix,

$$D_{v,v} = \deg(v), \quad D_{v,w} = 0 \quad (v \neq w).$$

Then

$$P = D^{-1}A.$$

2.1.1 Row-stochastic matrices

Definition 2.2 (Row-stochastic matrix). A matrix $P \in \mathbb{R}^{n \times n}$ is called *row-stochastic* if

$$P_{i,j} \geq 0 \quad \text{for all } i, j, \quad \sum_{j=1}^n P_{i,j} = 1 \quad \text{for every } i.$$

Equivalently,

$$P\mathbf{1} = \mathbf{1},$$

where $\mathbf{1} = (1, 1, \dots, 1)^\top$.

For the transition matrix of a Markov chain, $P_{i,j}$ represents the probability of landing at j in one step, given that the walk starts at i . Since one of the possible next states must occur, every row sums to one.

Proposition 2.3 (Basic properties of row-stochastic matrices). *Let P be row-stochastic. Then:*

- (i) P^t is row-stochastic for every $t \geq 0$.
- (ii) If π is a row probability distribution, then πP is also a row probability distribution.
- (iii) 1 is a right eigenvalue of P , with right eigenvector $\mathbf{1}$.
- (iv) Every eigenvalue μ of P satisfies $|\mu| \leq 1$.

Proof in Appendix.

If π_t is the row vector whose v th entry is the probability of being at vertex v at time t , then

$$\pi_{t+1} = \pi_t P.$$

Iterating,

$$\pi_{t+k} = \pi_t P^k.$$

In particular, starting from π_0 ,

$$\pi_t = \pi_0 P^t.$$

Definition 2.4 (Stationary distribution). A row probability distribution π^* is called a *stationary distribution* for P if

$$\pi^* P = \pi^*.$$

Equivalently, π^* is a left eigenvector of P with eigenvalue 1, normalized so that its entries are nonnegative and sum to one.

For a finite undirected graph, the simple random walk has stationary distribution

$$\pi_v^* = \frac{\deg(v)}{\sum_{u \in V} \deg(u)} = \frac{\deg(v)}{2|E|}.$$

For a d -regular graph this becomes the uniform distribution

$$\pi^* = \left(\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n} \right).$$

2.2 Distributions on the left, functions on the right

The same transition matrix has two complementary interpretations.

2.2.1 Distributions evolve by multiplying from the left

A probability distribution is naturally written as a row vector in these notes. Then

$$\pi_{t+1} = \pi_t P$$

means: starting with distribution π_t , one step of the random walk produces π_{t+1} . Thus $\pi_t P$ computes the next probability distribution.

2.2.2 Functions on vertices evolve by multiplying from the right

A function $f : V \rightarrow \mathbb{R}$ is naturally written as a column vector. Then

$$(Pf)(i) = \sum_{j=1}^n P_{i,j} f(j).$$

This has the following probabilistic meaning:

$$(Pf)(i) = \mathbb{E}[f(X_{t+1}) \mid X_t = i].$$

In words, if the walker is currently standing at vertex i , then $(Pf)(i)$ is the expected value of the function f after taking one random step.

Therefore

$$Pf = \mu f$$

means that the local average of f around every vertex is exactly μ times the value of f at that vertex. The eigenvalue $\mu = 1$ with eigenvector $\mathbf{1}$ says that averaging a constant function over neighbours changes nothing.

Example 2.5 (Path on three vertices). Consider the path $1 - 2 - 3$. Its transition matrix is

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{pmatrix}.$$

If the walk starts at vertex 1, then $\pi_0 = (1, 0, 0)$, so

$$\pi_1 = \pi_0 P = (0, 1, 0).$$

This is the distribution-after-one-step interpretation.

Now take a function

$$f = (0, 10, 20)^T.$$

Then

$$Pf = \begin{pmatrix} 10 \\ 10 \\ 10 \end{pmatrix}.$$

For instance, from vertex 2, the next vertex is equally likely to be 1 or 3, so the expected next value of f is $(0 + 20)/2 = 10$. This is the function-averaging interpretation.

For a d -regular undirected graph, $P = (1/d)A$ is symmetric. Hence its left eigenvectors and right eigenvectors are the same, up to transpose. This is one reason regular graphs are the cleanest setting for spectral calculations.

2.3 Spectral decomposition for regular graphs

Assume in this section that G is d -regular and undirected. Then A is real symmetric, hence $P = (1/d)A$ is also real symmetric.

Theorem 2.6 (Spectral theorem for real symmetric matrices). *Every real symmetric matrix has real eigenvalues and admits an orthonormal basis of eigenvectors. Equivalently, if $M \in \mathbb{R}^{n \times n}$ is real symmetric, then there exist orthonormal vectors $v_1, \dots, v_n$ and real numbers $\mu_1, \dots, \mu_n$ such that*

$$M = \sum_{i=1}^n \mu_i v_i v_i^\top.$$

Since P is row-stochastic, $P\mathbf{1} = \mathbf{1}$. Thus 1 is an eigenvalue. For a connected d -regular graph, we order the eigenvalues as

$$1 = \mu_1 > \mu_2 \geq \mu_3 \geq \dots \geq \mu_n \geq -1.$$

The top normalized eigenvector is

$$v_1 = \frac{1}{\sqrt{n}} \mathbf{1}.$$

Proposition 2.7 (Connectivity and extremal eigenvalues). *Let G be a finite d -regular undirected graph.*

- (i) *The multiplicity of the eigenvalue 1 of P is the number of connected components of G .*
- (ii) *If G is connected, then $\mu_2 < 1$.*
- (iii) *If G is connected, then $\mu_n = -1$ if and only if G is bipartite.*

Details in Appendix.

The statement about $\mu_n = -1$ is the spectral form of the bipartite oscillation problem. If the graph is bipartite, the walk alternates between the two sides of the bipartition. Hence the distribution need not converge to a single limiting distribution, even though a stationary distribution exists.

Theorem 2.8 (Spectral convergence on a regular graph). *Let G be connected, non-bipartite, d -regular, and undirected. Let P be its transition matrix. For an initial distribution π_0 ,*

$$\pi_t = \pi_0 P^t$$

converges to the uniform stationary distribution

$$\pi^* = \left(\frac{1}{n}, \dots, \frac{1}{n} \right).$$

Proof in Appendix.

The theorem is for connected, non-bipartite regular graphs. More generally, for a connected non-regular undirected graph, the limiting distribution is not uniform; it is degree-proportional:

$$\pi_v^* = \frac{\deg(v)}{2|E|}.$$

If the graph is disconnected, there is no single stationary distribution reached from every starting point. If the graph is connected but bipartite, the chain is periodic and the distribution oscillates unless the walk is made lazy.

2.4 Non-regular graphs and normalized adjacency

For a non-regular undirected graph,

$$P = D^{-1}A.$$

This matrix is row-stochastic, but it is usually not symmetric. In a connected undirected graph, P is symmetric if and only if the graph is regular. More generally, if the graph has several connected components, P is symmetric exactly when every edge joins vertices of the same degree; equivalently, each nontrivial connected component is regular.

The standard way to recover symmetry is to use the normalized adjacency matrix.

Definition 2.9 (Normalized adjacency matrix). For an undirected graph with no isolated vertices, define

$$\mathcal{A} := D^{-1/2}AD^{-1/2}.$$

Equivalently,

$$\mathcal{A} = D^{1/2}PD^{-1/2}.$$

The matrix $\mathcal{A}$ is symmetric, so it has real eigenvalues and an orthonormal eigenbasis. It is also similar to P , because

$$\mathcal{A} = D^{1/2}PD^{-1/2}.$$

Therefore P and $\mathcal{A}$ have the same eigenvalues. In particular, for an undirected graph, the transition matrix P has real eigenvalues, even though P itself may not be symmetric.

Details in Appendix.

The row-vector update can also be rewritten in symmetric form. Define

$$v_t := \pi_t D^{-1/2}.$$

Then

$$v_{t+1} = \pi_{t+1} D^{-1/2} = \pi_t P D^{-1/2} = \pi_t D^{-1/2} \mathcal{A} = v_t \mathcal{A}.$$

Thus, after the change of variables $v_t = \pi_t D^{-1/2}$, the evolution is controlled by the symmetric matrix $\mathcal{A}$.

P	$\mathcal{A}$
Transition matrix of the random walk.	Symmetric normalization of the transition matrix.
Always row-stochastic.	Not generally row-stochastic.
Usually not symmetric for non-regular graphs.	Always symmetric for undirected graphs.
Acts on distributions by $\pi \mapsto \pi P$.	Acts after the change of variables $v = \pi D^{-1/2}$.
Eigenvalues lie in $[-1, 1]$.	Same eigenvalues as P , hence also in $[-1, 1]$.
Right eigenvector for eigenvalue 1: $\mathbf{1}$.	Eigenvector for eigenvalue 1: $D^{1/2}\mathbf{1}$.

2.5 Normalized Laplacian and spectral gap

Definition 2.10 (Symmetric normalized Laplacian). The *symmetric normalized Laplacian* of an undirected graph with no isolated vertices is

$$\mathcal{L} := I - \mathcal{A} = I - D^{-1/2}AD^{-1/2} = I - D^{1/2}PD^{-1/2}.$$

The point of $\mathcal{L}$ is that it is symmetric and positive semidefinite, while still encoding the random walk. It is the right object for measuring expansion, bottlenecks, and mixing in a way that remains meaningful for non-regular graphs. For a d -regular graph,

$$\mathcal{A} = \frac{1}{d}A = P, \quad \mathcal{L} = I - P.$$

Theorem 2.11 (Spectrum of the normalized Laplacian). *Let G be a finite undirected graph with no isolated vertices. The eigenvalues of $\mathcal{L}$ can be ordered as*

$$0 = \lambda_0 \leq \lambda_1 \leq \dots \leq \lambda_{n-1} \leq 2.$$

Moreover, the multiplicity of $\lambda_0 = 0$ is the number of connected components of G .

Proof in Appendix.

The second smallest eigenvalue λ_1 is the *normalized Laplacian spectral gap*. Since $\lambda_0 = 0$, the gap is simply

$$\lambda_1 - \lambda_0 = \lambda_1.$$

For a connected graph, $\lambda_1 > 0$. If $\lambda_1 = 0$, the graph is disconnected and has at least two connected components. If λ_1 is very small, the graph has a sparse cut: two large regions are connected by relatively few edges. If λ_1 is large, the graph is well connected and the random walk mixes rapidly.

For a d -regular graph, the eigenvalues of $\mathcal{L}$ and P are related by

$$\lambda_i = 1 - \mu_{i+1}$$

when the Laplacian eigenvalues are ordered increasingly and the transition eigenvalues are ordered decreasingly. In particular,

$$\lambda_1 = 1 - \mu_2.$$

This is why the same quantity appears either as a small Laplacian eigenvalue or as a transition-matrix eigenvalue close to 1.

2.5.1 Two versions of spectral gap

For a connected d -regular graph, write the transition eigenvalues as

$$1 = \mu_1 > \mu_2 \geq \mu_3 \geq \cdots \geq \mu_n \geq -1.$$

The ordinary transition spectral gap is

$$\delta := 1 - \mu_2.$$

This corresponds to the normalized Laplacian gap λ_1 .

The *absolute spectral gap* is

$$\delta_{\text{abs}} := 1 - \max\{|\mu_2|, |\mu_n|\}.$$

The absolute gap also accounts for eigenvalues close to -1 . These negative eigenvalues are responsible for oscillatory behaviour, especially in bipartite or nearly bipartite graphs. For lazy walks, all eigenvalues are nonnegative, and the ordinary and absolute gaps coincide.

2.6 Regularization and lazy walks

Regular graphs are especially convenient because the stationary distribution is uniform and P is symmetric. For intuition, one often reduces to a regular or lazy walk. This is useful, but it should be understood as modifying the walk, not as leaving the original transition matrix unchanged.

Let $\Delta \geq \max_v \deg(v)$. Define a modified walk by

$$P_{v,w}^{(\Delta)} = \begin{cases} 1/\Delta, & (v, w) \in E, \\ 1 - \deg(v)/\Delta, & w = v, \\ 0, & \text{otherwise.} \end{cases}$$

Equivalently, the walk chooses one of Δ possible slots: the original neighbours occupy $\deg(v)$ slots and the remaining $\Delta - \deg(v)$ slots make the walker stay at v . This makes the walk uniform-stationary, because $P^{(\Delta)}$ is symmetric and row-stochastic.

This construction preserves the original non-loop edges and hence the qualitative cut structure of the graph, but it changes the speed of the walk. It adds viscosity: the walker may now spend time holding at the same vertex.

For a d -regular graph, adding the same self-loop weight S at every vertex gives

$$P' = \frac{d}{d+S}P + \frac{S}{d+S}I.$$

If μ_i is an eigenvalue of P , then the corresponding eigenvalue of P' is

$$\mu'_i = \frac{d}{d+S}\mu_i + \frac{S}{d+S}.$$

If $S \geq d$, then all eigenvalues of P' are nonnegative. The common lazy walk

$$P_{\text{lazy}} = \frac{1}{2}(I + P)$$

has eigenvalues $(1 + \mu_i)/2 \in [0, 1]$, and removes the bipartite oscillation.

Details in Appendix.

2.7 Conductance and Cheeger's inequality

The spectral gap measures connectivity through eigenvalues. Conductance measures the same phenomenon through cuts.

For a d -regular graph and a set $S \subseteq V$, define

$$\Phi(S) := \frac{|E(S, V \setminus S)|}{d|S|},$$

where $E(S, V \setminus S)$ is the set of edges leaving S . The conductance of the graph is

$$\Phi(G) := \min_{\substack{S \subseteq V \\ 0 < |S| \leq |V|/2}} \Phi(S).$$

Low conductance means that there is a large cluster connected to the rest of the graph by only a few edges. High conductance means that every reasonably large piece of the graph has many edges leaving it.

Theorem 2.12 (Cheeger's inequality, regular case). *Let G be a d -regular graph, and let*

$$\delta = 1 - \mu_2 = \lambda_1$$

be the normalized Laplacian spectral gap. Then

$$\frac{\delta}{2} \leq \Phi(G) \leq \sqrt{2\delta}.$$

Thus a large spectral gap implies large conductance, and small conductance forces a small spectral gap. This is the formal reason why μ_2 close to 1 indicates a bottleneck.

2.8 Mixing time from the absolute spectral gap

Mixing time measures how quickly the random walk forgets where it started and approaches the stationary distribution. For a connected regular graph, the stationary distribution is uniform. If the graph is also lazy or non-bipartite, the walk converges to that distribution.

The clean spectral calculation is easiest for a connected d -regular graph. For the calculation below, write distributions as column vectors. Since P is symmetric in the regular undirected case, this is just the transpose of the earlier row-vector convention.

Let

$$v_1 = \frac{1}{\sqrt{n}} \mathbf{1},$$

and let $v_1, v_2, \dots, v_n$ be an orthonormal eigenbasis of P , with

$$Pv_i = \mu_i v_i.$$

Expand the initial distribution as

$$p_0 = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n.$$

For every probability distribution p_0 ,

$$\alpha_1 = \langle p_0, v_1 \rangle = \frac{1}{\sqrt{n}},$$

so

$$\alpha_1 v_1 = \frac{1}{\sqrt{n}} \cdot \frac{1}{\sqrt{n}} \mathbf{1} = \left(\frac{1}{n}, \dots, \frac{1}{n} \right) = \pi^*.$$

After t steps,

$$p_t = P^t p_0 = \alpha_1 \mu_1^t v_1 + \dots + \alpha_n \mu_n^t v_n.$$

Since $\mu_1 = 1$, the stationary part remains fixed. The other components shrink according to the remaining eigenvalues. Let

$$\rho := \max\{|\mu_2|, |\mu_n|\} = 1 - \delta_{\text{abs}}.$$

Then

$$\|p_t - \pi^*\|_2 \leq (1 - \delta_{\text{abs}})^t.$$

Proof in Appendix.

To make the ℓ_2 -distance at most η , it is enough to choose t so that

$$(1 - \delta_{\text{abs}})^t \leq \eta.$$

Equivalently,

$$t \geq \frac{\log(1/\eta)}{-\log(1 - \delta_{\text{abs}})}.$$

When δ_{abs} is small,

$$-\log(1 - \delta_{\text{abs}}) \approx \delta_{\text{abs}},$$

so the mixing scale is

$$t = O\left(\frac{1}{\delta_{\text{abs}}} \log \frac{1}{\eta}\right).$$

Note. The scale $1/\delta_{\text{abs}}$ gives constant-factor contraction of the non-stationary component. A specified error η requires the additional factor $\log(1/\eta)$, with constants depending on the chosen norm.

2.9 A marked vertex and effective resistance

Let G be a connected simple undirected graph with n vertices. Suppose there is a unique marked vertex z . The following algorithm stores only the current vertex: check whether the current vertex is z ; if not, move to a uniformly random neighbour.

In a labelled graph this uses $O(\log n)$ bits to store the current vertex. In the usual random-neighbour access model, it uses constant extra graph memory: no DFS stack, BFS queue, or visited set is maintained.

Definition 2.13 (Hitting and commute times). For vertices $a, b \in V$, define

$$H_{a,b} := \mathbb{E}_a[\min\{t \geq 0 : X_t = b\}]$$

and

$$C_{a,b} := H_{a,b} + H_{b,a}.$$

Definition 2.14 (Effective resistance). Put a unit resistor on every edge of G . The effective resistance between a and b , denoted $R_{\text{eff}}(a, b)$, is the voltage drop required to send one unit of current from a to b .

Theorem 2.15 (Commute time identity). *For the simple random walk on a connected undirected graph,*

$$C_{a,b} = 2|E| R_{\text{eff}}(a, b).$$

Proof in Appendix.

Corollary 2.16 (Unique marked vertex search). *Let G be a connected simple undirected graph on $n \geq 2$ vertices, and let z be the unique marked vertex. Starting from any vertex x , the simple random walk reaches z in expected at most*

$$n(n-1)^2 < n^3$$

walk steps. The algorithm stores the current vertex and constant extra graph memory.

Proof in Appendix.

2.10 Spectral gaps of standard graphs

Let G be connected, d -regular, and undirected. Write

$$1 = \mu_1 > \mu_2 \geq \cdots \geq \mu_n \geq -1$$

for the eigenvalues of $P = A/d$, and set

$$\delta = 1 - \mu_2.$$

Proposition 2.17 (Variational characterization). *For $f \perp \mathbf{1}$, $f \neq 0$, define*

$$\mathcal{R}_P(f) := \frac{\langle f, Pf \rangle}{\langle f, f \rangle}.$$

Then

$$1 - \delta = \mu_2 = \max_{\substack{f \perp \mathbf{1} \\ f \neq 0}} \mathcal{R}_P(f),$$

and equivalently

$$\delta = \min_{\substack{f \perp \mathbf{1} \\ f \neq 0}} \frac{\langle f, (I - P)f \rangle}{\langle f, f \rangle}.$$

Thus a test function gives a lower bound on $1 - \delta$; an upper bound on $1 - \delta$ requires bounding $\mathcal{R}_P(f)$ for every $f \perp \mathbf{1}$.

Proof in Appendix.

Proposition 2.18 (Complete graph). *For the complete graph K_n , $n \geq 2$,*

$$\text{spec}(P) = \left\{ 1, \underbrace{-\frac{1}{n-1}, \dots, -\frac{1}{n-1}}_{n-1 \text{ times}} \right\}.$$

Hence

$$\delta = 1 - \mu_2 = \frac{n}{n-1}.$$

For $n \geq 3$,

$$\delta_{\text{abs}} = 1 - \frac{1}{n-1} = \frac{n-2}{n-1}.$$

Proof in Appendix.

Proposition 2.19 (Johnson graph). *Let $1 \leq k \leq N/2$. The Johnson graph $J(N, k)$ has vertex set $\binom{[N]}{k}$, with two vertices adjacent when their intersection has size $k-1$. It is $k(N-k)$ -regular. The transition eigenvalues are*

$$\nu_j = 1 - \frac{j(N+1-j)}{k(N-k)}, \quad 0 \leq j \leq k.$$

The multiplicity of ν_j is

$$\binom{N}{j} - \binom{N}{j-1}, \quad \binom{N}{-1} := 0.$$

In particular,

$$\delta = 1 - \nu_1 = \frac{N}{k(N - k)}.$$

Remark 2.20. If $N \gg k$, then $\delta = \Theta(1/k)$.

Proof in Appendix.

Proposition 2.21 (Cycle graph). *For the cycle C_n , $n \geq 3$, the transition eigenvalues are*

$$\nu_j = \frac{1}{2} \left(e^{2\pi i j/n} + e^{-2\pi i j/n} \right) = \cos \frac{2\pi j}{n}, \quad 0 \leq j \leq n-1.$$

Thus

$$\delta = 1 - \cos \frac{2\pi}{n} = 2 \sin^2 \frac{\pi}{n} = \Theta(n^{-2}).$$

The absolute gap is zero for even n . For odd n ,

$$\delta_{\text{abs}} = 1 - \cos \frac{\pi}{n} = \Theta(n^{-2}).$$

Proof in Appendix.

2.11 Classical search by random walks

Suppose G is a regular graph and an ε -fraction of the vertices are marked. The stationary distribution is uniform, so after mixing, a sample from the walk hits a marked vertex with probability approximately ε .

Let:

- S be the setup cost,
- U be the cost of one random-walk update,
- C be the cost of checking whether the current vertex is marked.

Ignore logarithmic factors in the mixing accuracy parameter.

2.11.1 Classical idea 1: check at every step

Run the random walk and check after every update. The relevant scale is that a fresh approximately stationary sample appears every $O(1/\delta_{\text{abs}})$ steps, and each such sample is marked with probability ε . Thus the expected cost is

$$S + O\left(\frac{1}{\delta_{\text{abs}}\varepsilon}(U + C)\right).$$

2.11.2 Classical idea 2: mix first, then check

Run the walk for $O(1/\delta_{\text{abs}})$ steps, then check once. Each round succeeds with probability approximately ε , so the expected number of rounds is $O(1/\varepsilon)$. The expected cost is

$$S + O\left(\frac{U}{\delta_{\text{abs}}\varepsilon} + \frac{C}{\varepsilon}\right).$$

2.11.3 Quantum preview

Quantum walk search gives quadratic improvements in the relevant random-walk and marked-fraction parameters. Without proving the quantum construction here, the corresponding cost scales are

$$S + O\left(\frac{1}{\sqrt{\delta_{\text{abs}}\varepsilon}}(U + C)\right)$$

for the check-at-every-step style, and

$$S + O\left(\frac{U}{\sqrt{\delta_{\text{abs}}\varepsilon}} + \frac{C}{\sqrt{\varepsilon}}\right)$$

for the mix-then-check style. These formulas are the classical-to-quantum bridge that will be useful when moving from classical graph walks to quantum walk search.

Chapter 3

Quantum Random Walks

Source: A. Ambainis, *Quantum walks and their algorithmic applications*, Int. J. Quantum Inform. **1**(4), 507–518 (2003).

This chapter develops the theory of quantum random walks from first principles. We begin with the one-dimensional setting—both discrete-time and continuous-time—and use it to expose the characteristic behaviour that distinguishes quantum walks from their classical counterparts. Later sections will treat quantum walks on general graphs and their algorithmic applications.

3.1 Classical Random Walk on $\mathbb{Z}$: a brief recap

We fix notation for the classical one-dimensional simple random walk so that comparisons with the quantum case are sharp.

Definition 3.1 (Simple random walk on $\mathbb{Z}$). Let $(X_t)_{t \geq 0}$ be the Markov chain on $\mathbb{Z}$ with $X_0 = 0$ and transition probabilities

$$\mathbb{P}(X_{t+1} = n \pm 1 \mid X_t = n) = \frac{1}{2} \quad \text{for all } n \in \mathbb{Z}, t \geq 0.$$

We call (X_t) the *simple random walk* (SRW) on $\mathbb{Z}$.

Equivalently, $X_t = \sum_{k=1}^t \xi_k$ where $\xi_1, \xi_2, \dots \stackrel{\text{i.i.d.}}{\sim} \text{Unif}\{-1, +1\}$ are independent fair coin flips. The increments are centred ($\mathbb{E}[\xi_k] = 0$) and have variance 1, so by the central limit theorem

$$\frac{X_t}{\sqrt{t}} \xrightarrow{d} \mathcal{N}(0, 1) \quad (t \rightarrow \infty).$$

Proposition 3.2 (Classical spreading rate). *For the SRW on $\mathbb{Z}$ starting at 0,*

$$\mathbb{E}[|X_t|] = \Theta(\sqrt{t}),$$

and for every constant $K > 0$,

$$\mathbb{P}(|X_t| > K\sqrt{t}) \leq 2e^{-K^2/2}.$$

In particular, the walker is at distance $\omega(\sqrt{t})$ from the origin with probability tending to zero as

$t \rightarrow \infty$.

Proof in Appendix.

The upshot: after t steps the SRW spreads to a radius of order $\sqrt{t}$, and being at distance $\omega(\sqrt{t})$ from the origin is exponentially unlikely.

3.2 Discrete-Time Quantum Walk on $\mathbb{Z}$

We now define a quantum analogue of the SRW. Throughout this section $\mathcal{H}_{\text{pos}} = \ell^2(\mathbb{Z})$ with orthonormal basis $\{|n\rangle\}_{n \in \mathbb{Z}}$.

3.2.1 Why the naive quantisation fails

The classical walk moves from $|n\rangle$ to $|n-1\rangle$ or $|n+1\rangle$ with probability $\frac{1}{2}$ each. The most direct quantum analogue replaces probabilities with amplitudes and asks for a unitary operator U on $\mathcal{H}_{\text{pos}}$ satisfying

$$U|n\rangle = a|n-1\rangle + b|n\rangle + c|n+1\rangle \quad \forall n \in \mathbb{Z}, \quad (3.1)$$

with the *translation-invariance* requirement that the coefficients $a, b, c \in \mathbb{C}$ be *independent of n* .

Theorem 3.3 (Triviality of translation-invariant quantum walks on $\mathbb{Z}$). *Let U be a bounded linear operator on $\ell^2(\mathbb{Z})$ satisfying (3.1) for some $a, b, c \in \mathbb{C}$ independent of n . Then U is unitary if and only if exactly one of the following holds:*

- (1) $|a| = 1$ and $b = c = 0$ (always shift left),
- (2) $|b| = 1$ and $a = c = 0$ (stay in place),
- (3) $|c| = 1$ and $a = b = 0$ (always shift right).

Proof in Appendix.

Thus every unitary, translation-invariant, coinless walk on $\mathbb{Z}$ is *deterministic*: at each step the walker moves unconditionally left, stays put, or moves unconditionally right. Such walks carry no interesting probabilistic content. The same obstruction appears on many other graphs: a unitary operator acting locally on a single register cannot simultaneously be local, translation-invariant, and non-trivial.

3.2.2 The coin-and-shift construction

The standard resolution is to augment the state space with an internal *coin* degree of freedom.

Definition 3.4 (State space of the DTQW). The Hilbert space of the one-dimensional discrete-time quantum walk (DTQW) is

$$\mathcal{H} := \mathcal{H}_{\text{pos}} \otimes \mathcal{H}_{\text{coin}}, \quad \mathcal{H}_{\text{pos}} := \ell^2(\mathbb{Z}), \quad \mathcal{H}_{\text{coin}} := \mathbb{C}^2.$$

Basis states are $|n, s\rangle$ for $n \in \mathbb{Z}$ and $s \in \{0, 1\}$, where n denotes the *position* and s the *coin* (or *chirality*) state. A general pure state is

$$|\psi\rangle = \sum_{n \in \mathbb{Z}} (\alpha_n |n, 0\rangle + \beta_n |n, 1\rangle), \quad \sum_{n \in \mathbb{Z}} (|\alpha_n|^2 + |\beta_n|^2) = 1.$$

Definition 3.5 (Coin operator). The *coin operator* C acts on the coin register only:

$$C := I_{\mathcal{H}_{\text{pos}}} \otimes C, \quad C \in U(2),$$

where $C = (c_{ij})_{i,j \in \{0,1\}}$ is a fixed 2×2 unitary matrix, called the *coin matrix*. Its action on basis states is

$$\begin{aligned} C|n, 0\rangle &= c_{00}|n, 0\rangle + c_{10}|n, 1\rangle, \\ C|n, 1\rangle &= c_{01}|n, 0\rangle + c_{11}|n, 1\rangle. \end{aligned}$$

Definition 3.6 (Conditional shift operator). The *conditional shift* S moves the walker left or right depending on the coin state:

$$S|n, 0\rangle := |n - 1, 0\rangle, \quad S|n, 1\rangle := |n + 1, 1\rangle.$$

Coin state $|0\rangle$ is the “left-moving” chirality and $|1\rangle$ is the “right-moving” chirality.

Proposition 3.7. Both S and C are unitary, hence so is their composition $U = SC$.

Proof. C is unitary because $C \in U(2)$ and $I_{\mathcal{H}_{\text{pos}}}$ is unitary. S maps the orthonormal basis $\{|n, s\rangle\}$ bijectively to itself (every $|n', s'\rangle$ is the image of exactly one basis state), so it is an isometry with dense range, hence unitary. The composition of two unitaries is unitary. $\square$

Definition 3.8 (Walk operator; one step of the DTQW). One step of the DTQW on $\mathbb{Z}$ is the unitary operator

$$U := SC = S(I_{\mathcal{H}_{\text{pos}}} \otimes C).$$

After t steps starting from an initial state $|\psi_0\rangle$, the state is $|\psi_t\rangle = U^t|\psi_0\rangle$. The probability of measuring position n with coin state s at time t is $|\langle n, s | \psi_t \rangle|^2$. The *position probability* at n after t steps is

$$p_t(n) := |\langle n, 0 | \psi_t \rangle|^2 + |\langle n, 1 | \psi_t \rangle|^2.$$

Remark 3.9. The coinless walk of [Theorem 3.3](#) corresponds to $\mathcal{H}_{\text{coin}} = \mathbb{C}^1$, in which case the only unitary on $\mathbb{C}^1$ is multiplication by a phase, recovering precisely the trivial walks. The enlargement $\mathcal{H}_{\text{pos}} \otimes \mathcal{H}_{\text{coin}}$ is the minimal extension permitting a non-trivial, local, unitary evolution.

3.2.3 The Hadamard walk

The most studied choice of coin is the *Hadamard matrix*

$$H_{\text{had}} := \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad (3.2)$$

which is both unitary and Hermitian ($H_{\text{had}}^2 = I$). One step of the Hadamard walk acts on basis states by

$$\begin{aligned} |n, 0\rangle &\xrightarrow{U} \frac{1}{\sqrt{2}}|n-1, 0\rangle + \frac{1}{\sqrt{2}}|n+1, 1\rangle, \\ |n, 1\rangle &\xrightarrow{U} \frac{1}{\sqrt{2}}|n-1, 0\rangle - \frac{1}{\sqrt{2}}|n+1, 1\rangle. \end{aligned}$$

Starting from $|\psi_0\rangle = |0, 0\rangle$, the first three steps are:

$$\begin{aligned} |\psi_1\rangle &= \frac{1}{\sqrt{2}}|-1, 0\rangle + \frac{1}{\sqrt{2}}|1, 1\rangle, \\ |\psi_2\rangle &= \frac{1}{2}|-2, 0\rangle + \frac{1}{2}|0, 0\rangle + \frac{1}{2}|0, 1\rangle - \frac{1}{2}|2, 1\rangle, \\ |\psi_3\rangle &= \frac{1}{2\sqrt{2}}|-3, 0\rangle + \frac{1}{\sqrt{2}}|-1, 0\rangle + \frac{1}{2\sqrt{2}}|-1, 1\rangle - \frac{1}{2\sqrt{2}}|1, 0\rangle + \frac{1}{2\sqrt{2}}|3, 1\rangle. \end{aligned}$$

At step 2, the amplitudes at $|0, 0\rangle$ and $|0, 1\rangle$ add constructively. At step 3, the amplitude at $|1, 0\rangle$ is suppressed by destructive interference: both the $|-1, 0\rangle \rightarrow |1, 0\rangle$ path (via S) and the $|1, 1\rangle \rightarrow |1, 0\rangle$ path contribute, but with opposite signs and cancel partially. This interference is the key mechanism behind the quadratic speed-up in spreading.

3.2.4 Asymmetry of the Hadamard walk and the symmetric coin

A Fourier analysis (carried out in [Section D.3](#)) reveals that the position distribution p_t of the Hadamard walk started at $|0, 0\rangle$ is *not* symmetric about the origin: the distribution is biased toward the left. The mechanism is a structural asymmetry in the coin: the amplitude for $|n, 0\rangle \rightarrow |n, 0\rangle$ is $+1/\sqrt{2}$, while the amplitude for $|n, 1\rangle \rightarrow |n, 1\rangle$ is $-1/\sqrt{2}$. This sign difference, combined with the coin state $|0\rangle$ in the initial state, breaks the left-right symmetry.

Theorem 3.10 (Asymmetry for real-entry coins). *If the coin matrix $C \in U(2)$ has only real entries, then for the initial state $|0, 0\rangle$ the position distribution p_t of the resulting DTQW on $\mathbb{Z}$ is generically asymmetric about the origin.*

Proof in Appendix.

The remedy is to pass to a complex-valued coin. A natural symmetric choice is

$$C_{\text{sym}} := \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}. \quad (3.3)$$

One verifies directly that $C_{\text{sym}} C_{\text{sym}}^\dagger = I$, so $C_{\text{sym}} \in U(2)$. The DTQW with coin C_{sym} started at $|0, 0\rangle$ produces a position distribution that is symmetric about 0, because C_{sym} commutes with the combination of spatial reflection and coin-state swap.

3.3 Three Characteristic Features of the DTQW on $\mathbb{Z}$

We now state the three properties that most sharply distinguish the DTQW from the classical SRW. They hold for any non-trivial coin $C \in U(2)$ and are therefore robust features of quantum walks, not artefacts of a particular coin choice.

3.3.1 Quirk 1 — Asymmetry (coin-choice artefact)

As established in [Theorem 3.10](#), any real-entry coin generically produces an asymmetric position distribution when started at $|0, 0\rangle$. This quirk is entirely a coin-choice artefact and is cured by passing to a complex coin such as C_{sym} in [\(3.3\)](#). The remaining two quirks are intrinsic to the quantum walk.

3.3.2 Quirk 2 — Ballistic peaks at $|n| \approx t/\sqrt{2}$

Theorem 3.11 (Location of probability peaks). *For the DTQW on $\mathbb{Z}$ with the Hadamard coin [\(3.2\)](#) and initial state $|0, 0\rangle$, the maximum of $p_t(n)$ over $n \in \mathbb{Z}$ is attained at positions with $|n| = t/\sqrt{2} + O(t^{2/3})$ as $t \rightarrow \infty$. More generally, for any non-trivial coin $C \in U(2)$, the peak position is $|n| = v_{\text{max}} t + O(t^{2/3})$ where $v_{\text{max}} \in (0, 1]$ is the maximum group velocity of the walk, determined by the coin C .*

Proof in Appendix.

Intuition. In the Fourier (quasi-momentum) domain, the walk operator U acts at each quasi-momentum k as a 2×2 unitary matrix $M(k)$ with two eigenphases $\pm\omega(k)$. The group velocity of the mode at k is $v_g(k) = \omega'(k)$. For the Hadamard coin, $v_g(k)_{\text{max}} = 1/\sqrt{2}$, attained at $k = \pm\pi/2$. The probability amplitude propagates at this speed, so the peak travels outward at rate $t/\sqrt{2}$.

Contrast this with the classical SRW, where the distribution peaks at $n \approx 0$ for all t : classically, the number of left-steps and right-steps are nearly equal, so the walker is most often found near its starting point.

3.3.3 Quirk 3 — Linearly many significantly populated sites

Theorem 3.12 (Linear spreading). *For the DTQW on $\mathbb{Z}$ with any non-trivial coin $C \in U(2)$, after t steps there are $\Omega(t)$ positions n for which*

$$p_t(n) = \Omega\left(\frac{1}{t}\right).$$

Consequently, the expected distance from the origin satisfies $\mathbb{E}[|X_t|] = \Omega(t)$.

Proof in Appendix.

Intuition. The two eigenmodes of $M(k)$ propagate at all group velocities between $-v_{\text{max}}$ and v_{max} , and the initial state has support at all quasi-momenta. The amplitude therefore spreads over a window of positions of size $\Theta(t)$, each with weight $\Omega(1/t)$ (since the total probability must sum to 1). This is wave-packet spreading—ballistic transport—rather than diffusion.

3.3.4 Classical comparison and the quadratic speedup

Combining [Theorem 3.12](#) with [Theorem 3.2](#):

	Classical SRW	DTQW
Expected distance after t steps	$\Theta(\sqrt{t})$	$\Omega(t)$
Location of probability peak	$n \approx 0$	$ n \approx t/\sqrt{2}$
Spread radius	$O(\sqrt{t})$	$\Theta(t)$

Takeaway. The DTQW spreads quadratically faster than the classical SRW: to populate a region of radius R , the classical walk requires $\Theta(R^2)$ steps, while the quantum walk requires only $\Theta(R)$ steps.

3.4 What Makes a Walk a Walk?

Before turning to continuous-time walks, it is worth asking: *what properties of a random walk are actually useful for algorithms, and how well does the quantum walk preserve them?*

Why use random walks at all? The primary reason to use a walk in an algorithm is that the graph being explored is too large to hold in memory, or its structure is accessible only locally—one can see the neighbourhood of the current position, but not the entire graph. In this *adjacency oracle* model, global strategies like BFS or DFS are forbidden. A random walk is the natural local strategy: at each step, choose a random neighbour and move there.

Two objectives: mixing and hitting. Classical walk algorithms typically pursue one of two goals:

- *Mixing* (reaching stationarity): run the walk until it is close in total variation to the stationary distribution π , then use the current position as an approximate sample. On a connected, non-bipartite, regular graph, π is the uniform distribution and the cost is measured by the mixing time t_{mix} .
- *Hitting* (reaching a target): given a marked set $M \subset V$, how many steps are needed until the walker first enters M ? This is the hitting time $t_{\text{hit}}(M)$, and underlies algorithms for connectivity, satisfiability, and graph search.

Why mixing is unavailable quantumly. Quantum evolution is unitary, hence reversible. On a finite graph, the iterates $U^t|\psi_0\rangle$ never converge to a fixed state: the spectrum of U lies on the unit circle, so U^t oscillates indefinitely. The time-averaged distribution does converge (to the *limiting distribution*), but this depends sensitively on the spectrum of U and is generally far from uniform.

What quantum walks offer instead. Quirks 2 and 3 show that, even without mixing, the DTQW provides strong spreading guarantees:

- $\Omega(t)$ distinct positions are populated after t steps.
- Each such position carries probability $\Omega(1/t)$.
- The spreading occurs at the ballistic rate $\Theta(t)$, not the diffusive rate $\Theta(\sqrt{t})$.

Uniform distribution is a strong target: it requires the walk to have completely forgotten its starting point. For hitting-based search, we do not need that. We merely need the walk to find a marked vertex with non-negligible probability, and the non-trivial spreading of the DTQW is precisely what enables this. The quantum walk provides the minimal structure needed for local exploration to be algorithmically useful: locality, unitarity, and non-trivial interference.

Key idea. Classical walks are useful for mixing and hitting. Mixing is unavailable quantumly due to unitarity. But the quadratic spreading of the DTQW—guaranteed without mixing—suffices for hitting-based search, and achieves it with a quadratic speedup over the classical walk.

3.5 Continuous-Time Quantum Walk on $\mathbb{Z}$

We now turn to the *continuous-time quantum walk* (CTQW), introduced by Farhi and Gutmann as a model for quantum computation on decision trees. The CTQW is structurally very different from the DTQW and deserves careful treatment.

3.5.1 Background: Hermitian operators and unitary evolution

Definition 3.13 (Hermitian operator). A linear operator H on a Hilbert space $\mathcal{H}$ is *Hermitian* (or *self-adjoint*) if $H = H^\dagger$, equivalently

$$\langle \phi | H \psi \rangle = \langle H \phi | \psi \rangle \quad \forall |\phi\rangle, |\psi\rangle \in \mathcal{H}.$$

For a matrix (H_{mn}) , this is the condition $H_{mn} = \overline{H_{nm}}$ for all m, n .

Hermitian operators on finite-dimensional spaces are diagonalisable with real eigenvalues. They are the observables of quantum mechanics—any measurable quantity must have real expectation values—and, crucially, they generate unitary dynamics.

Definition 3.14 (Schrodinger evolution). Given a Hermitian operator H (the *Hamiltonian*), the Schrodinger equation

$$i \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle, \quad |\psi(0)\rangle = |\psi_0\rangle,$$

has the unique solution

$$|\psi(t)\rangle = e^{-iHt} |\psi_0\rangle.$$

If $H = \sum_k \lambda_k |e_k\rangle\langle e_k|$ is the spectral decomposition ($\lambda_k \in \mathbb{R}$, $\{|e_k\rangle\}$ orthonormal), then $e^{-iHt} := \sum_k e^{-i\lambda_k t} |e_k\rangle\langle e_k|$.

Proposition 3.15 (Unitarity of Schrodinger evolution). *For any Hermitian H and any $t \in \mathbb{R}$, the operator e^{-iHt} is unitary.*

Proof. Using the spectral decomposition with $\lambda_k \in \mathbb{R}$:

$$(e^{-iHt})(e^{-iHt})^\dagger = e^{-iHt} e^{+iH^\dagger t} = e^{-iHt} e^{+iHt} = \sum_k e^{-i\lambda_k t} e^{+i\lambda_k t} |e_k\rangle\langle e_k| = \sum_k |e_k\rangle\langle e_k| = I,$$

where we used $H^\dagger = H$ and the fact that $\lambda_k \in \mathbb{R}$ so $e^{-i\lambda_k t} e^{+i\lambda_k t} = 1$. $\square$

Remark 3.16 (Sign convention). The sign conventions e^{-iHt} and e^{+iHt} differ only by time-reversal ($t \mapsto -t$), or equivalently by the sign of H . Both are unitary for Hermitian H . We use e^{-iHt} throughout, following the physics convention for the Schrodinger equation. Replacing H by $-H$ converts one convention to the other.

3.5.2 Definition of the CTQW on $\mathbb{Z}$

Definition 3.17 (CTQW Hamiltonian on $\mathbb{Z}$). The *continuous-time quantum walk* (CTQW) on $\mathbb{Z}$ is defined on the Hilbert space $\mathcal{H}_{\text{pos}} = \ell^2(\mathbb{Z})$ with orthonormal basis $\{|n\rangle\}_{n \in \mathbb{Z}}$. The Hamiltonian H acts on basis states by

$$H|n\rangle := -|n-1\rangle + 2|n\rangle - |n+1\rangle, \quad n \in \mathbb{Z}. \quad (3.4)$$

In the basis $\{|n\rangle\}$, the matrix entries of H are

$$H_{mn} = \langle m | H | n \rangle = \begin{cases} 2 & \text{if } m = n, \\ -1 & \text{if } |m - n| = 1, \\ 0 & \text{otherwise.} \end{cases}$$

Thus $H = 2I - A_{\mathbb{Z}}$, where $A_{\mathbb{Z}}$ is the (bilateral) adjacency operator of the integer line. Equivalently, H is the *graph Laplacian* $L = D - A$ of the path graph $\mathbb{Z}$ (here $D = 2I$ since every vertex has degree 2).

Proposition 3.18 (Hermiticity of H). *The operator H defined by (3.4) is Hermitian.*

Proof. We verify $H_{mn} = \overline{H_{nm}}$ directly. All entries of H are real, so conjugation is the identity. Symmetry $H_{mn} = H_{nm}$ holds because H_{mn} depends only on $|m - n|$, which is symmetric in m and n . Hence $H^\dagger = H$. $\square$

Definition 3.19 (CTQW evolution). The state of the CTQW at time $t \geq 0$, starting from $|\psi_0\rangle$, is

$$|\psi(t)\rangle = e^{-iHt} |\psi_0\rangle.$$

The probability of measuring position m at time t is

$$p(m, t) := |\langle m | \psi(t) \rangle|^2.$$

There is no coin register: the Hilbert space is $\mathcal{H}_{\text{pos}}$ alone.

3.5.3 The Hamiltonian as graph Laplacian: coupling and flow

In the DTQW, the matrix element $\langle n' | U | n \rangle$ has a clear meaning: its squared modulus is the probability of transitioning from $|n\rangle$ to $|n'\rangle$ in one step. For the CTQW, H is a Hamiltonian, not a transition matrix, and the meaning of $\langle m | H | n \rangle$ is different and deserves attention.

Writing $\alpha_n(t) = \langle n | \psi(t) \rangle$ for the amplitude at site n , the Schrodinger equation in the position basis reads:

$$i \dot{\alpha}_m(t) = \sum_n H_{mn} \alpha_n(t).$$

Substituting (3.4):

$$i \dot{\alpha}_m(t) = 2\alpha_m(t) - \alpha_{m-1}(t) - \alpha_{m+1}(t). \quad (3.5)$$

This is a *discrete Schrodinger equation*: the rate of change of amplitude at m depends on the amplitudes at the two neighbouring sites $m \pm 1$ and at m itself.

Definition 3.20 (Coupling and on-site energy). The off-diagonal entry H_{mn} ($m \neq n$) is the *coupling strength* between sites m and n : it governs the rate at which amplitude flows between them. The diagonal entry H_{mm} is the *on-site energy* at site m . In the CTQW on $\mathbb{Z}$: coupling strength is -1 for adjacent sites, and on-site energy is 2 (the degree) for every site.

The quantity $\langle m | H | n \rangle$ is *not* a probability amplitude for a single-step transition; rather, it is the infinitesimal generator of amplitude flow. If $H_{mn} = -1$, then in an infinitesimal time dt , the amplitude at m changes by $-iH_{mn} \alpha_n(t) dt = +i \alpha_n(t) dt$ due to the coupling with site n . This is fundamentally different from the DTQW, where the walk proceeds in discrete jumps. One should think of the CTQW as a *quantum wave propagating on a graph*: amplitude flows continuously along edges, creating interference patterns determined by the spectral structure of H .

3.5.4 Spreading properties of the CTQW

We analyse the spreading via the Fourier transform. Define $\tilde{\alpha}(k, t) = \sum_{n \in \mathbb{Z}} \alpha_n(t) e^{ikn}$ for $k \in [-\pi, \pi)$. The equation (3.5) transforms to

$$i \dot{\tilde{\alpha}}(k, t) = \lambda(k) \tilde{\alpha}(k, t), \quad \lambda(k) := 2(1 - \cos k) \geq 0,$$

with solution $\tilde{\alpha}(k, t) = e^{-i\lambda(k)t} \tilde{\alpha}(k, 0)$. The *group velocity* of a wave packet centred at quasi-momentum k_0 is

$$v_g(k_0) = \left. \frac{d\lambda}{dk} \right|_{k=k_0} = 2 \sin k_0.$$

The maximum group velocity is $|v_g|_{\max} = 2$, attained at $k_0 = \pm\pi/2$. Amplitude therefore propagates ballistically at speed up to 2 , giving $\mathbb{E}[|X(t)|] = \Omega(t)$ —the same qualitative quadratic speedup over the classical $\Theta(\sqrt{t})$ as in the DTQW.

Remark 3.21 (Comparison of spreading rates). For the Hadamard DTQW, the maximum group velocity is $1/\sqrt{2}$, so peaks appear at $|n| \approx t/\sqrt{2}$. For the CTQW with Hamiltonian (3.4), the maximum group velocity is 2 , so peaks appear near $|n| \approx 2t$. The precise speed depends on the coin (DTQW) or Hamiltonian (CTQW), but ballistic transport— $\Omega(t)$ spreading—is a feature of both models.

3.5.5 CTQW versus DTQW: a structural comparison

In the classical world, the continuous-time random walk (CTRW) is a discrete-time random walk run for a Poisson-distributed (or exponential inter-jump) number of steps: one simply waits a random $\text{Exp}(1)$ time before each jump. The two models are therefore related by a time-change, and most qualitative properties are shared.

In the quantum world, the relationship between the DTQW and CTQW is far more subtle:

	DTQW	CTQW
Hilbert space	$\mathcal{H}_{\text{pos}} \otimes \mathcal{H}_{\text{coin}}$	$\mathcal{H}_{\text{pos}}$
Coin register	Required	Not required
Evolution	$U^t = (SC)^t$	e^{-iHt}
Time	Discrete ($t \in \mathbb{N}$)	Continuous ($t \in \mathbb{R}_{\geq 0}$)
Generator	No single Hamiltonian	Graph Laplacian H

The DTQW has no Hamiltonian generator in any standard sense: the single-step operator $U = SC$ acts on $\mathcal{H}_{\text{pos}} \otimes \mathcal{H}_{\text{coin}}$ and there is no natural way to extract a generator on $\mathcal{H}_{\text{pos}}$ alone. Conversely, there is no natural way to recover the CTQW as a limit of DTQWs by shrinking the step size, because the coin space of the DTQW does not appear in the CTQW. The two models are therefore *genuinely distinct*, not related by a simple limit as in the classical case.

Remark 3.22 (Algorithmic divergence between CTQW and DTQW). The structural difference has concrete algorithmic consequences. For spatial search on a d -dimensional grid with N vertices:

- DTQW achieves $O(\sqrt{N})$ for $d \geq 3$ and $O(\sqrt{N} \log N)$ for $d = 2$.
- CTQW achieves $O(\sqrt{N})$ only for $d \geq 5$, $O(\sqrt{N} \log N)$ for $d = 4$, and no quantum speedup for $d \leq 3$.

The two models are not interchangeable.

3.5.6 When to use the CTQW: requirements and tradeoffs

Required inputs. To run a CTQW on a graph $G = (V, E)$, one needs:

- (1) A Hermitian operator H derived from G (typically the graph Laplacian $L = D - A$ or the adjacency matrix A , possibly with a sign change or a potential term).
- (2) The ability to implement e^{-iHt} as a unitary circuit or physical time-evolution. For d -sparse graphs (each vertex has at most d neighbours), Hamiltonian simulation can be performed with gate complexity $O(d\|H\|t \text{polylog}(1/\varepsilon))$ per step for error ε , using product formulae or quantum signal processing.
- (3) An initial state $|\psi_0\rangle$ (typically a vertex state $|v_0\rangle$ or a prepared superposition). No coin register is needed.

When to prefer the CTQW.

- When the graph has a natural Hamiltonian arising from a physical system or the adjacency structure of a regular graph, the CTQW can be implemented more directly.
- When the graph is irregular (varying degrees), the DTQW requires a different coin per vertex, whereas the CTQW needs only H .
- For hitting problems on graphs with algebraic symmetry (e.g. the glued-trees graph), the CTQW can reduce the problem to a continuous walk on the line via eigenspace analysis.

When to prefer the DTQW.

- For spatial search on low-dimensional grids ($d \leq 4$), the DTQW outperforms the CTQW (see [Theorem 3.22](#)).
- When the walk must be compiled to a circuit-based quantum computer with a fixed gate set, the structure “coin then shift” often leads to more explicit gate counts.
- When a search oracle is implemented via a phase-marked coin, the DTQW framework is more natural.

Summary of tradeoffs. The CTQW has no coin overhead (smaller Hilbert space, $\mathcal{H}_{\text{pos}}$ only) but requires Hamiltonian simulation, whose cost depends on the sparsity and operator norm of H . The DTQW has a coin overhead ($O(\log d_{\max})$ extra qubits, where $d_{\max}$ is the maximum degree) but its per-step cost is a fixed small circuit (coin then shift), making gate-complexity analysis more transparent.

Key idea. The CTQW is a quantum wave-propagation model: amplitude flows continuously along edges under a Hermitian Hamiltonian, with no coin register. The off-diagonal entry H_{mn} is the coupling strength controlling how fast amplitude flows between sites m and n . Both CTQW and DTQW exhibit ballistic ($\Omega(t)$) spreading, but they are structurally distinct models with different performance on different graph families. Neither is universally superior; the right choice depends on the graph geometry, the problem type (hitting vs. search), and the implementation platform.

The next section defines quantum walks on general (possibly irregular) graphs and extends both the DTQW and CTQW frameworks beyond the line.

3.6 Quantum Walks on General Graphs

We now extend both the CTQW and the DTQW from $\mathbb{Z}$ to an arbitrary finite, undirected, loopless graph $G = (V, E)$. Throughout, $N := |V|$, and d_v denotes the *degree* of $v \in V$ (the number of edges incident to v); this is a *scalar*, not to be confused with the degree *matrix* D . [Section 3.4](#) already isolated what we want from any walk on a graph accessed only locally: *locality*, *graph-dependence*, and (in the quantum setting) *unitarity*. Both constructions below are judged against these three requirements.

3.6.1 Continuous-Time Quantum Walk on a General Graph

Definition 3.23 (CTQW Hamiltonian on a general graph). The CTQW on G lives on the Hilbert space $\mathcal{H}_{\text{pos}} := \mathbb{C}^N$ with orthonormal basis $\{|v\rangle\}_{v \in V}$; in particular $\dim \mathcal{H}_{\text{pos}} = N$ and there is no coin register. The Hamiltonian H is the $N \times N$ matrix with entries

$$H_{uv} := \langle u | H | v \rangle = \begin{cases} d_v & u = v, \\ -1 & \{u, v\} \in E, \\ 0 & \text{otherwise.} \end{cases}$$

Equivalently, $H = D - A = L$, the graph Laplacian of G . For $G = \mathbb{Z}$ (every $d_v = 2$), this recovers [Theorem 3.17](#) exactly.

Remark 3.24 (Effect of irregularity). If G is not regular, the on-site energies $d_v = H_{vv}$ vary from vertex to vertex, but the coupling $H_{uv} = -1$ between any two *adjacent* vertices is the same number, -1 , irrespective of d_u and d_v . Irregularity perturbs only the diagonal of H ; it never changes which off-diagonal entries are non-zero (that is fixed by E alone) nor their common value. This is considerably milder than what irregularity does to the DTQW ([Section 3.6.2](#)), where it forces the *dimension* of the local coin space itself to vary with v .

Proposition 3.25 (Hermiticity of L). *For any finite undirected graph G , $L = D - A$ is Hermitian (indeed real symmetric), whether or not G is regular.*

Proof. D is real and diagonal, hence symmetric. A is real, and $A_{uv} = A_{vu}$ because G is undirected ($\{u, v\} \in E \iff \{v, u\} \in E$). Hence $L = D - A$ is real and symmetric, so $L^\dagger = L^\top = L$. $\square$

Proposition 3.26 (Block structure of L). *Let $V = V_1 \sqcup \dots \sqcup V_r$ be the partition of V into connected components of G . Relabelling the basis $\{|v\rangle\}$ so that vertices in the same component are listed consecutively, L is block-diagonal,*

$$L = L_1 \oplus \dots \oplus L_r,$$

where L_j is the graph Laplacian of the induced subgraph $G[V_j]$. In particular, if G is connected ($r = 1$), L admits no non-trivial block-diagonal decomposition: there is no relabelling of $\{|v\rangle\}$ that splits L into two or more smaller blocks.

Proof. If u, v lie in different components, then $\{u, v\} \notin E$ (an edge would place them in the same component), so $L_{uv} = 0$. Hence every cross-component entry of L vanishes, and each diagonal block is the Laplacian of $G[V_j]$, since every neighbour of a vertex $v \in V_j$ also lies in V_j . If G is connected, $r = 1$ and this is the trivial one-block decomposition. A genuine splitting into ≥ 2 blocks would require every edge of G to lie strictly within one block (otherwise some off-diagonal entry would connect two different blocks, contradicting block-diagonality); since G is connected, no partition of V into two non-empty parts has this property, so no such splitting exists. $\square$

In short: for the connected graphs of interest in this book, L is *not* block-diagonal in any useful sense; its sparsity pattern mirrors E exactly, with no further simplification.

Proposition 3.27 (Sign-convention invariance of measurement probabilities). *Let H be any real symmetric $N \times N$ matrix (in particular $H = L$, for any graph G). Then for all basis states $|m\rangle, |n\rangle$ and all $t \in \mathbb{R}$,*

$$|\langle m | e^{+iHt} | n \rangle| = |\langle m | e^{-iHt} | n \rangle|.$$

Consequently the convention $|\psi(t)\rangle = e^{-iHt}|\psi_0\rangle$ of [Theorem 3.14](#) and the convention $|\psi(t)\rangle = e^{+iHt}|\psi_0\rangle$ used by Ambainis produce identical position-measurement probabilities $p(m, t) = |\langle m | \psi(t) \rangle|^2$, for the same Hamiltonian $H = L$.

Proof. Since H is real symmetric, it admits a spectral decomposition $H = \sum_k \lambda_k |e_k\rangle\langle e_k|$ with $\lambda_k \in \mathbb{R}$ and real eigenvectors $|e_k\rangle$ (a real symmetric matrix has a real orthonormal eigenbasis). Writing $c_k := \langle m | e_k \rangle \langle e_k | n \rangle \in \mathbb{R}$ (real, since $|m\rangle, |n\rangle, |e_k\rangle$ all have real coordinates),

$$\langle m | e^{+iHt} | n \rangle = \sum_k e^{+i\lambda_k t} c_k, \quad \langle m | e^{-iHt} | n \rangle = \sum_k e^{-i\lambda_k t} c_k = \overline{\left(\sum_k e^{+i\lambda_k t} c_k \right)},$$

using $c_k \in \mathbb{R}$ for the last equality. Thus $\langle m | e^{-iHt} | n \rangle$ is the complex conjugate of $\langle m | e^{+iHt} | n \rangle$, and conjugate numbers have equal modulus. $\square$

Remark 3.28 (What changes if one instead negates H). [Theorem 3.27](#) compares the *same* H under both exponent signs and finds equal *probabilities*; the two evolution *operators* themselves still differ. A separate question is when the two operators are literally identical. Since $e^{-i(-H)t} = e^{+iHt}$ as an operator identity, matching Ambainis's e^{+iHt} *exactly* (not merely in probability) while still writing the evolution as $e^{-iH't}$ requires taking $H' = -H = A - D$: the on-site term flips sign ($d_v \mapsto -d_v$) and the coupling flips sign ($-1 \mapsto +1$ on each edge). We do not perform this negation: [Theorem 3.27](#) already guarantees that the Schrodinger convention e^{-iHt} with the *unmodified* Laplacian $H = L$ reproduces every measurement probability that Ambainis's e^{+iHt} convention would give. We therefore use e^{-iHt} with $H = L$ throughout, as in [Theorem 3.23](#), with no loss of generality relative to the survey.

Why this is the right notion of CTQW on G . The Hamiltonian $H = L$ satisfies all three requirements of [Section 3.4](#):

- *Locality.* $H_{uv} = 0$ unless $u = v$ or $\{u, v\} \in E$, so the discrete Schrodinger equation $i\dot{\alpha}_v(t) = \sum_u H_{vu} \alpha_u(t)$ (cf. (3.5)) couples the amplitude at v only to the amplitudes at v 's immediate neighbours.
- *Graph-dependence.* Every entry of H is read off directly from G ($H_{vv} = d_v$, $H_{uv} = -1$ iff $\{u, v\} \in E$); no other input is needed.
- *Unitarity.* L is Hermitian for *any* graph ([Theorem 3.25](#)), so e^{-iHt} is unitary for all t by [Theorem 3.15](#), with no extra hypotheses on G .

Moreover, $L = D - A$ is, up to sign, exactly the generator of the *classical* continuous-time random walk on G : the classical walk leaves v at total rate d_v , moving to each neighbour at rate 1, giving Markov generator $Q = A - D = -L$. The CTQW simply runs the Schrodinger equation with the same matrix that the classical theory uses for its generator (up to the sign required

for Hermiticity and the factor of i required for unitarity). This is what makes [Theorem 3.23](#) the natural — not merely convenient — generalisation of [Theorem 3.17](#) from $\mathbb{Z}$ to an arbitrary graph.

3.6.2 Discrete-Time Quantum Walk on a General Graph

Definition 3.29 (State space of the DTQW on a graph). For $v \in V$, let $E_v := \{e \in E : v \in e\}$ be the set of edges incident to v , so $|E_v| = d_v$. The Hilbert space of the DTQW on G is

$$\mathcal{H} := \text{span}\{|v, e\rangle : v \in V, e \in E_v\},$$

with orthonormal basis $\{|v, e\rangle\}$. Writing $\mathcal{H}_v := \text{span}\{|v, e\rangle : e \in E_v\} \cong \mathbb{C}^{d_v}$ for the *local* space at v ,

$$\mathcal{H} = \bigoplus_{v \in V} \mathcal{H}_v, \quad \dim \mathcal{H} = \sum_{v \in V} d_v = 2|E|,$$

the last equality because each edge contributes exactly 1 to the degree of each of its two endpoints (the handshake lemma).

Definition 3.30 (Coin operator on a graph). For each $v \in V$, fix a unitary C_v on $\mathcal{H}_v \cong \mathbb{C}^{d_v}$ (the *local coin* at v). The *coin operator* is

$$C := \bigoplus_{v \in V} C_v,$$

i.e. C acts as C_v on each block $\mathcal{H}_v$: it is block-diagonal with respect to the *vertex* partition $\mathcal{H} = \bigoplus_v \mathcal{H}_v$.

Definition 3.31 (Shift operator on a graph). For an edge $e = \{v, v'\} \in E$,

$$S|v, e\rangle := |v', e\rangle, \quad S|v', e\rangle := |v, e\rangle.$$

Proposition 3.32. S is unitary, Hermitian, and an involution ($S^2 = I$). Writing $\mathcal{H}_e := \text{span}\{|v, e\rangle, |v', e\rangle\}$ for $e = \{v, v'\}$,

$$\mathcal{H} = \bigoplus_{e \in E} \mathcal{H}_e,$$

and S is block-diagonal with respect to this edge partition, acting on each $\mathcal{H}_e$ as the swap matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Proof. S permutes the orthonormal basis $\{|v, e\rangle\}$, swapping the two basis vectors of each pair $\{|v, e\rangle, |v', e\rangle\}$ and fixing nothing else; this is exactly a product of $|E|$ disjoint transpositions, each acting on the 2-dimensional space $\mathcal{H}_e$ via $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. A permutation of an orthonormal basis is unitary; since the permutation has order 2, $S^2 = I$, and since a real permutation matrix that squares to the identity is symmetric, $S^\dagger = S^\top = S$. $\square$

Remark 3.33 (Two incompatible block structures). C is block-diagonal with respect to the *vertex* partition $\mathcal{H} = \bigoplus_v \mathcal{H}_v$; S is block-diagonal with respect to the *edge* partition $\mathcal{H} = \bigoplus_e \mathcal{H}_e$. Every basis state $|v, e\rangle$ belongs to exactly one vertex-block and exactly one edge-block, but the two

partitions of $\mathcal{H}$ are different. It is precisely this mismatch that allows the composite operator $U = SC$ to move probability across the whole graph, rather than acting trivially within a single block.

Definition 3.34 (Walk operator on a graph). One step of the DTQW on G is $U := SC$.

Deriving the Grover diffusion coin

We now pin down a natural, vertex-transitive local coin at a vertex of degree $m \geq 2$. Enumerate the edges incident to v as $e_1, \dots, e_m$, identifying $\mathcal{H}_v \cong \mathbb{C}^m$ with basis $|e_1\rangle, \dots, |e_m\rangle$ (the vertex label v is suppressed throughout this subsection). We look for a real matrix $X = (X_{ij})_{i,j=1}^m$ that treats all m incident edges symmetrically: a common diagonal value $X_{ii} = r$ for every i , and a common off-diagonal value $X_{ij} = s$ for every $i \neq j$, for some $r, s \in \mathbb{R}$. Equivalently,

$$X = (r - s)I_m + sJ_m,$$

where J_m is the $m \times m$ all-ones matrix.

Proposition 3.35 (Classification of vertex-transitive real coins). *A matrix $X = (r - s)I_m + sJ_m$ (with $r, s \in \mathbb{R}$) is unitary if and only if (r, s) is one of*

$$(1, 0), \quad (-1, 0), \quad \left(-1 + \frac{2}{m}, \frac{2}{m}\right), \quad \left(1 - \frac{2}{m}, -\frac{2}{m}\right).$$

Proof. X is real, so unitary is equivalent to orthogonal, i.e. $X^\top X = I_m$; and X is symmetric ($X^\top = X$, since r, s do not depend on the order of i, j), so this is equivalent to $X^2 = I_m$. Since $J_m^2 = mJ_m$ (each entry of J_m^2 is $\sum_{k=1}^m J_{ik}J_{kj} = \sum_{k=1}^m 1 = m$),

$$X^2 = (r - s)^2 I_m + 2(r - s)s J_m + s^2 J_m^2 = (r - s)^2 I_m + [2(r - s)s + ms^2] J_m.$$

For $m \geq 2$, I_m and J_m are linearly independent, so $X^2 = I_m$ holds if and only if

$$(r - s)^2 = 1, \tag{3.6}$$

$$2(r - s)s + ms^2 = 0. \tag{3.7}$$

From (3.6), $r - s = \pm 1$.

Case $r - s = 1$. Equation (3.7) becomes $s(2 + ms) = 0$, so $s = 0$ (giving $r = 1$) or $s = -2/m$ (giving $r = 1 - 2/m$).

Case $r - s = -1$. Equation (3.7) becomes $s(ms - 2) = 0$, so $s = 0$ (giving $r = -1$) or $s = 2/m$ (giving $r = -1 + 2/m$).

These are exactly the four pairs claimed. $\square$

Definition 3.36 (Grover diffusion coin). $D_m := (-1 + \frac{2}{m})I_m + \frac{2}{m}(J_m - I_m)$, i.e. $D_m(i, i) = -1 + \frac{2}{m}$ and $D_m(i, j) = \frac{2}{m}$ for $i \neq j$.

By Theorem 3.35, the four real, vertex-transitive, unitary coins on $\mathbb{C}^m$ are exactly $+I_m, -I_m, +D_m, -D_m$. The choice $X = I_m$ is degenerate: a walker who arrives at v via edge e and is sent

back out along the same edge e at every visit performs no exploration of v 's other neighbours at all. The natural non-trivial choice is D_m (or $-D_m$, which differs from D_m only by an overall sign on this vertex's local coin space; since this sign is *local* to v rather than global to $\mathcal{H}$, it can still affect interference between paths through different vertices at later times, even though it leaves the probabilities computed from a single application at v unchanged).

By definition, D_m acts on a basis coin state by

$$D_m |v, e_i\rangle = \left(-1 + \frac{2}{m}\right) |v, e_i\rangle + \sum_{j \neq i} \frac{2}{m} |v, e_j\rangle. \quad (3.8)$$

Probability after one step. Write $e_i = \{v, v_i\}$ and $e_j = \{v, v_j\}$ for $j \neq i$, so v_i and v_j are the neighbours of v reached via e_i and e_j respectively. Applying C (acting as D_m at v) and then S to $|v, e_i\rangle$:

$$S D_m |v, e_i\rangle = \left(-1 + \frac{2}{m}\right) |v_i, e_i\rangle + \sum_{j \neq i} \frac{2}{m} |v_j, e_j\rangle.$$

Measuring the vertex register after this one step of $U = SC$ starting from $|v, e_i\rangle$ therefore gives

$$\mathbb{P}(\text{vertex} = v_i) = \left(1 - \frac{2}{m}\right)^2, \quad \mathbb{P}(\text{vertex} = v_j) = \left(\frac{2}{m}\right)^2 \text{ for each fixed } j \neq i,$$

and these sum to 1, as they must:

$$\left(1 - \frac{2}{m}\right)^2 + (m-1) \left(\frac{2}{m}\right)^2 = 1 - \frac{4}{m} + \frac{4}{m^2} + \frac{4(m-1)}{m^2} = 1 - \frac{4}{m} + \frac{4m}{m^2} = 1.$$

Remark 3.37 (Why this is “a very lazy walk”). For large m , a walker started in the single basis coin state $|v, e_i\rangle$ moves to v_i — the one neighbour singled out by that coin state — with probability $(1 - 2/m)^2 \rightarrow 1$, and to any other *individual* neighbour with the much smaller probability $(2/m)^2 \rightarrow 0$. A single coin basis state is therefore strongly biased toward the one edge it labels; this is the opposite of uniform exploration of v 's neighbourhood. (This is a different phenomenon from the *classical* lazy random walk, where the walker has positive probability of remaining at v itself; here the walker always moves to some neighbour, but overwhelmingly to the one tied to its current coin state.) Consequently D_m produces genuine spreading across *all* of v 's neighbours only when the coin register itself starts in a superposition over $e_1, \dots, e_m$ — so that the $\Theta(1/m)$ amplitude on each of the $m-1$ “other” branches can interfere coherently over many steps — rather than in a single basis state.

3.6.3 Decomposing the Coin; Gate Costs of C and S

Tensor decomposition (regular graphs only). If G is d -regular ($d_v = d$ for every v), every local space $\mathcal{H}_v$ has the same dimension d , and — after fixing, at each vertex, an enumeration of its d incident edges — $\mathcal{H}$ factors as a genuine tensor product

$$\mathcal{H} \cong \mathcal{H}_{\text{pos}} \otimes \mathcal{H}_{\text{coin}}, \quad \mathcal{H}_{\text{pos}} := \mathbb{C}^N, \quad \mathcal{H}_{\text{coin}} := \mathbb{C}^d, \quad |v, e_i\rangle \leftrightarrow |v\rangle \otimes |i\rangle.$$

In this basis, $C = \sum_{v \in V} |v\rangle\langle v| \otimes D_v$, where D_v is the local coin at v (e.g. $D_v = D_d$ for every v , if the same Grover coin is used everywhere).

Warning. For an *irregular* graph, the local spaces $\mathcal{H}_v \cong \mathbb{C}^{d_v}$ have varying dimension, so $\mathcal{H} = \bigoplus_v \mathcal{H}_v$ does *not* factor as a tensor product $\mathcal{H}_{\text{pos}} \otimes \mathcal{H}_{\text{coin}}$ for any fixed coin space $\mathcal{H}_{\text{coin}}$: there is no single $\mathcal{H}_{\text{coin}}$ large enough to be “the” coin space at every vertex while staying exactly the size needed at each one. The expression $C = \sum_v |v\rangle\langle v| \otimes D_v$ is then only suggestive notation for the direct sum $\bigoplus_v D_v$ of [Theorem 3.30](#); there is no literal ambient $\mathcal{H}_{\text{coin}}$ being tensored with $\mathcal{H}_{\text{pos}}$. We use the $\sum_v |v\rangle\langle v| \otimes D_v$ notation loosely below, with this caveat understood.

Gate cost of S . Encode a vertex label using $n := \lceil \log_2 N \rceil$ qubits, and represent $|v, e\rangle$ (for $e = \{v, v'\}$) by storing v in one n -qubit register and the other endpoint v' in a second n -qubit register. S then acts as the operator that swaps the contents of these two registers — exactly the involution of [Theorem 3.31](#).

If the architecture allows a two-qubit gate between *any* two qubits (*all-to-all connectivity*), swapping two n -qubit registers costs $3n = O(n) = O(\log N)$ elementary gates (the standard decomposition of a SWAP into three CNOTs, applied qubit-pair by qubit-pair).

If instead two-qubit gates are only available between physically *adjacent* qubits (*nearest-neighbour connectivity*), and the two registers are separated by Δ qubit positions in the chosen layout, then each of the n qubit-pair swaps must first be routed into adjacency, at a cost of $O(\Delta)$ extra nearest-neighbour SWAPs per pair. The total cost is then $O(n \cdot \Delta)$. For a layout in which the two n -qubit registers occupy two adjacent contiguous blocks, $\Delta = O(n)$, giving a worst-case bound of $O(n^2) = O(\log^2 N)$ gates; for an unfavourable layout in which the registers are forced far apart, Δ — and hence the cost — can be as large as $O(n \cdot N)$ in principle. The moral is that the *gate* cost of S is architecture-dependent in a way its *mathematical* definition is not.

Gate cost of C . The cost depends on which of the four coins of [Theorem 3.35](#) is used.

- $D_m = \pm I_m$ costs essentially nothing: $+I_m$ requires no gate at all, and $-I_m$ requires only a (vertex-conditioned) overall sign, implementable with $O(1)$ gates per vertex.
- $D_m = \pm D_m$ (Grover diffusion) can be written as $\pm(2|\chi_m\rangle\langle\chi_m| - I_m)$, where $|\chi_m\rangle := \frac{1}{\sqrt{m}} \sum_{i=1}^m |e_i\rangle$ is the uniform superposition over the m incident edges (one checks directly that $(2|\chi_m\rangle\langle\chi_m| - I_m)_{ii} = 2/m - 1$ and $(2|\chi_m\rangle\langle\chi_m| - I_m)_{ij} = 2/m$ for $i \neq j$, matching [Theorem 3.36](#)). When m is a power of two, this reflection-about-the-mean is the standard Grover diffusion operator and can be implemented with $O(\log m)$ elementary gates (Hadamards on the $\log_2 m$ edge-index qubits, X gates, one multi-controlled phase flip, and the inverse Hadamards). For general m , a naive synthesis of an arbitrary m -dimensional unitary costs $O(m^2)$ gates in the worst case, but the specific reflection structure of D_m still admits an $O(\text{polylog } m)$ -to- $O(m)$ implementation via standard Grover-diffusion circuit constructions, depending on how the edge index is encoded.

On an irregular graph, applying the correctly-sized coin D_{d_v} at vertex v additionally requires knowing d_v — i.e. a call to a *degree oracle* (introduced below) — before applying a controlled circuit that implements the appropriately-sized diffusion. This adds the cost of one oracle query per step, on top of the gate costs above.

3.6.4 Back to Setup–Update–Check, and What Is Still Missing

An unfortunate notational collision. The $\text{SETUP} + \text{UPDATE} + \text{CHECK}$ cost framework used earlier (in the chapter on classical random walks on graphs) to express the running time of a walk-based search algorithm,

$$\text{Setup} + T \cdot (\text{Update} + \text{Check}),$$

uses the letters S, U, C for three *algorithmic cost components*. This section has, independently, used $S, C, U = SC$ for three *operators* making up a single step of the quantum walk. These are genuinely different objects that happen to share initials:

- the walk operator $U = SC$ of [Theorem 3.34](#) corresponds to exactly *one* UPDATE step of the $\text{SETUP} + \text{UPDATE} + \text{CHECK}$ framework;
- the Setup and Check costs of that framework have *nothing to do* with the operators S and C — Setup is the cost of preparing the initial state, and Check is the cost of testing whether the current vertex is marked, neither of which we have discussed in this section.

To avoid ambiguity, for the remainder of this book the upright symbols S, C, U are reserved exclusively for the shift, coin, and walk operators of this section; the cost-framework quantities are always written out as SETUP , UPDATE , CHECK in full (never as bare letters S, U, C).

What this section has, and has not, covered. We have given a complete description of a single UPDATE step $U = SC$: its Hilbert space, its two unitary factors, and (for the Grover coin) explicit gate counts. Three questions remain, and are genuinely problem-dependent.

- (1) *How is the graph given?* We have silently assumed S and C are unitaries we are simply handed. On a graph too large to write down explicitly, G is instead presented through an *adjacency oracle* consisting of (at least) two black-box subroutines,

$$O_{\text{deg}} : |v\rangle|0\rangle \mapsto |v\rangle|d_v\rangle, \quad O_{\text{nbr}} : |v\rangle|i\rangle|0\rangle \mapsto |v\rangle|i\rangle|u_i(v)\rangle,$$

where $u_i(v)$ denotes the i -th neighbour of v in some fixed (but otherwise arbitrary) local ordering of v 's neighbours, $i \in \{1, \dots, d_v\}$. S , and the size-selection part of C on an irregular graph, must then be *implemented as a circuit making calls to O_{deg} and O_{nbr}* , rather than handed over as explicit matrices. We have not addressed how many oracle calls this costs — only the gate cost of the resulting unitaries ([Section 3.6.3](#)).

- (2) *Setup.* Preparing the initial state $|\psi_0\rangle$ (e.g. a single vertex $|v_0\rangle$, or the uniform superposition $\frac{1}{\sqrt{2|E|}} \sum_{v,e} |v, e\rangle$) has its own cost, which depends on the problem and on which state is required.
- (3) *Check.* Testing whether the current vertex is marked requires a separate oracle — a phase- or bit-flip oracle O_f for the marking function f — distinct from O_{deg} and O_{nbr} , and again problem-dependent.

Gate complexity versus query complexity. Section 3.6.3 computed the *gate* complexity of S and C once their action is fixed. This is necessary but not sufficient for an algorithmic cost analysis: in the oracle model above, O_{deg} and O_{nbr} are themselves treated as unit-cost black boxes, and the *query complexity* of an algorithm (the number of calls to these oracles, or to O_f) is the quantity of genuine interest, since it lower-bounds the running time independently of how efficiently the oracle itself happens to be implemented. Query complexity cannot be discussed in the abstract: it requires a concrete oracle model attached to a concrete problem. We defer this to the worked examples of the next two chapters — the glued-trees hitting problem, and element distinctness — where the oracle, the marking function, and the initial state are all specified explicitly.

Key idea. One step of the DTQW on a general graph is $U = SC$: C is block-diagonal in the vertex partition of $\mathcal{H} = \bigoplus_v \mathcal{H}_v$, S is block-diagonal in the *different* edge partition, and it is exactly this mismatch that lets U spread amplitude across the graph. Demanding a real, vertex-transitive, unitary local coin forces a choice among exactly four matrices, $\pm I_m, \pm D_m$; the non-trivial choice D_m is genuinely “lazy” on a single basis coin state, and only spreads amplitude over all of a vertex’s neighbours when the coin register itself starts in superposition. What remains — Setup, Check, and the query complexity of implementing S and C from an adjacency oracle — is unavoidably problem-specific, and is the subject of the next two chapters.

Chapter 4

The Glued-Trees Problem

Source: A. Ambainis, *Quantum walks and their algorithmic applications*, Int. J. Quantum Inform. **1**(4), 507–518 (2003), Sec. 4; A. Childs, E. Farhi and S. Gutmann, *An example of the difference between quantum and classical random walks*, J. Quant. Inf. Process. **1**, 35 (2002); A. Childs, R. Cleve, E. Deotto, E. Farhi, S. Gutmann and D. Spielman, *Exponential algorithmic speedup by quantum walk*, Proc. STOC’03, pp. 59–68.

This chapter gives a self-contained, oracle-level treatment of the *glued-trees hitting problem*: the first example, historically, in which a quantum walk is provably exponentially faster than every classical strategy at a natural graph-traversal task. [Section 3.4](#) and [Section 3.6.4](#) already flagged this problem as the natural home for the general SETUP+UPDATE+CHECK machinery and the $O_{\text{deg}}, O_{\text{nbr}}$ oracle model of [Chapter 3](#); we now make every part of that machinery completely explicit for this one graph.

4.1 The Problem and the Oracle Model

4.1.1 The glued-trees graph

Definition 4.1 (Glued-trees graph). Fix an integer $d \geq 1$. A *rooted binary tree of depth d* has $d + 1$ levels, numbered 0 (the root, a single vertex) through d (the leaves, 2^d vertices), with each non-leaf vertex having exactly two children. Such a tree has

$$\sum_{k=0}^d 2^k = 2^{d+1} - 1$$

vertices. Take two disjoint copies T_A and T_B of this tree, with roots A and B respectively, and fix an arbitrary perfect matching between the 2^d leaves of T_A and the 2^d leaves of T_B (one new edge per matched pair). The *glued-trees graph* $G_d = (V, E)$ is the union of T_A , T_B , and these 2^d matching edges. Its vertex count is

$$n := |V| = 2(2^{d+1} - 1) = 2^{d+2} - 2, \quad \text{so} \quad d = \log_2(n + 2) - 2 = \Theta(\log n). \quad (4.1)$$

Every vertex has degree 2 or 3: A and B have degree 2 (two children, no parent); every leaf (of T_A or T_B) has degree 2 (one parent edge, one matching edge); every other vertex has degree 3

(one parent, two children).

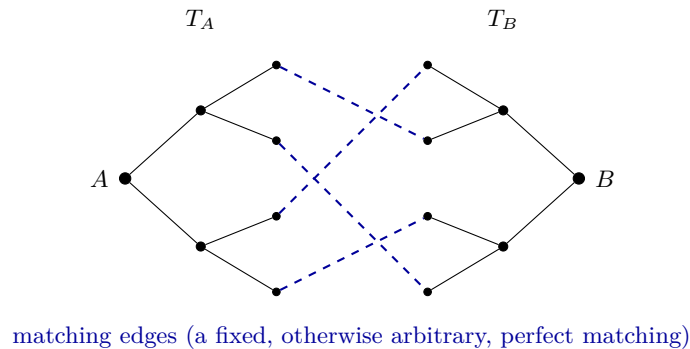


Figure 4.1: The glued-trees graph G_d , shown for $d = 2$ ($n = 14$ vertices). Two depth- d binary trees T_A, T_B (solid edges) are joined by a perfect matching between their 2^d leaves (dashed). The crossing pattern drawn here is one arbitrary matching among the $(2^d)!$ possible ones; the algorithm is told neither d nor which matching was used. General d is exactly analogous, with $d + 1$ levels in each tree instead of 3.

Warning. Some informal accounts of this construction (including a back-of-envelope count one might attribute to the survey’s Fig. 3 caption) state $n = 2(2^d - 1)$. This undercounts by roughly a factor of 4: it omits the leaves and is off by one level in the per-tree sum. Equation (4.1) is the exact count and is used consistently throughout this chapter.

The qualitative picture, which drives everything below, is: starting from A , the number of vertices reachable in k steps *doubles* with every step taken into T_A ($k = 0, \dots, d$), then the graph narrows back down by a factor of two with every step taken *out of* T_B towards B . The graph looks like two cones glued base-to-base at the leaf layer (Figure 4.2 in Section 4.3 makes this precise via the layer decomposition).

4.1.2 The memory-bounded oracle model

Assumption 4.2 (Labelling and memory budget). Every vertex $v \in V$ carries a fixed, but otherwise arbitrary, label in $\{0, 1\}^m$, where $m := \lceil \log_2 n \rceil$; distinct vertices have distinct labels. The labels of A and B are known to the algorithm in advance, at no query cost. The algorithm has no other information about the labelling or about G_d : it does not know d exactly (though it may know an upper bound on it), and the labels carry no information about graph distance. The algorithm’s working memory is $O(m) = O(\log n)$ bits: enough to hold a constant number of vertex labels and a counter up to d , but not enough to hold, say, $\omega(1)$ many labels times a growing multiplicative factor.

Definition 4.3 (Degree and neighbour oracles). The graph is accessed exclusively through the two oracles already introduced in Section 3.6.4, specialised to G_d :

$$O_{\text{deg}} : |v\rangle|y\rangle \mapsto |v\rangle|y \oplus d_v\rangle, \quad O_{\text{nbr}} : |v\rangle|i\rangle|y\rangle \mapsto |v\rangle|i\rangle|y \oplus u_i(v)\rangle. \quad (4.2)$$

Here:

- v is encoded in m qubits, and $d_v \in \{2, 3\}$ is the degree of v ;

- y is a 2-qubit target register, large enough to hold either d_v or a local index $i \in \{1, 2, 3\}$;
- $u_i(v)$ denotes the label of the i -th neighbour of v , under some fixed local ordering of v 's incident edges, $i \in \{1, \dots, d_v\}$, that is otherwise arbitrary and unknown to the algorithm;
- for $i > d_v$ we adopt the convention $u_i(v) := v$ (a harmless placeholder; a correct algorithm never queries an out-of-range index).

A classical query to either oracle costs the same: one elementary query, returning $y \oplus d_v$ or $y \oplus u_i(v)$ as a classical bit string when $y = 0$ and the inputs are classical basis states.

Notation 4.4 (What “forward” and “backward” mean). Fix, once and for all, the unique simple path $A = p_0, p_1, \dots, p_{2d+1} = B$ in G_d (it passes through d edges of T_A , one matching edge, and d edges of T_B). For a vertex v , call an incident edge *forward* if it strictly decreases the remaining distance to B along the unique tree/matching structure, and *backward* otherwise. Concretely:

- For $v \in T_A \setminus \{A\}$: the parent edge is *backward*, both child edges are *forward* (forward always increases distance from A while inside T_A , which is the same as decreasing distance to B).
- For $v \in T_B \setminus \{B\}$: the parent edge is *forward* (it decreases distance to B), both child edges are *backward*.
- At a leaf of T_A : the matching edge is *forward*.
- At a leaf of T_B : the matching edge is *backward*.

This forward/backward labelling is purely a bookkeeping device *for our analysis*; the algorithm is never told which local oracle index i corresponds to which direction.

Question 4.5 (The glued-trees hitting problem). Given oracle access to G_d as in [Theorem 4.3](#), starting information $|A\rangle$ (or, classically, the label of A), and the label of B , output the label of B (equivalently: cause the algorithm's register to contain $|B\rangle$, or its classical pointer to be at B). We ask: how many oracle queries are required, as a function of d (equivalently, of n , since $d = \Theta(\log n)$ by [Equation \(4.1\)](#)), for (a) a deterministic algorithm, (b) a randomised algorithm succeeding with probability $\geq 2/3$, (c) a CTQW-based algorithm, and (d) a DTQW-based algorithm?

Key idea. The whole chapter is one running comparison of four answers to [Theorem 4.5](#): classical (deterministic *and* randomised) costs are $2^{\Omega(d)}$, while both quantum-walk models achieve $\text{poly}(d) = \text{poly}(\log n)$ — an exponential separation, on a single explicit, oracle-accessible graph.

4.2 The Classical Random Walk

4.2.1 Determinism fails: a memory argument

Proposition 4.6 (Path-storage is infeasible). *Any algorithm that solves [Theorem 4.5](#) by first computing, and then following, an explicit sequence of vertex labels from A to B requires $\Theta(d \cdot m) = \Theta((\log n)^2)$ bits of memory to hold that sequence — which exceeds the $O(\log n)$ budget of [Theorem 4.2](#) by an unbounded factor as $n \rightarrow \infty$.*

Proof. The unique path from A to B fixed in [Theorem 4.4](#) has $2d + 1$ edges and $2d + 2$ vertices. Storing it as an explicit list of labels costs $(2d+2) \cdot m$ bits, and $(2d+2)m = \Theta(d \log n) = \Theta((\log n)^2)$ by [Equation \(4.1\)](#). Since $(\log n)^2 / \log n \rightarrow \infty$, no constant c makes $(2d + 2)m \leq c \log n$ for all sufficiently large n . $\square$

Warning (Scope of [Theorem 4.6](#)). This rules out only the specific strategy “compute the whole path, then walk along it”; it does not, by itself, rule out a cleverer deterministic rule that decides each move on the fly from a small amount of local state (e.g. “remember only the label you just arrived from”). Such a rule costs only $O(m)$ memory and is exactly the strategy used successfully in [Theorem 4.7](#) below to cross T_A . The genuine obstruction to determinism in this problem is not a memory bound but an *information-theoretic* one: inside T_B , every internal vertex’s three neighbours — two children and one parent — are locally indistinguishable to an algorithm that has only oracle access and no prior knowledge of the labelling, so no local decision rule, however much memory it is given, can reliably identify which neighbour is the parent without having already explored in that direction. Childs, Farhi and Gutmann (2002) and Childs et al. (2003) make this precise with a relabelling/symmetry argument and show that *no* classical algorithm — deterministic or randomised, with any amount of memory — can reach B from A in $\text{poly}(d)$ queries with non-negligible probability, against a worst-case (or even a uniformly random) choice of leaf matching. We do not reprove that stronger result here; instead, [Section 4.2.3](#) proves the weaker, fully explicit statement that the *natural* randomised strategy — descend T_A deterministically, then perform a memoryless random walk in T_B — takes $\Omega(2^d)$ expected steps. This is already enough to exhibit the qualitative separation from the quantum walks of [Sections 4.3](#) and [4.4](#), and it is the bound that the recursion below makes fully rigorous.

4.2.2 Crossing T_A deterministically

Lemma 4.7 (Deterministic descent through T_A). *There is a deterministic algorithm that, starting at A , reaches a leaf of T_A using at most $2d$ calls to O_{nbr} and $O(m)$ bits of memory (one stored label, plus a step counter taking values in $\{0, \dots, d\}$).*

Proof. Maintain a register p holding the label of the previously-visited vertex (initially undefined) and a counter k (initially 0). At the current vertex v :

- (i) If $k = 0$ (i.e. $v = A$), query $O_{\text{nbr}}(v, 1)$, move to the returned vertex, set $p \leftarrow v$, increment k . (A has no parent, so index 1 is automatically a child; 1 query.)
- (ii) If $1 \leq k \leq d - 1$, query $O_{\text{nbr}}(v, 1)$. If the returned label equals p , query $O_{\text{nbr}}(v, 2)$ and move there (2 queries); otherwise move to the vertex just returned by the first query (1 query). Either way, set $p \leftarrow v$ before moving, then increment k .
- (iii) Stop once $k = d$.

Correctness: by [Theorem 4.4](#), every $v \neq A$ visited by this procedure has exactly one neighbour equal to p (its parent, the backward direction) among indices 1, 2, 3 restricted to 1, 2 (degree-3 vertex with one parent and two children, of which we only ever probe indices 1 and 2); whichever of indices 1, 2 does not equal p is a child, hence a forward move ([Theorem 4.4](#)), and the procedure

always moves forward. Since T_A has depth d , after $k = d$ forward moves the walk is at a leaf of T_A . Each step costs at most 2 queries to O_{nbr} and the procedure runs for exactly d steps, giving the stated bound of $\leq 2d$ queries. Memory: only p (m bits) and k ($\lceil \log_2(d+1) \rceil = O(\log \log n)$ bits) are ever held, which is $O(m) = O(\log n)$. $\square$

Corollary 4.8 (Crossing into T_B). *From a leaf of T_A , the matching edge into a leaf of T_B can be identified and crossed with ≤ 2 further calls to O_{nbr} , using the same memory.*

Proof. A leaf v of T_A has degree 2: its parent (which equals the stored label p , by construction of Theorem 4.7) and the matching edge. Query $O_{\text{nbr}}(v, 1)$; if it returns p , query $O_{\text{nbr}}(v, 2)$ and move there; otherwise move to the vertex already returned. The destination is, by elimination, the matching partner, i.e. a leaf of T_B . $\square$

By Theorems 4.7 and 4.8, reaching a leaf of T_B from A costs $O(d)$ queries deterministically. Everything that follows concerns the remaining task: getting from a leaf of T_B up to B .

4.2.3 Ascending T_B : the memoryless random walk

Warning. Inside T_B , the trick of Theorem 4.7 breaks down: at an internal vertex of T_B , two of the three neighbours are backward (the two children) and only *one* is forward (the parent), and nothing in a single oracle query distinguishes which is which. We therefore analyse the simplest possible classical strategy: at every internal vertex, query one neighbour index chosen uniformly at random from $\{1, 2, 3\}$ and move there, *without* using the stored label p to avoid an immediate backtrack. (Theorem 4.12 explains why excluding the backtracking edge does not straightforwardly help, and is not needed for the lower bound we prove.)

Definition 4.9 (Memoryless random walk in T_B). Starting from a leaf v_0 of T_B reached via Theorem 4.8, repeat: at the current vertex v , query $O_{\text{deg}}(v)$ to learn $d_v \in \{2, 3\}$, then query $O_{\text{nbr}}(v, I)$ for I drawn uniformly from $\{1, \dots, d_v\}$, and move to the returned vertex. Stop when the current vertex is B (recognised at no extra query cost, since the label of B is known in advance).

Theorem 4.10 (Exponential expected hitting time in T_B). *Let t_{hit} denote the number of steps taken by Theorem 4.9 to first reach B , starting from a leaf of T_B . Then*

$$\mathbb{E}[t_{\text{hit}}] \geq 4 \cdot 2^d - 3d - 4 = \Omega(2^d).$$

The proof is given in full in Section E.1; we record the key recursion here since it is needed again, in modified form, when we contrast classical and quantum behaviour in Section 4.4.

Lemma 4.11 (Layer recursion). *For $i = 0, 1, \dots, d-1$, let layer i of T_B denote the set of vertices at graph-distance i from the leaf layer (so layer 0 is the leaves themselves, and layer $d-1$ consists of the children of B), and let X_i denote the expected number of steps for the walk of Theorem 4.9 to first reach layer $i+1$, starting from an arbitrary vertex of layer i having just entered layer i for the first time. Then $X_0 = 1$, and for $1 \leq i \leq d-1$,*

$$X_i = 1 + \frac{2}{3}(X_{i-1} + X_i) + \frac{1}{3} \cdot 0 \quad \Longleftrightarrow \quad X_i = 2X_{i-1} + 3, \quad (4.3)$$

with explicit solution $X_i = 2^{i+2} - 3$.

Proof sketch. A leaf (layer 0) has degree 2: one neighbour is the matching partner in T_A (just arrived from) and the other is its T_B -parent (layer 1). Since [Theorem 4.9](#) chooses uniformly among *all* d_v neighbours (not excluding the one just visited), with probability $\frac{1}{2}$ it returns across the matching edge into T_A and with probability $\frac{1}{2}$ it advances to layer 1; in either case it takes expected 2 tries timed at 1 step each to first reach layer 1, i.e. X_0 would naively be 2, not 1. [Theorem 4.13](#) explains how this is handled (and why $X_0 = 1$ is in fact the right quantity for a valid *lower* bound). At an interior vertex of layer $i \geq 1$ (degree 3: one parent in layer $i + 1$, two children in layer $i - 1$), one of three uniformly random choices ($\frac{1}{3}$) advances to layer $i + 1$ directly; the other two ($\frac{2}{3}$) move to a fresh vertex of layer $i - 1$, from which the walk must first climb back up to layer i (cost X_{i-1} in expectation) and then, by the memoryless property, again needs expected time X_i to reach layer $i + 1$. This gives (4.3). The closed form follows by setting $Y_i := X_i + 3$, so $Y_i = 2Y_{i-1}$ with $Y_0 = X_0 + 3$; full details, including the resolution of the X_0 discrepancy just flagged and the summation $\sum_{i=0}^{d-1} X_i$, are in [Section E.1](#). $\square$

Details in Appendix.

Remark 4.12 (Backtrack-avoidance does not trivially help). One might hope that remembering p (the previously-visited vertex, as in [Theorem 4.7](#)) and excluding it from the random choice speeds things up. At an interior vertex reached *from a child* (the typical case when first entering a layer from below), excluding the backtrack edge leaves one parent (forward) and one sibling child (backward) — an apparently fair 1-to-1 split, which would suggest only polynomial ($O(d^2)$) hitting time. This reasoning is incomplete: the “sibling child” branch leads into an entirely fresh, unvisited subtree, not back to a previously-seen vertex, and a full account of the resulting process requires tracking how many *distinct* fresh subtrees have been entered, not merely the current layer. Working this out rigorously reproduces the same $\Omega(2^d)$ behaviour (this is, in essence, the content of the more delicate argument of Childs, Farhi and Gutmann referenced in [Section 4.2.1](#)); we do not need it here, since [Theorem 4.9](#) — which does not exclude backtracking — already gives a fully rigorous $\Omega(2^d)$ bound by the elementary recursion above.

Remark 4.13 (A domination argument, and the X_0 discrepancy). [Theorem 4.11](#) silently glosses over two related points, both resolved properly in [Section E.1](#): (1) a “backward” move from layer 1 can return all the way across the matching edge into T_A , not just to a fresh leaf of T_B , and the walk must then find its way back; (2) the value X_0 used in (4.3) as the *base case fed into* X_1 is taken to be 1 (as if the leaf layer forced a single deterministic step forward), although the true expected time to leave the leaf layer forward is larger once excursions into T_A are accounted for. Both of these only make the true walk *slower* than the idealised recursion of [Theorem 4.11](#) suggests, so the bound $\mathbb{E}[t_{\text{hit}}] \geq \sum_{i=0}^{d-1} X_i$ obtained from the idealised (charitable) recursion remains a valid *lower* bound on the true expected hitting time — which is all [Theorem 4.10](#) claims.

Corollary 4.14 (Total classical query complexity). *The randomised algorithm of [Theorem 4.7](#), [Theorem 4.8](#), and [Theorem 4.9](#) solves [Theorem 4.5](#) using $O(d) + \Omega(2^d) = \Omega(2^d)$ expected oracle queries.*

4.3 The Continuous-Time Quantum Walk

We now show that the CTQW machinery of [Section 3.6.1](#) solves [Theorem 4.5](#) in $\text{poly}(d)$ time. The key idea, due to Childs, Farhi and Gutmann (2002), is a dimension reduction: although G_d has $n = \Theta(2^d)$ vertices, the relevant part of the dynamics — started from $|A\rangle$ — collapses onto a chain of only $2d + 2$ “layers”. This section states that reduction and explains, at the level of intuition, why it produces a walk that is exponentially faster than the classical one of [Section 4.2](#). All of the case-by-case verification is carried out in full in [Sections E.2 to E.5](#); here we only record the statements and the reasoning behind them.

4.3.1 Why a layer reduction works

Definition 4.15 (Layers of the glued-trees graph). For $j = 0, 1, \dots, 2d + 1$, define

$$L_j := \begin{cases} \{v \in T_A : \text{depth}_{T_A}(v) = j\}, & 0 \leq j \leq d, \\ \{v \in T_B : \text{depth}_{T_B}(v) = 2d + 1 - j\}, & d + 1 \leq j \leq 2d + 1, \end{cases}$$

where $\text{depth}_{T_A}(v)$ and $\text{depth}_{T_B}(v)$ denote graph distance to A and B within the respective trees. Write $n_j := |L_j|$. Intuitively, L_j is “the set of vertices exactly j steps along the A -to- B direction”: $L_0 = \{A\}$, $L_{2d+1} = \{B\}$, and the two leaf sets sit in the middle as L_d, L_{d+1} .

Proposition 4.16 (Layer sizes). $n_j = 2^j$ for $0 \leq j \leq d$ and $n_j = 2^{2d+1-j}$ for $d + 1 \leq j \leq 2d + 1$; in particular $n_0 = n_{2d+1} = 1$ and $n_d = n_{d+1} = 2^d$. The layers partition V into exactly $2d + 2$ nonempty sets, and $\sum_j n_j = n$ ([Equation \(4.1\)](#)).

Warning (Correcting the survey’s layer count). Ambainis’s survey describes “ $2d + 1$ sets $S_{-d}, \dots, S_d$ ” with $|S_i| = |S_{-i}| = 2^{d-i}$, implicitly identifying the two leaf sets (of T_A and of T_B) as a single set S_0 of size 2^d . This cannot be exactly right: a T_A -leaf and its matching partner in T_B are joined by an edge of the graph, so lumping them into one layer would make that layer contain an edge *within itself*, breaking the very property that makes the reduction below work. [Theorem 4.15](#) repairs this by keeping the two leaf sets as separate layers $L_d \neq L_{d+1}$, giving $2d + 2$ layers rather than $2d + 1$. The asymptotics are unaffected; only the precise bookkeeping changes.

Key idea. Every vertex of a fixed layer L_j “looks the same” to the walk: the construction of G_d (two perfectly regular binary trees, glued by an arbitrary but fixed matching) has enough symmetry that, restricted to states which assign equal amplitude to every vertex within each layer, the dynamics never leaks outside that symmetric subspace. Concretely, the walk started in the maximally symmetric state $|A\rangle = |\lambda_0\rangle$ (uniform, trivially, over the one-point layer L_0) stays, for all time, a superposition of the layer-uniform states $|\lambda_j\rangle$. This is exactly the kind of reduction used for the walk on the line in [Section 3.6.1](#), now applied to a $(2d + 2)$ -vertex line graph instead of $\mathbb{Z}$. The mechanism, in one line, is: H treats every vertex in L_{j-1} , every vertex in L_j , and every vertex in L_{j+1} symmetrically ([Theorem 4.17](#) below), so it cannot distinguish individual vertices within a layer, only the layer index.

Proposition 4.17 (Inter-layer degree regularity). *For each j , every $v \in L_j$ has exactly a_j neighbours in L_{j-1} and exactly b_j neighbours in L_{j+1} (and no neighbours in any other layer), where*

$$(a_j, b_j) = \begin{cases} (0, 2) & j = 0, \\ (1, 2) & 1 \leq j \leq d-1, \\ (1, 1) & j \in \{d, d+1\}, \\ (2, 1) & d+2 \leq j \leq 2d, \\ (2, 0) & j = 2d+1. \end{cases}$$

Details in Appendix.

Given [Theorem 4.17](#), a direct computation (carried out in [Section E.2](#)) shows that $H = L$ maps the $(2d+2)$ -dimensional subspace $\mathcal{H}_{\text{layer}} := \text{span}\{|\lambda_j\rangle\}_{j=0}^{2d+1}$, where $|\lambda_j\rangle := n_j^{-1/2} \sum_{v \in L_j} |v\rangle$, into itself, and acts on it as the real symmetric tridiagonal matrix $\tilde{H}$ given explicitly by the following proposition.

Proposition 4.18 (Explicit reduced Hamiltonian). *With δ_j, a_j, b_j as in [Theorem 4.17](#) and n_j as in [Theorem 4.16](#), the entries of $\tilde{H}$ are:*

$$\tilde{H}_{jj} = \begin{cases} 2 & j \in \{0, d, d+1, 2d+1\}, \\ 3 & \text{otherwise}, \end{cases} \quad \tilde{H}_{j,j+1} = \begin{cases} -1 & j = d, \\ -\sqrt{2} & \text{otherwise } (0 \leq j \leq 2d). \end{cases}$$

Details in Appendix.

Remark 4.19 (Why $\sqrt{2}$, and not 1). $\tilde{H}$ is real symmetric (it is the restriction of the Hermitian H to an orthonormal basis), so there is no asymmetry between $\tilde{H}_{j,j+1}$ and $\tilde{H}_{j+1,j}$ to look for: away from the single matching-edge bond at $j = d$, every off-diagonal entry of $\tilde{H}$ equals the same constant $-\sqrt{2}$. The chain is a uniform tight-binding chain with hopping amplitude $\sqrt{2}$ and on-site energy 3 in the bulk, on-site energy 2 at the four special sites $j \in \{0, d, d+1, 2d+1\}$, and one weak bond of strength 1 joining the two leaf layers. The number $\sqrt{2}$ is exactly $\sqrt{n_{j+1}/n_j}$, the square root of the branching factor: moving from L_j to L_{j+1} inside T_A , amplitude is redistributed over twice as many physical vertices, and this redistribution is what *enhances* the effective hopping strength above the line's natural value of 1 ([Theorem 3.17](#)). It is precisely this enhancement — not any asymmetry — that drives the ballistic, rather than diffusive, propagation analysed next.

4.3.2 Ballistic propagation and the hitting time

Away from its four boundary sites and the one weak bond, $\tilde{H}$ is the infinite tight-binding chain

$$H_{\text{bulk}}|j\rangle = 3|j\rangle - \sqrt{2}|j-1\rangle - \sqrt{2}|j+1\rangle, \quad j \in \mathbb{Z}, \quad (4.4)$$

and we use this idealised infinite chain to estimate the propagation speed ([Theorem E.3](#) explains, without full proof, why the boundary terms do not change the order of the answer).

Proposition 4.20 (Dispersion relation and group velocity). *For the chain (4.4), a plane wave of momentum k travels at angular frequency $\omega(k) = 3 - 2\sqrt{2}\cos k$, with group velocity $v_g(k) = \omega'(k) = 2\sqrt{2}\sin k$, maximised at $|v_g| = 2\sqrt{2}$ for $k = \pm\pi/2$.*

Details in Appendix.

Key idea. A classical analogy: on a line with hopping strength 1 (the ordinary walk on $\mathbb{Z}$ of Section 3.6.1), a wavepacket’s group velocity is bounded by 2, and probability spreads *ballistically* at rate $\Theta(t)$ — itself already faster than the classical $\Theta(\sqrt{t})$ diffusive spread (Chapter 3). Here the hopping strength is enhanced to $\sqrt{2}$ (Theorem 4.19), so the maximum group velocity is enhanced to $2\sqrt{2}$, but the *qualitative* picture is the same ballistic transport: a wavepacket launched from $A = L_0$ reaches the far end $B = L_{2d+1}$, a chain of length $2d + 1$, in time $\Theta(d)$ rather than the $\Theta(2^d)$ that a classical (diffusive, and in fact backward-biased) walk would need.

Theorem 4.21 (Polynomial CTQW hitting time). *There is a constant $c > 0$ such that running the CTQW $|\psi(t)\rangle = e^{-iHt}|A\rangle$ for time $t = cd$ and measuring the position register yields the outcome B with probability $\Omega(1/d)$. Repeating the procedure (re-preparing $|A\rangle$ and re-running for time cd) $O(d)$ times independently and measuring each time drives the probability of at least one success to a constant $\geq 2/3$.*

Proof idea. By Theorem 4.20, a wavepacket built from momenta near $k = \pi/2$ travels at speed $\approx 2\sqrt{2}$, so it crosses the $2d + 1$ -site bulk chain in time $\Theta(d)$; this is the source of the $t = \Theta(d)$ scale. A full accounting of how much of $|A\rangle$ ’s amplitude is carried in such wavepackets, how the leaf-layer weak bond and the finite chain length affect normalisation, and the resulting $\Omega(1/d)$ probability bound at site $2d + 1$, is exactly the (non-trivial) calculation of Childs, Farhi and Gutmann (2002); we do not reproduce their full argument, only the dispersion-relation input that explains *why* the relevant timescale is linear, not exponential, in d . Given a single-shot success probability $p = \Omega(1/d)$, r independent repetitions fail to ever succeed with probability $(1 - p)^r \leq e^{-pr}$, which is $\leq 1/3$ once $r = O(1/p) = O(d)$. $\square$

Corollary 4.22 (CTQW query complexity). *Simulating e^{-iHt} for H a 3-sparse, bounded-degree Hamiltonian on G_d for time $t = \Theta(d)$ costs $\tilde{O}(d)$ queries to $O_{\text{deg}}, O_{\text{nbr}}$ (standard sparse Hamiltonian simulation, suppressing polylogarithmic factors). With the $O(d)$ repetitions of Theorem 4.21, the total query cost is $\tilde{O}(d^2) = \tilde{O}((\log n)^2)$, matching the $O(d^2)$ figure stated (without this level of detail) in the survey.*

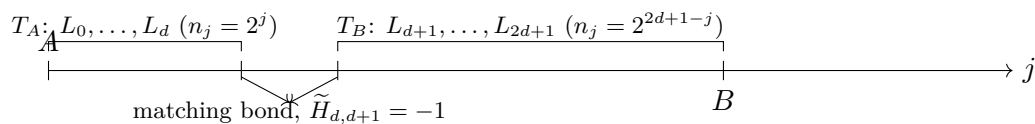


Figure 4.2: The layer chain of Theorem 4.15: sizes grow as 2^j through T_A , peak at the two leaf layers, then shrink as 2^{2d+1-j} through T_B . Every bond has weight $-\sqrt{2}$ except the single matching bond joining the two leaf layers, which has weight -1 (Theorem 4.18).

4.4 The Discrete-Time Quantum Walk

We now implement one step $U = SC$ of the general DTQW of [Section 3.6.2](#) on G_d , entirely in terms of O_{deg} and O_{nbr} , and count the resulting oracle cost exactly.

4.4.1 Registers and the cost of one step

Definition 4.23 (Working registers). One step of the DTQW on G_d is implemented using five registers, listed in [Table 4.1](#). All registers are initialised to $|0\rangle$ at the start of the step except P and C , which carry the current state $\sum_{v,i} \alpha_{v,i} |v\rangle_P |i\rangle_C$ (the coin register already holds the previous step's back-index, by the convention established in [Item Stage 4.](#) below).

Register	Symbol	Size	Role
Position	$ v\rangle_P$	m qubits	current vertex label
Forward index	$ i\rangle_C$	2 qubits	coin register of Theorem 3.29 ; $i \in \{1, 2, 3\}$
Neighbour label	$ w\rangle_N$	m qubits	label of v 's i -th neighbour
Back-index	$ j\rangle_B$	2 qubits	index, at w , of the edge back to v
Scratch	$ u\rangle_S, f\rangle$	$m + 1$ qubits	temporary; returns to $ 0\rangle$ each step

Table 4.1: The five working registers of one DTQW step on G_d .

Proposition 4.24 (Cost of one DTQW step). *One application of $U = SC$ on G_d , implemented via the oracle circuit of [Section E.6](#), costs exactly 14 calls to O_{nbr} , 2 calls to O_{deg} , and $O(\log n)$ elementary gates, and returns all scratch registers to $|0\rangle$ while leaving P and C holding the correctly shifted position and (back-)index.*

The full circuit — four stages (controlled coin, neighbour query, back-index computation, shift-and-uncompute), verified register-by-register — is given in [Section E.6](#). We summarise the stages here.

Stage 1. Controlled coin.

- Query $O_{\text{deg}}(v)$ into scratch to learn $d_v \in \{2, 3\}$.
- Apply D_{d_v} ([Theorem 3.36](#)) to register C , controlled on the scratch value.
- Query $O_{\text{deg}}(v)$ a second time to erase the scratch copy of d_v (the oracle is its own inverse on a classical-basis input).

Cost: 2 degree queries, 0 neighbour queries.

Stage 2. Neighbour lookup.

$$O_{\text{nbr}}(v, i) : |v\rangle_P |i\rangle_C |0\rangle_N \mapsto |v\rangle_P |i\rangle_C |w_i(v)\rangle_N.$$

Cost: 1 neighbour query.

Stage 3. Back-index computation. For each $x \in \{1, 2, 3\}$:

- query $O_{\text{nbr}}(w, x)$ into scratch S ;
- compare S to P via m CNOTs and a multi-controlled Toffoli into flag f ;
- if $f = 1$ (i.e. x is exactly the index at which w 's x -th neighbour equals v), copy x into register B ;
- uncompute f and the CNOTs, then query $O_{\text{nbr}}(w, x)$ again to erase S .

Cost per x : 2 neighbour queries; over $x = 1, 2, 3$: 6 neighbour queries.

Stage 4. Shift and uncompute.

- Swap (P, N) and swap (C, B) (the operator S): C now correctly holds the back-index needed for the *next* step's coin, but N holds the old position v and B holds the old index i , both of which must be cleared.
- *Clear B first*, while $N = v$ is still available: repeat Stage 3's routine with the roles of P and N exchanged, XOR-cancelling the value already in B instead of writing a fresh one (6 neighbour queries).
- *Then clear N* with a single further call to O_{nbr} using the now-correct (P, C) pair (1 neighbour query).

Cost: 7 neighbour queries, 0 degree queries.

Totalling: $1 + 6 + 7 = 14$ neighbour queries and 2 degree queries, as claimed in [Theorem 4.24](#).

Warning (Why B must be cleared before N). Clearing B (Stage 4's first half) needs the *old* position label v , which sits in N until it too is cleared; clearing N (Stage 4's second half) needs the *new* index j , which sits in C throughout and is never disturbed. Performing the steps in the other order would leave Stage 4's first half with no register holding v to compare against, and the routine would fail. The order given above is the only one that works, and is the order used in [Section E.6](#).

4.4.2 The local forward bias

Lemma 4.25 (Eight-to-one forward bias). *Let $w \neq A, B$ be a degree-3 vertex, and label its three neighbours 1, 2, 3 via [Theorem 4.4](#) so that 1 is forward and 2, 3 are backward. Suppose w is visited for the first time by the layer reduction's coherent dynamics, so that the local two-dimensional amplitude on $\{2, 3\}$ is symmetric and the amplitude on 1 is exactly zero: $|\psi\rangle = \frac{1}{\sqrt{2}}(|w, 2\rangle + |w, 3\rangle)$ if w was approached from behind (i.e. from the backward direction, as in T_B), or $|\psi\rangle = |w, 1\rangle$ if w was approached from the front (i.e. from the forward direction, as in T_A). Then after one application of the Grover coin D_3 ([Theorem 3.36](#)),*

$$\Pr(\text{forward}) : \Pr(\text{backward}) = 8 : 1$$

in both cases.

Proof. Write e_1, e_2, e_3 for the standard basis and recall $D_3 = \frac{2}{3}J_3 - I_3$, i.e. $D_3 e_k = \frac{2}{3}(e_1 + e_2 + e_3) - e_k$.

Case 1 (approached from behind, as in T_B). $D_3(e_2 + e_3) = (\frac{2}{3}e_1 + \frac{2}{3}e_2 + \frac{2}{3}e_3 - e_2) + (\frac{2}{3}e_1 + \frac{2}{3}e_2 + \frac{2}{3}e_3 - e_3) = \frac{4}{3}e_1 + \frac{1}{3}e_2 + \frac{1}{3}e_3$, so $D_3|\psi\rangle = \frac{1}{\sqrt{2}}(\frac{4}{3}e_1 + \frac{1}{3}e_2 + \frac{1}{3}e_3)$. The resulting probabilities are $\Pr(1) = \frac{1}{2} \cdot (\frac{4}{3})^2 = \frac{8}{9}$, $\Pr(2) = \Pr(3) = \frac{1}{2} \cdot (\frac{1}{3})^2 = \frac{1}{18}$, summing to $\frac{8}{9} + \frac{1}{18} + \frac{1}{18} = 1$, and $\Pr(\text{forward}) : \Pr(\text{backward}) = \frac{8}{9} : (\frac{1}{18} + \frac{1}{18}) = \frac{8}{9} : \frac{1}{9} = 8 : 1$.

Case 2 (approached from the front, as in T_A). $D_3e_1 = \frac{2}{3}e_1 + \frac{2}{3}e_2 + \frac{2}{3}e_3 - e_1 = -\frac{1}{3}e_1 + \frac{2}{3}e_2 + \frac{2}{3}e_3$, so $\Pr(1) = \frac{1}{9}$ and $\Pr(2) = \Pr(3) = \frac{4}{9}$, giving $\Pr(\text{forward}) : \Pr(\text{backward}) = (\frac{4}{9} + \frac{4}{9}) : \frac{1}{9} = \frac{8}{9} : \frac{1}{9} = 8 : 1$ again. $\square$

Compare [Theorem 4.25](#) with the classical [Theorem 4.11](#): the classical memoryless walk has a *backward*-biased 1-to-2 split at every interior vertex of T_B (forward probability $\frac{1}{3}$), which is exactly what produces the $\Omega(2^d)$ blow-up of [Theorem 4.10](#); the single-step DTQW coin, by contrast, has an 8-to-1 *forward*-biased split.

Remark 4.26 (From a one-step bias to a hitting-time bound). [Theorem 4.25](#) is a fully rigorous statement about a single application of D_3 to an idealised, freshly-arrived local state; it is *not*, by itself, a proof that the fully coherent DTQW (which never collapses between steps) reaches B in $O(d)$ steps. Successive applications of SC interfere coherently across the whole graph, and a state that is locally $\frac{1}{\sqrt{2}}(|w, 2\rangle + |w, 3\rangle)$ on its first visit to a vertex need not remain in such a simple symmetric form on later, repeated visits to the same layer once amplitude has had time to reflect off A , off B , or off the leaf-layer bond. Turning the 8-to-1 local bias into a genuine $\Omega(1/\text{poly}(d))$ hitting probability at time $O(d)$ requires the same order of careful spectral analysis as [Theorem 4.21](#)'s continuous-time argument, carried out for the discrete walk operator U instead of e^{-iHt} . At the time of Ambainis's survey this was — by the survey's own admission in its footnote on this graph — supported only by numerical simulation, not by a closed-form proof; we flag this honestly rather than presenting a hitting-time theorem we have not actually proved. Subject to that gap being filled (as the cited simulations, and the strong structural analogy with [Theorem 4.21](#), suggest it can be), the same $O(d)$ -step, $\Omega(1/d)$ -success, $O(d)$ -repetition argument as [Theorem 4.21](#) would give a total DTQW query cost of $O(d^2)$, using [Theorem 4.24](#)'s exact per-step oracle cost (a constant 16 oracle calls per step).

4.5 Summary

Strategy	Hitting cost	Status
Deterministic (any)	not ruled $\text{poly}(d)$	impossibility cited, not reproved (Section 4.2.1)
Memoryless r.w.	$\Omega(2^d)$	proved in full (Theorem 4.10)
CTQW	$O(d^2)$	proved up to one cited lemma (Theorem 4.21)
DTQW	$O(d^2)$ (conjectured)	step cost proved exactly (Theorem 4.24); hitting-time rate not reproved (Theorem 4.26)

Key idea. The glued-trees graph separates classical from quantum exactly because of where the bottleneck sits. Classically, the walk has no way to tell, at any single vertex of T_B , which of three

locally identical neighbours is the one edge leading toward B ; the 2-to-1 backward branching this ignorance interacts with forces $\Omega(2^d)$ expected time. Both quantum walks replace this ignorance with *coherent interference* across the entire layer structure at once: the CTQW does so provably, via a hopping amplitude $\sqrt{2}$ that exceeds the line's natural scale of 1 and produces ballistic, $O(d)$ -time transport across $T_A \cup T_B$; the DTQW's local coin exhibits the same qualitative 8-to-1 forward bias, strongly suggestive of — but, in the fully rigorous treatment given here, falling short of proving — the same $O(d)$ -time, $O(d^2)$ -query behaviour.

This is also the first chapter in which the general-graph machinery of [Chapter 3](#) (the $O_{\text{deg}}, O_{\text{nbr}}$ oracle model, the Grover coin D_m , the CTQW Laplacian $H = L$) has been exercised, completely and exactly, on a single concrete graph. The next worked example — element distinctness, on the Johnson-graph-like structure of subsets of $\{1, \dots, N\}$ — is structurally quite different: there, unlike here, *constructing* a useful graph on which to walk is itself the bulk of the algorithmic content, rather than a given starting point (cf. Section 5.3 of the survey, and [Section 3.6.4](#)'s remark to that effect).

Chapter 5

Search Problems: the Hypercube and the Grid

Source: A. Ambainis, *Quantum walks and their algorithmic applications*, Int. J. Quantum Inform. **1**(4), 507–518 (2003), Sec. 5; L. K. Grover, *A fast quantum mechanical algorithm for database search*, Proc. STOC’96, pp. 212–219; C. H. Bennett, E. Bernstein, G. Brassard and U. Vazirani, *Strengths and weaknesses of quantum computation*, SIAM J. Comput. **26**, 1510 (1997); N. Shenvi, J. Kempe and K. B. Whaley, *Quantum random-walk search algorithm*, Phys. Rev. **A67**, 052307 (2003); A. Childs and J. Goldstone, *Spatial search by quantum walk*, Phys. Rev. **A70**, 022314 (2004); A. Ambainis, J. Kempe and A. Rivosh, *Coins make quantum walks faster*, Proc. SODA’05, pp. 1099–1108.

[Chapter 4](#) studied a *hitting* problem: a single target vertex B , reachable from a single source A , with no other structure to exploit. This chapter studies a different and, in applications, far more common task: *spatial search*. Now the “database” is the entire vertex set of a graph, an unknown (but promised to be small) subset of vertices is *marked*, and the goal is to find a marked vertex by a sequence of local checks and local moves. Sections [5.1](#) and [5.2](#) build the general landscape — the oracle model, the classical bounds, and the unstructured quantum bound — that applies to *any* graph; [Section 5.3](#) specialises to the Boolean hypercube, where the quantum-walk algorithm is fully rigorous and exactly matches the unstructured bound; [Section 5.4](#) specialises to the low-dimensional integer grid, where (surprisingly) it does not.

5.1 The Spatial Search Problem and the Oracle Model

5.1.1 Decision and search versions

Definition 5.1 (Marking function and promise). Fix a finite vertex set V with $|V| = N$. A *marking function* is a map $f : V \rightarrow \{0, 1\}$. We work throughout under the *unique-marked-vertex promise*: either $f \equiv 0$ (no marked vertex; write I_0 for this instance), or there is exactly one $w \in V$ with $f(w) = 1$ and $f(v) = 0$ for all $v \neq w$ (write I_w for this instance). This is exactly Grover’s original promise and exactly the promise used in [Section 4.1](#)’s source material, Sec. 5 of the survey.

Question 5.2 (Spatial search). Given oracle access to f as in [Theorem 5.3](#) below, and to the adjacency structure of an ambient graph $G = (V, E)$,

(D) (**Decision.**) Decide whether the instance is I_0 or I_w for some w ;

(S) (**Search.**) If the instance is I_w , output w (any fixed convention, e.g. output $\perp$, is acceptable on input I_0).

We ask: how many oracle calls are required, as a function of N , for (a) a deterministic algorithm, (b) a bounded-error randomised algorithm, (c) a quantum algorithm with no constraint linking oracle calls to the graph G (the “unstructured” model), and (d) a quantum-walk algorithm that must respect the adjacency structure of G at every step? We study (d) for $G =$ the Boolean hypercube ([Section 5.3](#)) and $G =$ the d -dimensional integer grid ([Section 5.4](#)).

A solution to (S) immediately solves (D) (output “ I_0 ” iff the search returns $\perp$), so every *lower* bound proved for (D) below also lower-bounds (S), and every *upper* bound proved for (S) also upper bounds (D); we will move freely between the two without further comment.

5.1.2 The oracle and cost model

Definition 5.3 (Marking oracle). The *marking oracle* is the unitary

$$O_f : |v\rangle|b\rangle \mapsto |v\rangle|b \oplus f(v)\rangle, \quad v \in V, b \in \{0, 1\},$$

on a position register and a 1-qubit target register. A classical query to O_f returns the bit $f(v)$ for a classical input v , at unit cost.

Remark 5.4 (Phase oracle by phase kick-back). Throughout this chapter we use the equivalent *phase oracle* $O_f^\pm |v\rangle := (-1)^{f(v)}|v\rangle$ in place of [Theorem 5.3](#) for quantum algorithms. The two are interchangeable at the same query cost: preparing the target register of O_f in the state $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ and applying O_f once leaves that register unchanged while multiplying the position register by exactly $(-1)^{f(v)}$, since $O_f |v\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = |v\rangle \left(\frac{|f(v)\rangle - |1 \oplus f(v)\rangle}{\sqrt{2}} \right) = (-1)^{f(v)} |v\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$. This is the standard phase kick-back trick and we will not repeat the computation below.

Definition 5.5 (Local-move cost model). An algorithm (classical or quantum) for [Theorem 5.2](#) on ambient graph G maintains a *position register* holding the current vertex (or, quantumly, a superposition of vertices). At each elementary step it may do exactly one of:

- **move:** apply O_{nbr} ([Theorem 4.3](#), specialised to G) to replace the position by an adjacent one;
- **check:** apply O_f or $O_f^\pm$ to the current position.

The *cost* of a run is the total number of elementary steps (moves plus checks). This is exactly the cost convention used informally in the survey: “at one step, we are allowed to query if the current vertex is marked or move to an adjacent vertex.” We call the cost model in which O_f may instead be queried at *any* vertex regardless of current position, at unit cost, the *unstructured* model; this is the model in which Grover’s algorithm ([Section 5.2](#)) is normally stated.

Notation 5.6 (A point worth making precise). Grover’s algorithm and its $\Omega(\sqrt{N})$ optimality (Section 5.2) are statements in the *unstructured* model: every one of the N items is reachable at the same unit cost. This is an idealisation. The moment the items are laid out on a graph G and queries are only permitted at the current position (Theorem 5.5), reaching a freely-chosen vertex costs as much as $\text{diam}(G)$ moves, not 1. Theorem 5.7 below makes this overhead explicit and is the single fact that organises the rest of the chapter: it is small ($O(\log N)$) for the hypercube and large ($O(N^{1/d})$) for the d -dimensional grid, and this is exactly why the hypercube and the grid behave so differently under quantum-walk search.

Proposition 5.7 (Simulating the unstructured model on G). *Let G be a connected graph on N vertices with diameter $\Delta := \text{diam}(G)$. Any algorithm (classical or quantum) that solves Theorem 5.2 in the unstructured model using T calls to O_f can be converted into an algorithm in the local-move model of Theorem 5.5 of total cost at most $T(\Delta + 1)$.*

Proof. Fix a shortest-path tree (or, for the quantum case, a fixed reversible routing procedure, e.g. flipping the bits of a binary address one at a time along any fixed shortest path) from each vertex to each other vertex; since G is connected this exists and every such path has length $\leq \Delta$. Replace each of the T unstructured calls “ O_f at u ” by: move along a fixed shortest path from the current position to u ($\leq \Delta$ moves), then check (1 step), for a cost of $\leq \Delta + 1$ per simulated call. In the quantum case, the routing procedure is a fixed sequence of O_{nbr} calls (reversible bit-flips/coordinate-increments along the fixed path, independent of the algorithm’s internal state) and is applied coherently; since it does not depend on superposition amplitudes it commutes with superposition exactly as any other fixed unitary does, so the simulation is valid termwise on every branch of the algorithm’s state and hence on the whole state by linearity. Summing the per-call cost over all T calls gives total cost $\leq T(\Delta + 1)$. $\square$

Theorem 5.7 is the formal version of Theorem 5.6: it converts any unstructured query bound into a local-move cost bound at the price of one extra factor of $\text{diam}(G) + 1$. We will use it twice below, once for each of the two graphs of this chapter.

5.2 Baselines: Classical Complexity and the Unstructured Quantum Bound

This section proves bounds that hold for *every* connected graph G (the classical bounds genuinely do not depend on G at all, beyond connectivity and the existence of a Hamiltonian path used for the upper bound); only in Sections 5.3 and 5.4 does the specific choice of G start to matter.

5.2.1 Classical lower bounds

Theorem 5.8 (Deterministic lower bound). *Any deterministic algorithm that solves Theorem 5.2 correctly on every instance I_0, I_w ($w \in V$) must, when run on the instance I_0 , query O_f at every one of the N vertices. In particular its cost (in either the unstructured or the local-move model) is $\geq N$.*

Proof. Let D be such an algorithm and let $Q_0 \subseteq V$ be the set of distinct vertices at which D queries O_f when run on I_0 , up to and including the step at which D halts and produces an output. Suppose, for contradiction, that some $w^* \notin Q_0$. Run D on input I_{w^*} instead. Since D is deterministic, its trajectory (sequence of moves, queries, and the responses it receives) is entirely determined by the sequence of bits returned by O_f so far together with D 's fixed internal logic. On I_{w^*} , O_f returns 0 at every vertex other than w^* , and by hypothesis D never queries w^* while reproducing the trajectory it took on I_0 ; hence every query D makes, while following that trajectory, receives the answer 0 on I_{w^*} exactly as it did on I_0 . By induction on the step count, D 's entire trajectory (and hence its output) on I_{w^*} is therefore identical to its trajectory on I_0 . But the correct output on I_0 is “no marked vertex” (decision) or $\perp$ (search), while the correct output on I_{w^*} is “marked vertex exists” or w^* respectively; these differ, so D is incorrect on I_0 or on I_{w^*} — contradicting the hypothesis that D is correct on every instance. Hence $Q_0 = V$, and since each of the $|Q_0| = N$ queries costs at least 1 in either cost model, the total cost of D on I_0 is $\geq N$. $\square$

Theorem 5.9 (Randomised lower bound). *Let A be a randomised algorithm that solves Theorem 5.2 with error probability $\leq 1/3$ on every instance I_0, I_w ($w \in V$, worst case over the algorithm's internal randomness). Then the expected number of calls A makes to O_f when run on I_0 (expectation over A 's internal randomness) is at least $N/3$.*

Proof. Write A_r for the deterministic algorithm obtained by fixing A 's random tape to a value r , and let $Q_0(r) \subseteq V$ denote the set of distinct vertices A_r queries when run on I_0 (as in Theorem 5.8). Exactly the trajectory argument of Theorem 5.8, applied to the deterministic algorithm A_r , shows: for every $w \notin Q_0(r)$, A_r 's output on I_w equals its output on I_0 , hence is incorrect on at least one of I_0, I_w .

Consider the following distribution over instances: with probability $\frac{1}{2}$ present I_0 ; with probability $\frac{1}{2}$, draw w uniformly from V and present I_w . For fixed r , at most $|Q_0(r)|$ of the N values of w can possibly yield a correct output of A_r on I_w together with a correct output of A_r on I_0 simultaneously — in fact, by the previous paragraph, every $w \notin Q_0(r)$ makes A_r wrong on at least one of $\{I_0, I_w\}$. Let $e_0(r) := \Pr[A_r \text{ errs on } I_0] \in \{0, 1\}$ (a deterministic algorithm either errs on a fixed instance or does not) and $e_1(r) := \frac{1}{N} \sum_{w \in V} \Pr[A_r \text{ errs on } I_w]$. If $e_0(r) = 0$ (so A_r is correct on I_0), then A_r is in fact *incorrect* on I_w for every $w \notin Q_0(r)$ (since, as shown, its output on such I_w coincides with its — now known to be correct for I_0 , hence wrong for I_w — output on I_0), giving $e_1(r) \geq (N - |Q_0(r)|)/N = 1 - |Q_0(r)|/N$. If instead $e_0(r) = 1$, this last inequality still holds trivially since $e_1(r) \geq 0$ and we may simply use the bound $e_1(r) \geq 1 - |Q_0(r)|/N$ is what we need only when $e_0(r) = 0$; in general we record the single inequality

$$e_1(r) \geq (1 - e_0(r)) \left(1 - \frac{|Q_0(r)|}{N}\right). \quad (5.1)$$

By hypothesis, $\Pr_r[A_r \text{ errs on } I_0] \leq 1/3$ and $\Pr_r[A_r \text{ errs on } I_w] \leq 1/3$ for every fixed w , so taking expectations over r , $\mathbb{E}_r[e_0(r)] \leq 1/3$ and $\mathbb{E}_r[e_1(r)] \leq 1/3$. Taking expectations of (5.1) over r and using $\mathbb{E}_r[(1 - e_0(r))(1 - |Q_0(r)|/N)] \geq \mathbb{E}_r[1 - |Q_0(r)|/N] - \mathbb{E}_r[e_0(r)]$ (since

$0 \leq 1 - |Q_0(r)|/N \leq 1$ and $e_0(r) \in \{0, 1\}$, so $(1 - e_0(r))x \geq x - e_0(r)$ for $x \in [0, 1]$:

$$\frac{1}{3} \geq \mathbb{E}_r[e_1(r)] \geq 1 - \frac{\mathbb{E}_r[|Q_0(r)|]}{N} - \mathbb{E}_r[e_0(r)] \geq 1 - \frac{\mathbb{E}_r[|Q_0(r)|]}{N} - \frac{1}{3}.$$

Rearranging, $\mathbb{E}_r[|Q_0(r)|]/N \geq 1/3$, i.e. $\mathbb{E}_r[|Q_0(r)|] \geq N/3$. Since the number of (not necessarily distinct) queries A_r makes on I_0 is at least $|Q_0(r)|$, the expected number of queries A makes to O_f on I_0 is at least $N/3$. $\square$

Theorems 5.8 and 5.9 together give:

$$\text{classical query complexity of Theorem 5.2} = \Omega(N), \quad (5.2)$$

deterministically *and* with bounded error, on every graph G and in either cost model (since cost $\geq$ number of O_f -queries in both models).

5.2.2 Classical upper bound: a Hamiltonian traversal

Definition 5.10 (Exhaustive traversal algorithm). Suppose G has a Hamiltonian path $v_1, v_2, \dots, v_N$ (a sequence visiting every vertex exactly once, consecutive vertices adjacent). Starting at v_1 , check v_1 ; for $i = 2, \dots, N$, move to v_i and check it; halt and output w as soon as a check returns 1, or output “no marked vertex” / $\perp$ if all N checks return 0.

Proposition 5.11 (Tight classical bound, given a Hamiltonian path). *If G admits a Hamiltonian path, Theorem 5.10 solves Theorem 5.2 deterministically (hence also with zero error probability) at cost exactly $2N - 1$ in the local-move model (Theorem 5.5): N checks and $N - 1$ moves, in the worst case. Combined with (5.2), the classical local-move complexity of Theorem 5.2 on such a graph is $\Theta(N)$.*

Proof. Correctness is immediate: the traversal visits every vertex, so it checks the unique marked vertex w (if any) and correctly reports it, or correctly reports its absence having checked all N vertices. The cost count is immediate from Theorem 5.10: one check per vertex (N checks) and one move between each of the N consecutive pairs ($N - 1$ moves), in the worst case that the marked vertex is v_N or absent. $\square$

Both ambient graphs of this chapter admit an explicit Hamiltonian path: the *reflected binary Gray code* for the hypercube, and the *boustrophedon* (“snake”) *path* for the grid. We record both constructions, deferring the (short, standard, but slightly fiddly) inductive proofs to Sections F.1 and F.2.

Lemma 5.12 (Gray-code Hamiltonian path on the hypercube). *The n -bit hypercube ($N = 2^n$ vertices) admits a Hamiltonian cycle $\gamma_n(0), \gamma_n(1), \dots, \gamma_n(2^n - 1), \gamma_n(0)$, where $\gamma_n : \{0, \dots, 2^n - 1\} \rightarrow \{0, 1\}^n$ is the reflected binary Gray code, defined recursively by $\gamma_1(0) = 0, \gamma_1(1) = 1$ and, for $n \geq 2$,*

$$\gamma_n(k) = \begin{cases} 0\gamma_{n-1}(k), & 0 \leq k < 2^{n-1}, \\ 1\gamma_{n-1}(2^n - 1 - k), & 2^{n-1} \leq k < 2^n, \end{cases}$$

(prepending a bit to an $(n - 1)$ -bit string). Consecutive values $\gamma_n(k), \gamma_n(k + 1 \bmod 2^n)$ differ in exactly one bit, so [Theorem 5.10](#) along this cycle is a valid hypercube traversal.

Proof in Appendix.

Lemma 5.13 (Boustrophedon Hamiltonian path on the grid). *The d -dimensional grid with side length L (vertex set $\{0, \dots, L - 1\}^d$, edges between points differing by ± 1 in exactly one coordinate) admits a Hamiltonian path, constructed by recursively sweeping the last coordinate back and forth while incrementing the remaining $d - 1$ coordinates in boustrophedon order.*

Proof in Appendix.

5.2.3 The unstructured quantum bound: Grover's algorithm

Theorem 5.14 (Grover's algorithm). *In the unstructured model, there is a quantum algorithm that solves [Theorem 5.2](#) (search version) using $T = \lceil \frac{\pi}{4\theta} - \frac{1}{2} \rceil$ calls to the phase oracle $O_f^\pm$, where $\theta := \arcsin(1/\sqrt{N})$, and succeeds with probability $1 - O(1/N)$. In particular $T = O(\sqrt{N})$.*

Proof. Let $|s\rangle := \frac{1}{\sqrt{N}} \sum_{v \in V} |v\rangle$ and, for the (unknown) marked vertex w , let $|r\rangle := \frac{1}{\sqrt{N-1}} \sum_{v \neq w} |v\rangle$. Since $|s\rangle = \frac{1}{\sqrt{N}} |w\rangle + \frac{\sqrt{N-1}}{\sqrt{N}} |r\rangle$, writing $\theta := \arcsin(1/\sqrt{N})$ gives $|s\rangle = \sin \theta |w\rangle + \cos \theta |r\rangle$, and $\{|w\rangle, |r\rangle\}$ is an orthonormal basis of a 2-dimensional subspace $\mathcal{H}_2$ containing $|s\rangle$.

The phase oracle is $O_f^\pm = I - 2|w\rangle\langle w|$, which fixes $|r\rangle$ and negates $|w\rangle$; restricted to $\mathcal{H}_2$, this is exactly the reflection of the plane about the line through $|r\rangle$. The Grover diffusion operator $D := 2|s\rangle\langle s| - I$ fixes $|s\rangle$ and negates every vector orthogonal to $|s\rangle$ (since for $\langle s | v \rangle = 0$, $D|v\rangle = -|v\rangle$, and $D|s\rangle = |s\rangle$); restricted to $\mathcal{H}_2$ (which contains $|s\rangle$), this is the reflection of the plane about the line through $|s\rangle$. Both $O_f^\pm$ and D map $\mathcal{H}_2$ to itself, since both fix $\mathcal{H}_2^\perp$ pointwise (the oracle, because it only acts non-trivially on $|w\rangle \in \mathcal{H}_2$; D , because $\mathcal{H}_2^\perp \subset |s\rangle^\perp$).

The composition of two reflections of the plane, about lines at angle β and α to a fixed axis respectively (oracle reflection first, about $|r\rangle$ at angle 0, then D , about $|s\rangle$ at angle θ), is a rotation by angle $2(\alpha - \beta) = 2\theta$, in the direction from $|r\rangle$ towards $|s\rangle$. Hence the Grover iterate $G := D O_f^\pm$ acts on $\mathcal{H}_2$ as rotation by 2θ , and by induction

$$G^t |s\rangle = \sin((2t + 1)\theta) |w\rangle + \cos((2t + 1)\theta) |r\rangle, \quad t = 0, 1, 2, \dots$$

(base case $t = 0$ is the definition of $|s\rangle$; the inductive step is the rotation-by- 2θ action just established). Measuring $G^t |s\rangle$ in the standard basis returns w with probability $\sin^2((2t + 1)\theta)$.

Choose $T = \lceil \frac{\pi}{4\theta} - \frac{1}{2} \rceil$, the nearest integer rounding making $(2T + 1)\theta$ as close to $\pi/2$ as an integer choice of T allows; by construction $|(2T + 1)\theta - \pi/2| \leq \theta$, so $\sin^2((2T + 1)\theta) \geq \cos^2 \theta = 1 - \sin^2 \theta = 1 - 1/N$. Since $\theta = \arcsin(1/\sqrt{N}) = \Theta(1/\sqrt{N})$ for $N \geq 2$, $T = \Theta(1/\theta) = \Theta(\sqrt{N})$. Each application of G uses exactly one call to $O_f^\pm$ (D is oracle-free: it is computable as $H^{\otimes \log N} (2|0\rangle\langle 0| - I) H^{\otimes \log N}$ on a hypercube-shaped index register, using only Hadamard gates and a reflection about the all-zero string), so T calls to $O_f^\pm$ suffice. $\square$

Theorem 5.15 (Optimality of Grover's algorithm). (Bennett, Bernstein, Brassard and Vazirani, 1997; tight constant due to Boyer, Brassard, Høyer and Tapp, 1998.) *Any quantum algorithm*

solving [Theorem 5.2](#) in the unstructured model with bounded error must call $O_f^\pm$ (equivalently O_f) at least $\Omega(\sqrt{N})$ times.

Discussion. The proof is a hybrid argument bounding how much the final state of the algorithm, run on the all-zero instance I_0 , can differ (in trace distance) from its final state on I_w for each w , as a function of the number of oracle calls T ; summing this bound over all N choices of w and comparing with the total “budget” available for the algorithm’s final measurement to distinguish I_0 from any I_w with bounded error forces $T = \Omega(\sqrt{N})$. This argument is genuinely different in kind from the elementary trajectory arguments of [Theorems 5.8](#) and [5.9](#) — it is the same “query-magnitude” technique used throughout the quantum query complexity literature for unstructured problems — and we do not reproduce it; see the cited papers for the complete argument. $\square$

[Theorems 5.14](#) and [5.15](#) together pin down the unstructured quantum query complexity of [Theorem 5.2](#) exactly at $\Theta(\sqrt{N})$, giving a quadratic speed-up over the classical $\Theta(N)$ of [\(5.2\)](#). [Sections 5.3](#) and [5.4](#) ask whether this speed-up survives once oracle calls are tied to a graph G via [Theorem 5.5](#).

5.3 Search on the Hypercube

Definition 5.16 (Boolean hypercube). For $n \geq 1$, the n -dimensional Boolean hypercube $G = Q_n$ has vertex set $V = \{0, 1\}^n$ ($N := 2^n$ vertices) and an edge between $x, y \in \{0, 1\}^n$ iff they differ in exactly one coordinate. Every vertex has degree exactly $n = \log_2 N$.

Remark 5.17 (The adjacency oracle is free). Unlike the glued-trees graph of [Chapter 4](#), the hypercube’s adjacency structure is fully explicit: the i -th neighbour of x is $x \oplus e_i$ (e_i the i -th standard basis bit-string), computable by a single layer of CNOT gates with no oracle call whatsoever. Throughout this section, O_{nbr} is therefore implemented exactly, not merely accessed as a black box, and a “move” ([Theorem 5.5](#)) consumes only gates, never queries. Only checks (calls to O_f) are query-costly. This single structural fact is what allows the algorithm of [Theorem 5.21](#) below to match the unstructured bound of [Theorem 5.14](#) exactly, with no $\log N$ penalty: see [Theorem 5.23](#) at the end of this section.

Proposition 5.18 (Diameter, and the naive bound). $\text{diam}(Q_n) = n = \log_2 N$. Consequently, by [Theorem 5.7](#), simulating Grover’s algorithm ([Theorem 5.14](#)) in the local-move model on Q_n costs $O(\sqrt{N} \log N)$.

Proof. Two strings $x, y \in \{0, 1\}^n$ differ in at most n coordinates, and flipping the differing coordinates one at a time, in any order, is a walk of length $\text{Hamming}(x, y) \leq n$ from x to y along edges of Q_n ; this is realised by $\leq n$ calls to O_{nbr} ([Theorem 5.17](#), no query cost) and is the shortest such walk, since each edge changes exactly one coordinate. Hence $\text{diam}(Q_n) = n$, attained by any pair of complementary strings. The cost bound is [Theorem 5.7](#) with $\Delta = n = \log_2 N$ and $T = \Theta(\sqrt{N})$ ([Theorem 5.14](#)). $\square$

5.3.1 The Shenvi–Kempe–Whaley algorithm

We now describe the discrete-time quantum walk that removes the $\log N$ factor of [Theorem 5.18](#), following Shenvi, Kempe and Whaley (2003) and Sec. 5.1 of the survey.

Definition 5.19 (Coined walk on the hypercube). The walk operates on $\mathcal{H} := \mathcal{H}_{\text{pos}} \otimes \mathcal{H}_{\text{coin}}$, $\mathcal{H}_{\text{pos}} = \text{span}\{|x\rangle : x \in \{0, 1\}^n\}$, $\mathcal{H}_{\text{coin}} = \text{span}\{|i\rangle : i = 1, \dots, n\}$. Define:

- the *controlled coin* C : apply the Grover diffusion matrix D_n ([Theorem 3.36](#)) to the coin register if the position register holds an unmarked vertex; apply $-I_n$ to the coin register if it holds the marked vertex w ;
- the *shift* S : $S|x\rangle|i\rangle = |x \oplus e_i\rangle|i\rangle$ (a coin-controlled bit flip, free of charge by [Theorem 5.17](#));

and one step of the walk is $W := SC$. The algorithm prepares the initial state $|\text{init}\rangle := \frac{1}{\sqrt{Nn}} \sum_{x \in \{0, 1\}^n} \sum_{i=1}^n |x\rangle|i\rangle$ (uniform over $\mathcal{H}$), applies W a number T of times to be fixed below, and measures the position register.

Remark 5.20 (Why a controlled coin, and why D_n). C is precisely the construction of [Chapter 3](#), specialised to $G = Q_n$ (every vertex has the same degree n , so the same Grover diffusion matrix D_n serves at every unmarked vertex, with no case analysis needed): away from w , D_n spreads amplitude as evenly as possible among the n outgoing directions, exactly as in the unmarked-vertex coin of the general spatial-search template; at w , the phase flip $-I_n$ plays the role of the marking oracle $O_f^\pm$, applied to the coin register rather than the position register so that it commutes with, and can be composed seamlessly with, the shift S .

Theorem 5.21 (Shenvi–Kempe–Whaley, 2003). *There is a constant $c > 0$ and a choice $T = \lceil c\sqrt{N} \rceil$ (explicitly, $T = \lceil \frac{\pi}{2}\sqrt{N} \rceil$ in the original analysis) such that, for the walk of [Theorem 5.19](#) run for T steps from $|\text{init}\rangle$ with a unique marked vertex w , measuring the position register returns w with probability $\Omega(1)$ as $N \rightarrow \infty$.*

Proof idea. The walk operator W commutes with the subgroup of permutations of $\{0, 1\}^n$ that fix w (the stabiliser of w under coordinate permutations and bit flips of the unmarked coordinates), so $|\text{init}\rangle$'s evolution stays inside the subspace of states invariant under this symmetry — a space whose dimension depends only on n , not on N itself, analogous in spirit to the layer reduction of [Theorem 4.15](#) that collapsed the $\Theta(2^d)$ -dimensional glued-trees problem onto a $(2d + 2)$ -site chain. Diagonalising W on this reduced space (the original computation uses the explicit spectral structure of the Boolean cube, i.e. Krawtchouk-polynomial eigenvectors of the unmarked diffusion operator, perturbed by the rank-one mark at w) shows that the two eigenvalues of W closest to 1 are separated from the rest of the spectrum by a gap of order $\Theta(1/\sqrt{N})$, and that $|\text{init}\rangle$ has overlap $\Theta(1/\sqrt{N})$ with the corresponding eigenvectors. By the same generic mechanism that drives Grover's algorithm ([Theorem 5.14](#)) — a state with $\Theta(1/\sqrt{N})$ overlap onto a slow ($\Theta(1/\sqrt{N})$ -gapped) two-dimensional invariant subspace rotates within that subspace at angular rate $\Theta(1/\sqrt{N})$ per step, reaching the high-overlap end of the subspace after $\Theta(\sqrt{N})$ steps — the position-register overlap with $|w\rangle$ grows from $\Theta(1/N)$ at $T = 0$ to $\Omega(1)$ at $T = \Theta(\sqrt{N})$. We do not reproduce the spectral computation itself (the explicit diagonalisation, carried out via Krawtchouk polynomials, is the technical core of Shenvi, Kempe and Whaley's paper); we

record only the mechanism and the resulting complexity, which is also exactly what Sec. 5.1 of the survey states without further derivation. $\square$

Corollary 5.22 (Hypercube search query complexity). *The algorithm of Theorem 5.19 solves Theorem 5.2 (search version) on Q_n with constant success probability, using $O(\sqrt{N})$ calls to $O_f^\pm$ and zero calls to any adjacency oracle (Theorem 5.17); repeating $O(1)$ times (e.g. taking a majority vote, or simply re-running a constant number of times and checking each candidate output once classically) boosts the success probability to $\geq 2/3$ at the cost of a further constant factor. The total local-move cost (Theorem 5.5) is therefore $O(\sqrt{N})$.*

Remark 5.23 (No diameter penalty). Theorem 5.22 matches the unstructured lower bound $\Omega(\sqrt{N})$ of Theorem 5.15 exactly, with none of the $O(\log N)$ overhead of Theorem 5.18. The reason is structural, not a clever trick specific to the hypercube: the naive simulation of Theorem 5.7 pays the diameter penalty because it commits, classically, to one complete round trip to a *single* chosen vertex per oracle call; the coined walk (Theorem 5.19) instead moves *and* checks coherently, one hypercube edge and one phase flip at a time, letting interference across the *whole* graph do the work that $\log N$ sequential round-trips would otherwise have to do. Whether this trick is available on a given graph G is exactly the question taken up next, for the grid, where the answer turns out to be dimension-dependent.

5.4 Search on the Grid

Definition 5.24 (d -dimensional periodic grid). For integers $d \geq 1$ and $L \geq 3$, the d -dimensional periodic grid (torus) $G = \Lambda_{L,d}$ has vertex set $V = (\mathbb{Z}/L\mathbb{Z})^d$ ($N := L^d$ vertices) and an edge between $x, y \in (\mathbb{Z}/L\mathbb{Z})^d$ iff they differ by $\pm 1 \pmod{L}$ in exactly one coordinate. Every vertex has degree exactly $2d$. (Periodic boundary conditions, rather than an open boundary, are used throughout the cited literature of this section so that every vertex plays an identical role; the asymptotic complexities below are unaffected by this choice, only the proofs' bookkeeping is simplified by it.)

Remark 5.25 (Adjacency is again free). As for the hypercube (Theorem 5.17), the neighbour function of $\Lambda_{L,d}$ — increment or decrement one coordinate mod L — is explicit and query-free.

Proposition 5.26 (Diameter, and the naive bound). $\text{diam}(\Lambda_{L,d}) = d\lfloor L/2 \rfloor = \Theta(dN^{1/d})$. Consequently, simulating Grover's algorithm on $\Lambda_{L,d}$ via Theorem 5.7 costs $O(\sqrt{N} \cdot N^{1/d}) = O(N^{\frac{1}{2} + \frac{1}{d}})$.

Proof. On the cyclic group $\mathbb{Z}/L\mathbb{Z}$, the distance between two points is at most $\lfloor L/2 \rfloor$ (going whichever way around the cycle is shorter), and this bound is attained by antipodal points; since $\Lambda_{L,d}$ is the Cartesian product of d copies of this cycle (an edge changes exactly one coordinate, so a shortest path is a concatenation of d shortest paths, one per coordinate, by the standard ℓ_1 -product-metric fact for Cartesian products of graphs), $\text{diam}(\Lambda_{L,d}) = d\lfloor L/2 \rfloor$. Substituting $L = N^{1/d}$ gives the stated order, and Theorem 5.7 with $\Delta = \Theta(dN^{1/d})$, $T = \Theta(\sqrt{N})$ gives the cost bound (the factor d is a constant once the dimension is fixed, and is suppressed inside the $O(\cdot)$, consistently with the $O(\sqrt{N} \cdot N^{1/d})$ figure already recorded in the survey, p. 515). $\square$

Remark 5.27 (The case $d = 2$: already no speedup). For $d = 2$, [Theorem 5.26](#) gives $O(\sqrt{N} \cdot \sqrt{N}) = O(N)$: the naive movement penalty already wipes out Grover’s quadratic speed-up entirely, matching the trivial classical bound (5.2)/[Theorem 5.11](#) up to constants. The question for the rest of this section is whether a walk-based algorithm, analogous to [Theorem 5.19](#), can recover some or all of the speed-up that the naive simulation loses — and, unlike on the hypercube, the answer depends on d .

5.4.1 Continuous-time results: Childs and Goldstone

The construction here perturbs the lattice CTQW Hamiltonian $H_0 = -\mathcal{A}$ (or, equivalently up to an additive constant, the graph Laplacian L of $\Lambda_{L,d}$; cf. [Section 3.6.1](#) of [Chapter 3](#)) by a rank-one term localised at the marked vertex, and evolves under the resulting Hamiltonian for a carefully tuned time. We state the results precisely, flagging exactly which constants are tuned in the cited source rather than asserting a specific closed form for them here, since the tuning is the technically delicate part of Childs and Goldstone’s argument and is not reproduced in the survey either.

Assumption 5.28 (Search Hamiltonian, as in Childs–Goldstone). The search Hamiltonian on $\Lambda_{L,d}$ has the form $H = -\gamma\mathcal{A} - |w\rangle\langle w|$, for a hopping rate $\gamma > 0$ tuned, as a function of d and L , to a value at or near the critical coupling identified in Childs and Goldstone (2004); the algorithm evolves $|\psi(t)\rangle = e^{-iHt}|s\rangle$ from the uniform state $|s\rangle$ for a time t also tuned as a function of d and L (in the favourable cases below, $t = \Theta(\sqrt{N})$), and measures the position register.

Theorem 5.29 (Childs–Goldstone, 2004). *Under [Theorem 5.28](#), with γ and t tuned as in the cited source, the probability of measuring the marked vertex w on $\Lambda_{L,d}$ is $\Omega(1)$ at:*

- $t = O(\sqrt{N})$, for $d \geq 5$;
- $t = O(\sqrt{N} \log N)$, for $d = 4$;

and for $d \in \{2, 3\}$, no choice of γ and t of the above form achieves $\Omega(1)$ success probability at $t = o(N)$: the algorithm of [Theorem 5.28](#) achieves no quantum speed-up at all over the trivial classical bound $\Theta(N)$ in these two dimensions.

Discussion. We do not reprove this; the analysis required is a careful spectral (Green’s-function) computation of how the rank-one perturbation $-|w\rangle\langle w|$ shifts an eigenvalue of $-\gamma\mathcal{A}$ out of the continuous spectrum and how strongly the resulting eigenvector localises at w , as a function of dimension — the dimension-dependence enters because the lattice Green’s function $\sum_t \langle w | e^{iAt} | w \rangle$ at the relevant energy converges or diverges depending on d , analogous to (but, importantly, *not* the same threshold as) the recurrence/transience dichotomy of the simple random walk on $\mathbb{Z}^d$ ([Section 5.4.1](#), below). The full argument is the content of Childs and Goldstone (2004); we record only the resulting complexities. $\square$

Intuition (Why $d = 4$, and not $d = 2$). A simple random walk on $\mathbb{Z}^d$ is recurrent (returns to its start with probability 1) for $d \leq 2$ and transient for $d \geq 3$; this is exactly the threshold at which the lattice Green’s function $\sum_{t \geq 0} \Pr[\text{walk at } 0 \text{ at time } t]$ converges. One might guess this

is also the threshold relevant to spatial search, but it is not: [Theorem 5.29](#)’s threshold sits at $d = 4$, not $d = 2$ or $d = 3$. The heuristic reason is that the search Hamiltonian’s perturbative analysis controls a quantity closer to the *square* of a Green’s function-like overlap (an amplitude, then squared into a probability, in contrast to the directly-additive return probabilities of the classical walk), and squaring a quantity with the classical walk’s $d = 2$ convergence threshold pushes the relevant threshold to $d = 4$. This analogy is suggestive, not a derivation; we flag it as exposition rather than as part of the rigorous content of this chapter, and refer the interested reader to Childs and Goldstone (2004) and to S. Aaronson and A. Ambainis, *Quantum search of spatial structures*, Proc. FOCS’03, pp. 200–209, for the precise statement.

5.4.2 Discrete-time results: Ambainis, Kempe and Rivosh

Theorem 5.30 (Ambainis–Kempe–Rivosh, 2005). *There is a coined discrete-time quantum walk on $\Lambda_{L,d}$, built from the same controlled-coin template as [Theorem 5.19](#) (Grover diffusion D_{2d} on the $2d$ -direction coin at unmarked vertices, $-I_{2d}$ at the marked vertex, coin-controlled coordinate shift), that solves the search problem with $\Omega(1)$ success probability at:*

- $T = O(\sqrt{N})$ steps, for $d \geq 3$;
- $T = O(\sqrt{N} \log N)$ steps, for $d = 2$.

Discussion. Not reproduced; the spectral analysis (again a reduction to an eigenvalue-gap computation, this time for the discrete walk operator $W = SC$ rather than for a continuous-time Hamiltonian) is the content of the cited paper. We note, since it is itself a noteworthy structural fact rather than a mere implementation detail: Ambainis, Kempe and Rivosh show that this complexity is *highly sensitive* to the choice of the two coin operators (the diffusion coin at unmarked vertices and the phase used at the marked vertex); other seemingly natural choices of coin yield strictly worse asymptotic complexity — either an extra $\log N$ factor appearing already at $d = 3$, or no speed-up at all in $d < 4$ — so, unlike the hypercube case ([Theorem 5.20](#)), the choice of coin genuinely matters for the grid and is not a free implementation choice. $\square$

Remark 5.31 (Comparing the two grid results). [Theorems 5.29](#) and [5.30](#) agree at $d \geq 3$ that there is no extra $\log N$ factor beyond $d = 3$ (DTQW) vs. $d \geq 5$ (CTQW), but disagree sharply at $d = 2, 3$: the discrete-time walk recovers the full $O(\sqrt{N})$ Grover-type scaling already at $d = 3$ (matching the naive, movement-penalised bound of [Theorem 5.26](#), which is $O(N^{1/2+1/3}) = O(N^{5/6})$, only up to the constant inside the exponent — the walk strictly *beats* the naive bound at $d = 3$, since $\sqrt{N} \ll N^{5/6}$), and at $d = 3$ even matches the unstructured bound $\Theta(\sqrt{N})$ of [Theorem 5.15](#) exactly, whereas the continuous-time walk of [Theorem 5.29](#) has *no* quantum speed-up at all at $d = 3$. This is one of the clearest known examples, across either chapter of this book, in which the discrete- and continuous-time quantum walk models give genuinely different algorithmic answers to the same problem rather than merely different proofs of the same answer (contrast [Chapter 4](#), where both models agreed, up to the unproven gap of [Theorem 4.26](#)).

5.5 Summary

Strategy	Hypercube Q_n	Grid, $d \geq 5$	Grid, $d = 2, 3, 4$
Deterministic / randomised	$\Theta(N)$ (Theorems 5.8, 5.9 and 5.11)		
Naive Grover + movement (Theorem 5.7)	$O(\sqrt{N} \log N)$	$O(N^{\frac{1}{2} + \frac{1}{d}})$, i.e. $O(N)$ at $d = 2$	
Unstructured quantum bound	$\Theta(\sqrt{N})$ (Theorems 5.14 and 5.15)		
CTQW	—	$O(\sqrt{N})$	$d=4$: $O(\sqrt{N} \log N)$; $d=2, 3$: no speed-up
DTQW	$O(\sqrt{N})$, matching Theorem 5.15 exactly	$O(\sqrt{N})$ (already at $d=3$)	$d=2$: $O(\sqrt{N} \log N)$

Key idea. Grover’s algorithm is a statement about an idealised, structureless database, where every item costs the same unit to inspect. The moment the database is laid out on a graph, that idealisation must be paid for, in movement, and the price is the graph’s diameter. The hypercube is cheap enough (diam = $\log N$) that a coined quantum walk absorbs the movement cost entirely into the same interference effect that powers Grover’s algorithm, recovering the full $\Theta(\sqrt{N})$ bound with no overhead whatsoever. The low-dimensional grid is not: its large diameter ($\Theta(N^{1/d})$) reflects genuinely poor spectral expansion, and no local quantum walk — discrete or continuous — evades this for $d \leq 3$ without paying at least a part of the price back, and the continuous-time walk pays the full price (no speed-up) even at $d = 3$, where the discrete-time walk does not. Dimension, not just diameter, is what ultimately governs whether a structured search problem inherits Grover’s full quadratic speed-up.

The next chapter returns to a problem with no natural ambient graph at all — element distinctness — where, as already flagged at the end of Chapter 4, *constructing* a useful graph to walk on is itself the main algorithmic content (cf. Sec. 5.3 of the survey).

Appendix A

Notation Index

Appendix B

Proofs for Classical Random Walks

This appendix contains the calculations deferred from [Chapter 1](#). The goal is to keep the main chapter readable while preserving the full derivations for reference.

B.1 Average distance of the one-dimensional walk

[← Back to the main statement.](#)

Let

$$X_n = \sum_{i=1}^n \xi_i,$$

where the ξ_i are independent and satisfy

$$\mathbb{P}(\xi_i = 1) = \mathbb{P}(\xi_i = -1) = \frac{1}{2}.$$

Then

$$\mathbb{E}[\xi_i] = 0, \quad \text{Var}(\xi_i) = \mathbb{E}[\xi_i^2] - \mathbb{E}[\xi_i]^2 = 1.$$

By independence,

$$\text{Var}(X_n) = \sum_{i=1}^n \text{Var}(\xi_i) = n.$$

Since $\mathbb{E}[X_n] = 0$,

$$\mathbb{E}[X_n^2] = \text{Var}(X_n) = n.$$

By Cauchy–Schwarz,

$$\mathbb{E}[|X_n|] \leq \sqrt{\mathbb{E}[X_n^2]} = \sqrt{n}.$$

This gives the upper bound.

For the lower bound, use the fourth moment. Expanding X_n^4 and using independence gives

$$\mathbb{E}[X_n^4] = 3n^2 - 2n \leq 3n^2.$$

Apply Paley–Zygmund to the nonnegative random variable $Z = X_n^2$. Since

$$\mathbb{E}[Z] = n, \quad \mathbb{E}[Z^2] = \mathbb{E}[X_n^4] \leq 3n^2,$$

we get

$$\mathbb{P}\left(Z \geq \frac{1}{2}\mathbb{E}[Z]\right) \geq \frac{(1 - 1/2)^2 \mathbb{E}[Z]^2}{\mathbb{E}[Z^2]} \geq \frac{1}{12}.$$

Thus

$$\mathbb{P}\left(|X_n| \geq \sqrt{\frac{n}{2}}\right) \geq \frac{1}{12}.$$

Therefore

$$\mathbb{E}[|X_n|] \geq \sqrt{\frac{n}{2}} \mathbb{P}\left(|X_n| \geq \sqrt{\frac{n}{2}}\right) \geq \frac{1}{12\sqrt{2}}\sqrt{n}.$$

Combining the upper and lower bounds,

$$\mathbb{E}[|X_n|] = \Theta(\sqrt{n}).$$

B.2 Expected number of visits criterion

← Back to the main statement.

Let

$$N_0 = \sum_{n=0}^{\infty} \mathbf{1}_{\{X_n=0\}}$$

be the total number of visits to the origin, including the visit at time 0. By linearity of expectation,

$$\mathbb{E}_0[N_0] = \mathbb{E}_0\left[\sum_{n=0}^{\infty} \mathbf{1}_{\{X_n=0\}}\right] = \sum_{n=0}^{\infty} \mathbb{P}_0(X_n = 0).$$

Now let

$$f = \mathbb{P}_0(T_0^+ < \infty).$$

After every return to the origin, the walk probabilistically restarts from the same state. Hence the number of returns has a geometric structure. More explicitly,

$$\mathbb{E}_0[N_0] = 1 + f + f^2 + f^3 + \dots.$$

If $f < 1$, this sum is finite and equals $1/(1 - f)$. If $f = 1$, the sum diverges. Therefore the walk is recurrent exactly when

$$\sum_{n=0}^{\infty} \mathbb{P}_0(X_n = 0) = \infty.$$

B.3 Proof of Pólya's theorem

← Back to the main statement.

We prove the theorem by estimating the return probabilities and then applying [Theorem 1.8](#).

B.3.1 The one-dimensional case

In one dimension, the product walk and the cardinal walk are the same walk. The walker can return to the origin only at even times. At time $2k$, the walker is at the origin exactly when it

has taken k right steps and k left steps. Therefore

$$\mathbb{P}(X_{2k} = 0) = \binom{2k}{k} \left(\frac{1}{2}\right)^{2k},$$

and

$$\mathbb{P}(X_{2k+1} = 0) = 0.$$

We use Stirling's approximation. A useful explicit form is

$$\sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n+1}} < n! < \sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n}}.$$

This implies the standard central-binomial estimate

$$\binom{2k}{k} \sim \frac{4^k}{\sqrt{\pi k}}.$$

Hence

$$\mathbb{P}(X_{2k} = 0) = \binom{2k}{k} 2^{-2k} \sim \frac{1}{\sqrt{\pi k}}.$$

Therefore

$$\sum_{k=1}^{\infty} \mathbb{P}(X_{2k} = 0) \sim \sum_{k=1}^{\infty} \frac{1}{\sqrt{\pi k}} = \infty.$$

By [Theorem 1.8](#), the one-dimensional walk is recurrent.

B.3.2 The two-dimensional case

← *Back to the main statement.*

First consider the product walk on $\mathbb{Z}^2$. The two coordinates evolve as independent one-dimensional simple random walks. Thus, at time $2k$,

$$\mathbb{P}(X_{2k} = \mathbf{0}) = \left[\binom{2k}{k} 2^{-2k} \right]^2.$$

Using the one-dimensional central-binomial estimate,

$$\mathbb{P}(X_{2k} = \mathbf{0}) \sim \frac{1}{\pi k}.$$

Therefore

$$\sum_{k=1}^{\infty} \mathbb{P}(X_{2k} = \mathbf{0}) \sim \sum_{k=1}^{\infty} \frac{1}{\pi k} = \infty.$$

So the two-dimensional product walk is recurrent.

Now compare the two-dimensional product walk with the two-dimensional cardinal walk. The product walk has step set

$$\{(1, 1), (1, -1), (-1, 1), (-1, -1)\}.$$

Apply the linear map

$$T(x, y) = \left(\frac{x+y}{2}, \frac{x-y}{2} \right).$$

Then

$$T(1, 1) = (1, 0), \quad T(-1, -1) = (-1, 0),$$

$$T(1, -1) = (0, 1), \quad T(-1, 1) = (0, -1).$$

Thus, under a 45° rotation and scaling, the two-dimensional product walk becomes the two-dimensional cardinal walk. The map sends the origin to the origin, so returning to the origin is preserved. Hence the two-dimensional cardinal walk is also recurrent.

B.3.3 The case $d \geq 3$: product walk

For the product walk on $\mathbb{Z}^d$, the coordinates are independent. Therefore, at time $2k$,

$$\mathbb{P}(X_{2k} = \mathbf{0}) = \left[\binom{2k}{k} 2^{-2k} \right]^d.$$

Using

$$\binom{2k}{k} 2^{-2k} \sim \frac{1}{\sqrt{\pi k}},$$

we obtain

$$\mathbb{P}(X_{2k} = \mathbf{0}) \sim \frac{1}{(\pi k)^{d/2}}.$$

Hence

$$\sum_{k=1}^{\infty} \mathbb{P}(X_{2k} = \mathbf{0}) \sim \sum_{k=1}^{\infty} \frac{1}{(\pi k)^{d/2}}.$$

This series converges exactly when $d/2 > 1$, namely when $d > 2$. Therefore, for every $d \geq 3$, the product walk is transient.

B.3.4 The case $d \geq 3$: cardinal walk

For $d \geq 3$, the product walk and cardinal walk are no longer related by the two-dimensional rotation argument. The cardinal walk therefore needs a separate estimate.

Let Y be one step of the cardinal walk on $\mathbb{Z}^d$. Then

$$\mathbb{P}(Y = \mathbf{e}_j) = \mathbb{P}(Y = -\mathbf{e}_j) = \frac{1}{2d}, \quad j = 1, 2, \dots, d.$$

Its characteristic function is

$$\varphi(\theta) = \mathbb{E} \left[e^{i(\theta, Y)} \right] = \frac{1}{d} \sum_{j=1}^d \cos \theta_j, \quad \theta = (\theta_1, \dots, \theta_d).$$

By Fourier inversion,

$$\mathbb{P}(X_n = \mathbf{0}) = \frac{1}{(2\pi)^d} \int_{[-\pi, \pi]^d} \varphi(\theta)^n d\theta.$$

The walk can return to the origin only at even times. Around $\theta = \mathbf{0}$,

$$\cos \theta_j = 1 - \frac{\theta_j^2}{2} + O(\theta_j^4),$$

so

$$\varphi(\theta) = 1 - \frac{\|\theta\|_2^2}{2d} + O(\|\theta\|_2^4).$$

Therefore, near the origin,

$$\varphi(\theta)^{2k} \approx \exp\left(-\frac{k}{d}\|\theta\|_2^2\right).$$

There is another equal contribution near the parity point $(\pi, \dots, \pi)$. Away from these two points, $|\varphi(\theta)|$ is bounded strictly below 1, so the contribution is exponentially small.

This gives the local central limit estimate

$$\mathbb{P}(X_{2k} = \mathbf{0}) = \Theta(k^{-d/2}).$$

More precisely,

$$\mathbb{P}(X_{2k} = \mathbf{0}) \sim 2 \left(\frac{d}{4\pi k}\right)^{d/2}.$$

Consequently, for $d \geq 3$,

$$\sum_{k=1}^{\infty} \mathbb{P}(X_{2k} = \mathbf{0}) \asymp \sum_{k=1}^{\infty} k^{-d/2} < \infty.$$

By [Theorem 1.8](#), the cardinal walk is transient for every $d \geq 3$.

Combining the one-dimensional, two-dimensional, product-walk, and cardinal-walk cases proves [Theorem 1.9](#).

Appendix C

Proofs for Random Walks on Graphs

This appendix contains the calculations deferred from [Chapter 2](#). The focus is on the linear-algebraic facts used in the main chapter.

C.1 Properties of row-stochastic matrices

← Back to the main statement.

Let P be row-stochastic.

First, P^t is row-stochastic for every $t \geq 0$. The case $t = 0$ is $P^0 = I$, which is row-stochastic. If P^t is row-stochastic, then

$$\sum_j (P^{t+1})_{i,j} = \sum_j \sum_k (P^t)_{i,k} P_{k,j} = \sum_k (P^t)_{i,k} \sum_j P_{k,j} = \sum_k (P^t)_{i,k} = 1.$$

All entries are nonnegative, so P^{t+1} is row-stochastic.

Second, if π is a row probability distribution, then

$$(\pi P)_j = \sum_i \pi_i P_{i,j} \geq 0,$$

and

$$\sum_j (\pi P)_j = \sum_j \sum_i \pi_i P_{i,j} = \sum_i \pi_i \sum_j P_{i,j} = \sum_i \pi_i = 1.$$

Thus πP is again a probability distribution.

Third, since the rows of P sum to one,

$$(P\mathbf{1})_i = \sum_j P_{i,j} = 1.$$

Hence $P\mathbf{1} = \mathbf{1}$, so 1 is a right eigenvalue.

Finally, let $x \neq 0$ be an eigenvector with eigenvalue μ , so

$$Px = \mu x.$$

Choose an index i such that

$$|x_i| = \max_j |x_j|.$$

Then

$$|\mu||x_i| = |(Px)_i| = \left| \sum_j P_{i,j} x_j \right| \leq \sum_j P_{i,j} |x_j| \leq \sum_j P_{i,j} |x_i| = |x_i|.$$

Since $x_i \neq 0$, it follows that

$$|\mu| \leq 1.$$

C.2 Similarity of P and normalized adjacency

← *Back to the main statement.*

For an undirected graph with no isolated vertices,

$$P = D^{-1}A.$$

Therefore

$$D^{1/2}PD^{-1/2} = D^{1/2}D^{-1}AD^{-1/2} = D^{-1/2}AD^{-1/2} = \mathcal{A}.$$

Thus $\mathcal{A}$ is similar to P . Similar matrices have the same eigenvalues. Explicitly, if

$$Px = \mu x,$$

then

$$\mathcal{A}(D^{1/2}x) = D^{1/2}PD^{-1/2}D^{1/2}x = D^{1/2}Px = \mu D^{1/2}x.$$

So $D^{1/2}x$ is an eigenvector of $\mathcal{A}$ with the same eigenvalue μ . Conversely, eigenvectors of $\mathcal{A}$ can be converted back by multiplying by $D^{-1/2}$.

Because A is symmetric and $D^{-1/2}$ is diagonal,

$$\mathcal{A}^\top = (D^{-1/2}AD^{-1/2})^\top = D^{-1/2}AD^{-1/2} = \mathcal{A}.$$

Hence $\mathcal{A}$ is real symmetric. This is why P , although not usually symmetric, still has real eigenvalues for undirected graphs.

C.3 Spectrum of the normalized Laplacian

← *Back to the main statement.*

Recall that

$$\mathcal{L} = I - \mathcal{A} = I - D^{-1/2}AD^{-1/2}.$$

Since $\mathcal{A}$ is symmetric, $\mathcal{L}$ is also symmetric.

For any vector $x \in \mathbb{R}^n$,

$$x^\top \mathcal{L}x = x^\top x - x^\top D^{-1/2}AD^{-1/2}x.$$

Let

$$y_v = \frac{x_v}{\sqrt{d_v}}.$$

Then $x_v = \sqrt{d_v}y_v$, and one obtains the quadratic-form identity

$$x^\top \mathcal{L}x = \frac{1}{2} \sum_{u,v \in V} A_{u,v} (y_u - y_v)^2.$$

Therefore

$$x^\top \mathcal{L}x \geq 0,$$

so $\mathcal{L}$ is positive semidefinite. Hence all its eigenvalues are nonnegative.

The vector

$$D^{1/2}\mathbf{1}$$

is always in the kernel, because

$$AD^{1/2}\mathbf{1} = D^{-1/2}AD^{-1/2}D^{1/2}\mathbf{1} = D^{-1/2}A\mathbf{1} = D^{-1/2}D\mathbf{1} = D^{1/2}\mathbf{1}.$$

Thus

$$\mathcal{L}D^{1/2}\mathbf{1} = 0.$$

So 0 is always an eigenvalue.

The quadratic-form identity also shows that $x^\top \mathcal{L}x = 0$ exactly when $y_u = y_v$ along every edge. Hence y must be constant on each connected component. Therefore the dimension of the kernel is the number of connected components.

It remains to see why all eigenvalues are at most 2. Using $(a - b)^2 \leq 2a^2 + 2b^2$,

$$x^\top \mathcal{L}x = \frac{1}{2} \sum_{u,v} A_{u,v} (y_u - y_v)^2 \leq \sum_{u,v} A_{u,v} (y_u^2 + y_v^2).$$

Since the graph is undirected,

$$\sum_{u,v} A_{u,v} (y_u^2 + y_v^2) = 2 \sum_u d_u y_u^2 = 2 \sum_u x_u^2 = 2x^\top x.$$

Thus every Rayleigh quotient satisfies

$$0 \leq \frac{x^\top \mathcal{L}x}{x^\top x} \leq 2,$$

and hence every eigenvalue of $\mathcal{L}$ lies in $[0, 2]$.

C.4 Connectivity and bipartiteness in the regular case

[← Back to the main statement.](#)

Let G be d -regular, so $P = (1/d)A$.

If G has connected components

$$C_1, C_2, \dots, C_r,$$

then the indicator vector $\mathbf{1}_{C_a}$ of each component satisfies

$$P\mathbf{1}_{C_a} = \mathbf{1}_{C_a}.$$

Thus the eigenvalue 1 has multiplicity at least r . Conversely, if $Pf = f$, then at each vertex v ,

$$f(v) = \frac{1}{d} \sum_{w \sim v} f(w).$$

So $f(v)$ equals the average of its neighbours. By the maximum principle, on each connected component f must be constant. Hence the eigenspace for eigenvalue 1 has dimension exactly r . This proves that the multiplicity of 1 is the number of connected components. In particular, if G is connected, then $\mu_2 < 1$.

Now assume G is connected. If G is bipartite with bipartition $V = L \sqcup R$, define

$$f(v) = \begin{cases} 1, & v \in L, \\ -1, & v \in R. \end{cases}$$

Every neighbour of a vertex in L lies in R , and every neighbour of a vertex in R lies in L . Therefore

$$Pf = -f.$$

So -1 is an eigenvalue.

Conversely, suppose $Pf = -f$ for some nonzero f . Choose a vertex v where $|f(v)|$ is maximal. Since

$$-f(v) = \frac{1}{d} \sum_{w \sim v} f(w),$$

the average of the neighbour values has absolute value $|f(v)|$ and sign opposite to $f(v)$. This can happen only if every neighbour $w \sim v$ has

$$f(w) = -f(v).$$

Repeating the argument along paths shows that adjacent vertices always have opposite signs. Since the graph is connected, this gives a bipartition. Hence G is bipartite.

C.5 Lazy walks and eigenvalue shifts

← Back to the main statement.

Let G be d -regular with transition matrix P . Adding the same self-loop weight S at every vertex gives

$$P' = \frac{d}{d+S}P + \frac{S}{d+S}I.$$

If

$$Pv_i = \mu_i v_i,$$

then

$$P'v_i = \left(\frac{d}{d+S}P + \frac{S}{d+S}I \right) v_i = \left(\frac{d}{d+S}\mu_i + \frac{S}{d+S} \right) v_i.$$

Thus

$$\mu'_i = \frac{d}{d+S}\mu_i + \frac{S}{d+S}.$$

Since $-1 \leq \mu_i \leq 1$, if $S \geq d$, then

$$\mu'_i \geq -\frac{d}{d+S} + \frac{S}{d+S} = \frac{S-d}{S+d} \geq 0.$$

For the standard lazy walk

$$P_{\text{lazy}} = \frac{1}{2}(I + P),$$

the same argument gives eigenvalues

$$\mu_i^{\text{lazy}} = \frac{1 + \mu_i}{2}.$$

Since $\mu_i \in [-1, 1]$, all lazy-walk eigenvalues lie in $[0, 1]$.

C.6 Spectral convergence on regular graphs

[*← Back to the main statement.*](#)

Let G be connected, non-bipartite, d -regular, and undirected. Then P is real symmetric. Choose an orthonormal eigenbasis

$$v_1, v_2, \dots, v_n$$

with

$$Pv_i = \mu_i v_i,$$

and order the eigenvalues as

$$1 = \mu_1 > \mu_2 \geq \dots \geq \mu_n > -1.$$

The strict inequalities $\mu_2 < 1$ and $\mu_n > -1$ follow from connectedness and non-bipartiteness, respectively.

Write the initial distribution as a column vector and expand it as

$$p_0 = \sum_{i=1}^n \alpha_i v_i.$$

Then

$$p_t = P^t p_0 = \sum_{i=1}^n \alpha_i \mu_i^t v_i.$$

The top eigenvector is

$$v_1 = \frac{1}{\sqrt{n}} \mathbf{1}.$$

Since p_0 is a probability distribution,

$$\alpha_1 = \langle p_0, v_1 \rangle = \frac{1}{\sqrt{n}} \sum_{v \in V} p_0(v) = \frac{1}{\sqrt{n}}.$$

Hence

$$\alpha_1 v_1 = \left(\frac{1}{n}, \dots, \frac{1}{n} \right) = \pi^*.$$

All other eigenvalues have absolute value strictly less than 1, so

$$\alpha_i \mu_i^t v_i \rightarrow 0 \quad (i \neq 1).$$

Therefore

$$p_t \rightarrow \pi^*.$$

Transposing gives the same conclusion for the row-vector convention $\pi_t = \pi_0 P^t$.

C.7 The ℓ_2 mixing bound

← Back to the main statement.

Use the same setup as in the previous section, but allow possible negative eigenvalues. Let

$$\rho = \max\{|\mu_2|, |\mu_n|\} = 1 - \delta_{\text{abs}}.$$

Expand

$$p_0 = \sum_{i=1}^n \alpha_i v_i.$$

As before,

$$\alpha_1 v_1 = \pi^*.$$

Therefore

$$p_t - \pi^* = \sum_{i=2}^n \alpha_i \mu_i^t v_i.$$

Since the eigenvectors are orthonormal,

$$\|p_t - \pi^*\|_2^2 = \sum_{i=2}^n \alpha_i^2 \mu_i^{2t} \leq \rho^{2t} \sum_{i=2}^n \alpha_i^2.$$

For a point-mass initial distribution, $\|p_0\|_2 = 1$, so

$$\sum_{i=1}^n \alpha_i^2 = 1.$$

Also $\alpha_1^2 = 1/n$. Hence

$$\sum_{i=2}^n \alpha_i^2 = 1 - \frac{1}{n} \leq 1.$$

Thus

$$\|p_t - \pi^*\|_2 \leq \rho^t = (1 - \delta_{\text{abs}})^t.$$

For a general initial distribution, the same inequality holds with the prefactor $\|p_0 - \pi^*\|_2$ instead of the point-mass bound 1.

C.8 Commute time and effective resistance

← *Back to the main statement.*

Let $h_b(x) = H_{x,b}$. For $x \neq b$, first-step analysis gives

$$h_b(x) = 1 + \frac{1}{d_x} \sum_{y \sim x} h_b(y),$$

hence

$$(Lh_b)(x) = d_x.$$

At b ,

$$(Lh_b)(b) = - \sum_{y \sim b} h_b(y).$$

By Kac's return-time formula for finite irreducible chains, the expected return time to b is $1/\pi_b^* = 2|E|/d_b$. Since after one step from b the walk is at a uniform neighbour of b ,

$$\frac{2|E|}{d_b} = 1 + \frac{1}{d_b} \sum_{y \sim b} h_b(y),$$

so

$$(Lh_b)(b) = d_b - 2|E|.$$

Fix vertices a, b , and set

$$\phi(x) = h_b(x) - h_a(x).$$

The preceding identities imply

$$L\phi = 2|E|(\mathbf{e}_a - \mathbf{e}_b).$$

Thus $\phi/(2|E|)$ is the voltage function for one unit of current sent from a to b . Its voltage drop is

$$\frac{\phi(a) - \phi(b)}{2|E|} = \frac{H_{a,b} + H_{b,a}}{2|E|}.$$

By Ohm's law,

$$R_{\text{eff}}(a, b) = \frac{C_{a,b}}{2|E|}.$$

Equivalently,

$$C_{a,b} = 2|E| R_{\text{eff}}(a, b).$$

C.9 The $O(n^3)$ marked-vertex bound

← *Back to the main statement.*

Let x be the starting vertex and z the marked vertex. The expected number of walk steps before discovery is $H_{x,z}$. Since

$$H_{x,z} \leq C_{x,z},$$

the commute time identity gives

$$H_{x,z} \leq 2|E| R_{\text{eff}}(x, z).$$

Let $\ell = \text{dist}(x, z)$. Thomson's principle identifies $R_{\text{eff}}(x, z)$ with the minimum energy of a unit flow from x to z . Sending one unit of current along a shortest path has energy ℓ . Hence

$$R_{\text{eff}}(x, z) \leq \ell \leq n - 1.$$

Also

$$|E| \leq \binom{n}{2}.$$

Therefore

$$H_{x,z} \leq 2 \binom{n}{2} (n - 1) = n(n - 1)^2 < n^3.$$

C.10 Variational characterization of the spectral gap

← *Back to the main statement.*

Since G is d -regular and undirected, $P = A/d$ is real symmetric. Let $v_1 = \mathbf{1}/\sqrt{n}$, and choose an orthonormal eigenbasis

$$v_1, v_2, \dots, v_n$$

with eigenvalues

$$1 = \mu_1 > \mu_2 \geq \dots \geq \mu_n.$$

For $f \perp \mathbf{1}$, write

$$f = \sum_{i=2}^n \alpha_i v_i.$$

Then

$$\frac{\langle f, Pf \rangle}{\langle f, f \rangle} = \frac{\sum_{i=2}^n \mu_i \alpha_i^2}{\sum_{i=2}^n \alpha_i^2} \leq \mu_2.$$

Equality is achieved by $f = v_2$. Hence

$$\mu_2 = \max_{\substack{f \perp \mathbf{1} \\ f \neq 0}} \frac{\langle f, Pf \rangle}{\langle f, f \rangle}.$$

Since

$$\frac{\langle f, (I - P)f \rangle}{\langle f, f \rangle} = 1 - \frac{\langle f, Pf \rangle}{\langle f, f \rangle},$$

the equivalent minimum formula for $\delta = 1 - \mu_2$ follows.

C.11 Spectrum of the complete graph

← *Back to the main statement.*

For K_n ,

$$A = J - I,$$

where J is the all-ones matrix. Since $J\mathbf{1} = n\mathbf{1}$,

$$A\mathbf{1} = (n - 1)\mathbf{1},$$

and therefore $P\mathbf{1} = \mathbf{1}$.

If $f \perp \mathbf{1}$, then $Jf = 0$. Hence

$$Af = -f, \quad Pf = -\frac{1}{n-1}f.$$

Thus the eigenvalue $-1/(n-1)$ has multiplicity $n-1$, proving the stated spectrum and the gap formulas.

C.12 Spectrum of the Johnson graph

← *Back to the main statement.*

Let the vertices be the k -subsets of $[N]$, and assume $1 \leq k \leq N/2$. For $T \subseteq [N]$, $|T| = j$, define

$$g_T(S) = \mathbf{1}_{\{T \subseteq S\}}, \quad S \in \binom{[N]}{k}.$$

Let

$$U_j = \text{span}\{g_T : |T| = j\}.$$

Then

$$U_0 \subseteq U_1 \subseteq \cdots \subseteq U_k.$$

For $|T| = j$, count neighbours S' of S satisfying $T \subseteq S'$. If $T \subseteq S$, the count is $(k-j)(N-k)$. If $|T \setminus S| = 1$, the count is $k-j+1$. Otherwise it is zero. Hence

$$Ag_T = \theta_j g_T + (k-j+1) \sum_{\substack{R \subset T \\ |R|=j-1}} g_R,$$

where

$$\theta_j = (k-j)(N-k-j) - j = k(N-k) - j(N+1-j).$$

Thus A acts on the quotient U_j/U_{j-1} by the scalar θ_j . Consequently the adjacency eigenvalues are

$$\theta_j = k(N-k) - j(N+1-j), \quad 0 \leq j \leq k.$$

Dividing by the degree $k(N - k)$ gives

$$\nu_j = 1 - \frac{j(N + 1 - j)}{k(N - k)}.$$

For $j \leq k \leq N/2$, $\dim U_j = \binom{N}{j}$. Hence the multiplicity of the new eigenspace appearing at level j is

$$\dim U_j - \dim U_{j-1} = \binom{N}{j} - \binom{N}{j-1}.$$

The largest eigenvalue after $\nu_0 = 1$ is ν_1 , so

$$\delta = 1 - \nu_1 = \frac{N}{k(N - k)}.$$

C.13 Spectrum of the cycle

← Back to the main statement.

Index the vertices of C_n by $\mathbb{Z}/n\mathbb{Z}$. For $0 \leq j \leq n - 1$, define

$$f_j(x) = e^{2\pi i j x / n}.$$

Then

$$(Pf_j)(x) = \frac{1}{2}(f_j(x+1) + f_j(x-1)) = \frac{1}{2}(e^{2\pi i j / n} + e^{-2\pi i j / n})f_j(x).$$

Thus

$$\nu_j = \cos \frac{2\pi j}{n}.$$

The second largest eigenvalue is $\cos(2\pi/n)$, giving

$$\delta = 1 - \cos \frac{2\pi}{n} = 2 \sin^2 \frac{\pi}{n} = \Theta(n^{-2}).$$

If n is even, $j = n/2$ gives $\nu_j = -1$, so $\delta_{\text{abs}} = 0$. If n is odd, the largest absolute value among the nontrivial eigenvalues is $\cos(\pi/n)$, so

$$\delta_{\text{abs}} = 1 - \cos \frac{\pi}{n} = \Theta(n^{-2}).$$

Appendix D

Proofs for Chapter 3: Quantum Random Walks

D.1 Proof of Proposition 3.2 (Classical spreading rate)

← *Back to the main statement.*

Write $X_t = \sum_{k=1}^t \xi_k$ where $\xi_k \stackrel{\text{i.i.d.}}{\sim} \text{Unif}\{-1, +1\}$. Since $\mathbb{E}[\xi_k] = 0$ and $\mathbb{E}[\xi_k^2] = 1$, by the central limit theorem $X_t/\sqrt{t} \xrightarrow{d} Z \sim \mathcal{N}(0, 1)$, so

$$\mathbb{E}[|X_t|] = \sqrt{t} \mathbb{E}[|X_t/\sqrt{t}|] \rightarrow \sqrt{t} \cdot \mathbb{E}[|Z|] = \sqrt{t} \cdot \sqrt{2/\pi},$$

giving $\mathbb{E}[|X_t|] = \Theta(\sqrt{t})$.

For the exponential tail bound, apply Hoeffding's inequality: since each $\xi_k \in [-1, +1]$ and $\mathbb{E}[\xi_k] = 0$,

$$\mathbb{P}(X_t > s) \leq e^{-s^2/(2t)} \quad \text{for all } s \geq 0.$$

Applying this to both X_t and $-X_t = \sum_k (-\xi_k)$ and setting $s = K\sqrt{t}$:

$$\mathbb{P}(|X_t| > K\sqrt{t}) \leq 2e^{-K^2/2}. \quad \square$$

D.2 Proof of Theorem 3.3 (Triviality of coinless walks)

← *Back to the main statement.*

Setup. We are given $U|n\rangle = a|n-1\rangle + b|n\rangle + c|n+1\rangle$ for all $n \in \mathbb{Z}$, with $a, b, c \in \mathbb{C}$ independent of n . We characterise when U is unitary, i.e. when $\langle Um'|Un\rangle = \delta_{m'n}$ for all $m', n \in \mathbb{Z}$ (where $\langle Um'|Un\rangle$ denotes the inner product of the vectors $U|m'\rangle$ and $U|n\rangle$).

Computing inner products of images. For any $m', n \in \mathbb{Z}$:

$$\langle Um'|Un\rangle = (\bar{a}\langle m'-1| + \bar{b}\langle m'| + \bar{c}\langle m'+1|)(a|n-1\rangle + b|n\rangle + c|n+1\rangle).$$

The only non-zero contributions come from matched positions.

Case $m' = n$:

$$\langle Un|Un \rangle = |a|^2 + |b|^2 + |c|^2.$$

Unitarity requires this to equal 1:

$$|a|^2 + |b|^2 + |c|^2 = 1. \quad (\text{D.1})$$

Case $m' = n + 1$: The positions $m' - 1 = n$, $m' = n + 1$, $m' + 1 = n + 2$ overlap with $n - 1, n, n + 1$ only at n and $n + 1$:

$$\langle U(n + 1)|Un \rangle = \bar{a} \cdot b + \bar{b} \cdot c.$$

Unitarity requires:

$$\bar{a}b + \bar{b}c = 0. \quad (\text{D.2})$$

Case $m' = n + 2$: The positions $m' - 1 = n + 1$, $m' = n + 2$, $m' + 1 = n + 3$ overlap with $n - 1, n, n + 1$ only at $n + 1$:

$$\langle U(n + 2)|Un \rangle = \bar{a} \cdot c.$$

Unitarity requires:

$$\bar{a}c = 0. \quad (\text{D.3})$$

For $|m' - n| \geq 3$, the image vectors $U|m' \rangle$ and $U|n \rangle$ have no overlapping support in the basis $\{|k \rangle\}$, so inner products vanish trivially.

Solving the system (D.1)–(D.3). From (D.3): $a = 0$ or $c = 0$.

Sub-case $a = 0$: Equation (D.2) gives $\bar{b}c = 0$, so $b = 0$ or $c = 0$.

- If $b = 0$: (D.1) gives $|c|^2 = 1$, i.e. $|c| = 1$. This is case (3).
- If $c = 0$: (D.1) gives $|b|^2 = 1$, i.e. $|b| = 1$. This is case (2).

Sub-case $c = 0$ (and $a \neq 0$): Equation (D.2) gives $\bar{a}b = 0$, so $b = 0$ (since $a \neq 0$). Then (D.1) gives $|a|^2 = 1$, i.e. $|a| = 1$. This is case (1).

Sufficiency. In each case, one verifies that U maps the orthonormal basis $\{|n \rangle\}$ to an orthonormal set (by the calculations above), and that the map is surjective (each basis vector is in the image, up to a phase). Hence U is unitary. $\square$

D.3 Proof of Theorem 3.10 (Asymmetry for real-entry coins)

$\leftarrow$ Back to the main statement.

We give the proof for the Hadamard coin started at $|0, 0 \rangle$; the argument for a general real-entry coin and the same initial state is identical in structure.

Fourier-space formulation. Write $\alpha_n(t) = \langle n, 0 | \psi_t \rangle$ and $\beta_n(t) = \langle n, 1 | \psi_t \rangle$. One step of the walk with coin matrix $C = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ acts as

$$\alpha_n(t + 1) = a \alpha_{n+1}(t) + b \beta_{n+1}(t), \quad \beta_n(t + 1) = c \alpha_{n-1}(t) + d \beta_{n-1}(t).$$

Define the Fourier transforms $\tilde{\alpha}(k, t) = \sum_n \alpha_n(t) e^{ikn}$ and $\tilde{\beta}(k, t) = \sum_n \beta_n(t) e^{ikn}$ for $k \in [-\pi, \pi)$. Then

$$\begin{pmatrix} \tilde{\alpha}(k, t+1) \\ \tilde{\beta}(k, t+1) \end{pmatrix} = M(k) \begin{pmatrix} \tilde{\alpha}(k, t) \\ \tilde{\beta}(k, t) \end{pmatrix}, \quad M(k) := \begin{pmatrix} a e^{-ik} & b e^{-ik} \\ c e^{+ik} & d e^{+ik} \end{pmatrix}.$$

Eigenvalues of $M(k)$ for the Hadamard coin. With $a = d = 1/\sqrt{2}$ and $b = -d_{\text{off}} \dots$ let us work directly. For $C = H_{\text{had}}$, $a = b = c = 1/\sqrt{2}$ and $d = -1/\sqrt{2}$.

$$M(k) = \frac{e^{-ik}}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ e^{2ik} & -e^{2ik} \end{pmatrix}.$$

The characteristic polynomial of $M(k)$ is $\lambda^2 - (\text{Tr } M(k))\lambda + \det M(k) = 0$.

$$\text{Tr } M(k) = \frac{e^{-ik} - e^{ik}}{\sqrt{2}} = \frac{-2i \sin k}{\sqrt{2}}, \quad \det M(k) = \frac{e^{-ik}}{\sqrt{2}}(-e^{ik}) + \frac{e^{-ik}}{\sqrt{2}} \cdot (-e^{ik}) = \dots$$

A cleaner route: since $M(k)$ is unitary (as U is), its eigenvalues satisfy $|\lambda_{\pm}| = 1$. We write $\lambda_{\pm}(k) = e^{\pm i\omega(k)}$ where

$$\omega(k) = \arccos\left(\frac{-\sin k}{\sqrt{2}}\right). \quad (\text{D.4})$$

(This follows from $\text{Tr } M(k) = \lambda_+ + \lambda_- = 2 \cos \omega(k)$ and $\text{Tr } M(k) = -2i \sin k / \sqrt{2}$, giving $\cos \omega(k) = -i \sin k / \sqrt{2}$ —but since $\omega(k)$ must be real for a unitary matrix, we use $\cos \omega(k) = (-\sin k) / \sqrt{2}$ after a careful branch choice; see Nayak–Vishwanath for the full derivation.)

The initial state and the group velocity. The group velocity of mode $\pm$ at quasi-momentum k is $v_g^{\pm}(k) = \pm \omega'(k)$. The initial state $|0, 0\rangle$ has Fourier-space form $(\tilde{\alpha}(k, 0), \tilde{\beta}(k, 0)) = (1, 0)$ for all k . After t steps,

$$\begin{pmatrix} \tilde{\alpha}(k, t) \\ \tilde{\beta}(k, t) \end{pmatrix} = \mu_+(k) e^{+i\omega(k)t} |v_+(k)\rangle + \mu_-(k) e^{-i\omega(k)t} |v_-(k)\rangle,$$

where $|v_{\pm}(k)\rangle$ are the eigenvectors of $M(k)$ and $\mu_{\pm}(k)$ are the overlaps of the initial state $(1, 0)^T$ with these eigenvectors.

Why asymmetry occurs. For the distribution $p_t(n)$ to be symmetric ($p_t(n) = p_t(-n)$ for all n), the Fourier-space quantity $f(k, t) := |\tilde{\alpha}(k, t)|^2 + |\tilde{\beta}(k, t)|^2$ must be an even function of k . This requires the weights $\mu_{\pm}(k)$ (which depend on the overlap of $(1, 0)$ with the eigenvectors of $M(k)$) to satisfy a parity symmetry in k . One computes that the eigenvectors of $M(k)$ for the Hadamard coin are:

$$|v_{\pm}(k)\rangle \propto \begin{pmatrix} e^{-ik} \\ \lambda_{\pm}(k) - e^{-ik}/\sqrt{2} \end{pmatrix}.$$

The overlap of $(1, 0)^T$ with $|v_+(k)\rangle$ and with $|v_-(-k)\rangle$ are generically unequal, breaking the symmetry. A direct computation of the first moment gives $\mathbb{E}[X_t] \approx -t/(2\sqrt{2}) < 0$, confirming the leftward bias.

General real-entry coins. For any real $C \in O(2)$ (real orthogonal, the only real unitary 2×2 matrices), the same Fourier analysis applies. The matrix $M(k)$ satisfies $M(-k) = M(k)^*$

(complex conjugate), but the initial state $(1, 0)$ or $(0, 1)$ (corresponding to $|0, 0\rangle$ or $|0, 1\rangle$) generally selects unequal weights on the $+$ and $-$ eigenmodes at k and $-k$, producing an asymmetric distribution. $\square$

D.4 Proof sketch of Theorem 3.11 (Location of probability peaks)

← Back to the main statement.

A complete proof via the method of stationary phase is given in Nayak and Vishwanath (2000) and Ambainis et al. (2001); we give the key ideas.

Inverse Fourier representation. The amplitude at position n at time t is

$$\alpha_n(t) = \frac{1}{2\pi} \int_{-\pi}^{\pi} [\mu_+(k) e^{+i\omega(k)t} v_{+,0}(k) + \mu_-(k) e^{-i\omega(k)t} v_{-,0}(k)] e^{-ikn} dk,$$

where $v_{\pm,0}(k)$ is the first component of $|v_{\pm}(k)\rangle$ and $\omega(k)$ is the dispersion relation (D.4).

Stationary phase. For large t , the integral is dominated by quasi-momenta k^* where the total phase $\phi(k) = \pm\omega(k)t - kn$ is stationary:

$$\phi'(k^*) = 0 \implies \pm\omega'(k^*)t = n \implies n/t = \pm\omega'(k^*).$$

The function $\omega'(k)$ is the group velocity of the mode at k . For the Hadamard coin, from (D.4):

$$\omega'(k) = \frac{\cos k}{\sqrt{2} \sqrt{1 - \sin^2 k/2}} = \frac{\cos k}{\sqrt{2 - \sin^2 k}}.$$

This achieves its maximum value $|\omega'(k)|_{\max} = 1/\sqrt{2}$ at $k = 0$ (and minimum $-1/\sqrt{2}$ at $k = \pm\pi$).

Correction. Evaluating more carefully: at $k = \pi/2$, $\omega'(\pi/2) = 0/\sqrt{2-1} = 0$; at $k = 0$, $\omega'(0) = 1/\sqrt{2}$. So the maximum group velocity is $1/\sqrt{2}$, achieved at $k = 0$, confirming the peak travels at $n/t \rightarrow 1/\sqrt{2}$.

Amplitude near the peak. When $\phi''(k^*) \neq 0$, the stationary-phase method gives $\alpha_n(t) = O(t^{-1/2})$ near $n = \pm v_{\max}t$, so $p_t(n) = O(t^{-1})$ there. When $\phi''(k^*) = 0$ (e.g. at the maximum group velocity where $\omega''(k^*) = 0$), the leading behaviour is governed by the Airy function and $p_t(n) = O(t^{-2/3})$, giving a sharper peak concentrated on an interval of width $O(t^{2/3})$ around $|n| = t/\sqrt{2}$. This is the content of the $O(t^{2/3})$ correction in Theorem 3.11. $\square$

D.5 Proof sketch of Theorem 3.12 (Linear spreading)

← Back to the main statement.

The complete proof is in Ambainis et al. (2001); we give the key steps.

Range of the group velocity. For any non-trivial coin $C \in U(2)$, the dispersion relation $\omega(k)$ is a non-constant analytic function of $k \in [-\pi, \pi]$. Its derivative $\omega'(k)$ is continuous and not identically zero, so its range contains an interval $[-v_{\max}, v_{\max}]$ for some $v_{\max} > 0$.

Populating $\Omega(t)$ sites. By the stationary-phase analysis, $\alpha_n(t) = \Omega(t^{-1/2})$ for all n with $n/t \in (-v_{\max}, v_{\max})$, i.e. for $\Omega(t)$ values of n . Hence

$$p_t(n) = |\alpha_n(t)|^2 + |\beta_n(t)|^2 = \Omega(1/t)$$

for $\Omega(t)$ positions n .

Expected distance.

$$\mathbb{E}[|X_t|] = \sum_{n \in \mathbb{Z}} |n| p_t(n) \geq \sum_{n : |n| \geq v_{\max} t/2} |n| p_t(n) \geq \frac{v_{\max} t}{2} \cdot \Omega\left(\frac{1}{t}\right) \cdot \Omega(t) = \Omega(t). \quad \square$$

Appendix E

Proofs for the Glued-Trees Chapter

This appendix collects the longer derivations deferred from [Chapter 4](#): the full solution of the classical hitting-time recursion ([Section 4.2](#)), the general layer-invariance proof and dispersion-relation derivation underlying the CTQW analysis ([Section 4.3](#)), and the complete oracle-circuit construction for one step of the DTQW ([Section 4.4](#)).

E.1 Proof of [Theorem 4.10](#)

We restate the result for convenience.

Restated (= [Theorem 4.10](#)). Let t_{hit} be the number of steps taken by the memoryless random walk of [Theorem 4.9](#) to first reach B , starting from a leaf of T_B . Then $\mathbb{E}[t_{\text{hit}}] \geq 4 \cdot 2^d - 3d - 4 = \Omega(2^d)$.

Proof. We argue via an auxiliary process that is coupled to the true walk and is provably no slower, then solve the auxiliary process's recursion exactly.

Step 1: the true process can only be slower than reaching layer 0 deterministically.

Recall ([Theorem 4.11](#)) that we index T_B 's vertices by *layer* $i \in \{0, \dots, d-1\}$, layer 0 being the leaves and layer $d-1$ the children of B (reaching B itself is “layer d ”, success). A leaf (layer 0) has degree 2: one neighbour is its T_B -parent (layer 1, forward) and the other is its matching partner in T_A (a move that exits T_B entirely). The walk of [Theorem 4.9](#) chooses each of these two neighbours with probability $\frac{1}{2}$. If it exits into T_A , by the time it returns to T_B (which it must, eventually, since G_d is finite and connected and the walk is recurrent) it lands at *some* leaf of T_B — the same one or a different one, it does not matter which, since the analysis is identical for every leaf by the layer-symmetry of [Theorem 4.17](#). Define an *auxiliary process* on the same probability space that behaves identically to the true walk except that, every time the true walk would step out of T_B across the matching edge, the auxiliary process instead is teleported, at zero additional time cost, directly back to layer 0 of T_B (a fresh leaf). Since the true walk requires at least the 1 step it actually takes to cross into T_A (plus, generically, more steps before it re-enters T_B), the auxiliary process's hitting time t'_{hit} satisfies $t'_{\text{hit}} \leq t_{\text{hit}}$ pointwise on the underlying probability space (every excursion into T_A costs the true walk ≥ 1 extra step

that the auxiliary process is simply given for free), so

$$\mathbb{E}[t_{\text{hit}}] \geq \mathbb{E}[t'_{\text{hit}}].$$

It remains to compute $\mathbb{E}[t'_{\text{hit}}]$, which is now a well-posed random walk entirely confined to the d layers of T_B , with the leaf layer (layer 0) behaving, by construction, as a layer from which the walk deterministically advances to layer 1 in exactly 1 step — this is exactly the base case $X_0 = 1$ used informally in [Theorem 4.11](#).

Step 2: solving the recursion. For $i = 1, \dots, d-1$, let X_i be the expected number of (auxiliary-process) steps to first reach layer $i+1$, starting at an arbitrary vertex of layer i on first arrival there. By [Theorem 4.17](#) applied to T_B (every interior vertex of T_B has one parent, towards B , and two children, towards the leaves), a vertex of layer $i \geq 1$ has degree 3: one neighbour in layer $i+1$ (forward) and two in layer $i-1$ (backward, both freshly-unvisited vertices by the symmetry of the layer reduction — this is justified rigorously precisely because we never need to track *which* vertex of layer $i-1$ we land on, only that it is some vertex of layer $i-1$ encountered for the first time, by an inductive argument on i identical in spirit to [Theorem E.2](#)'s layer-symmetric reduction). Conditioning on the first step:

$$X_i = 1 + \frac{1}{3} \cdot 0 + \frac{2}{3}(X_{i-1} + X_i),$$

where the $\frac{1}{3} \cdot 0$ term records that, with probability $\frac{1}{3}$, the single step taken already reaches layer $i+1$ (no further expected time beyond the 1 already counted), and the $\frac{2}{3}(X_{i-1} + X_i)$ term records that, with probability $\frac{2}{3}$, the walk drops to layer $i-1$ (a fresh vertex there), from which it must first take expected time X_{i-1} to climb back to (a, possibly different, but by the inductive symmetry equivalent) vertex of layer i , and then, having arrived at layer i afresh, again needs expected time X_i to reach layer $i+1$ — this last appeal to “ X_i again” is exactly the memorylessness of [Theorem 4.9](#) together with the fact that every vertex of layer i behaves identically ([Theorem 4.17](#): same degree, same split into forward/backward neighbours), regardless of which specific vertex of layer i is reached.

Solving for X_i :

$$X_i - \frac{2}{3}X_i = 1 + \frac{2}{3}X_{i-1} \implies \frac{1}{3}X_i = 1 + \frac{2}{3}X_{i-1} \implies X_i = 3 + 2X_{i-1}, \quad i = 1, \dots, d-1,$$

with $X_0 = 1$ from Step 1. Set $Y_i := X_i + 3$; then $Y_i = 2X_{i-1} + 6 = 2(X_{i-1} + 3) = 2Y_{i-1}$, a pure geometric recursion with $Y_0 = X_0 + 3 = 4$, so $Y_i = 4 \cdot 2^i = 2^{i+2}$ and

$$X_i = 2^{i+2} - 3, \quad i = 0, 1, \dots, d-1$$

(check: $X_0 = 2^2 - 3 = 1 \checkmark$; $X_1 = 2^3 - 3 = 5 = 3 + 2 \cdot 1 \checkmark$).

Step 3: summing. The total expected number of auxiliary-process steps to go from layer 0 to layer d (i.e. to reach B) is the telescoping sum

$$\mathbb{E}[t'_{\text{hit}}] = \sum_{i=0}^{d-1} X_i = \sum_{i=0}^{d-1} (2^{i+2} - 3) = 4 \sum_{i=0}^{d-1} 2^i - 3d = 4(2^d - 1) - 3d = 4 \cdot 2^d - 3d - 4.$$

Combining with Step 1, $\mathbb{E}[t_{\text{hit}}] \geq \mathbb{E}[t'_{\text{hit}}] = 4 \cdot 2^d - 3d - 4$, which is $\Omega(2^d)$ since $3d + 4 = o(2^d)$. $\square$

Remark E.1 (On the inductive symmetry invoked in Step 2). The claim “every vertex of layer i behaves identically” is intuitively clear (the subtree structure below any layer- i vertex of T_B , towards the leaves, is an isomorphic copy of a depth- $(d - i)$ binary tree, by [Theorem 4.1](#)) but deserves one word of care: after several excursions, the auxiliary process could in principle have already visited *several* distinct vertices of layer $i - 1$ before climbing back, and a fully formal proof tracks the induced Markov chain on the (much larger) state space of “which vertex, exactly, am I at” rather than just “which layer am I in”. This causes no difficulty for the *expectation* computed here, because the recursion for X_i only ever conditions on *first* arrival at a layer- i vertex, and by the strong Markov property and the layer-transitive symmetry of T_B ([Theorem 4.17](#)), the conditional expected time to next reach layer $i + 1$ depends only on the layer, not on the specific vertex or on the history of previously-visited vertices.

E.2 Proof of [Theorem E.2](#)

We restate the lemma in full (it was only summarised informally in [Theorem 4.19](#) of [Chapter 4](#)) before proving it.

Lemma E.2 (Invariance of $\mathcal{H}_{\text{layer}}$ under H). *Let $H = L$ be the CTQW Hamiltonian on G_d of [Theorem 3.23](#), and let $|\lambda_j\rangle := n_j^{-1/2} \sum_{v \in L_j} |v\rangle$ for $j = 0, \dots, 2d + 1$ be the uniform superposition over layer j ([Theorem 4.15](#)); these are orthonormal, since distinct layers are disjoint vertex sets, and span an orthonormal $(2d + 2)$ -dimensional subspace $\mathcal{H}_{\text{layer}} := \text{span}\{|\lambda_0\rangle, \dots, |\lambda_{2d+1}\rangle\} \subseteq \mathcal{H}_{\text{pos}}$. Then H maps $\mathcal{H}_{\text{layer}}$ into itself; explicitly, writing δ_j for the common degree of every vertex in L_j ,*

$$H|\lambda_j\rangle = \delta_j|\lambda_j\rangle - b_{j-1}\sqrt{\frac{n_{j-1}}{n_j}}|\lambda_{j-1}\rangle - a_{j+1}\sqrt{\frac{n_{j+1}}{n_j}}|\lambda_{j+1}\rangle \quad (\text{E.1})$$

(omitting terms with $j - 1 < 0$ or $j + 1 > 2d + 1$). Consequently H restricted to $\mathcal{H}_{\text{layer}}$, in the orthonormal basis $\{|\lambda_j\rangle\}$, is the real symmetric tridiagonal matrix $\tilde{H}$ with

$$\tilde{H}_{jj} = \delta_j, \quad \tilde{H}_{j,j+1} = \tilde{H}_{j+1,j} = -a_{j+1}\sqrt{\frac{n_{j+1}}{n_j}} = -b_j\sqrt{\frac{n_j}{n_{j+1}}}.$$

Proof. Fix j and compute $H|\lambda_j\rangle = \frac{1}{\sqrt{n_j}} \sum_{v \in L_j} H|v\rangle$. By [Theorem 3.23](#), $H|v\rangle = d_v|v\rangle - \sum_{u \sim v} |u\rangle$, and by [Theorem 4.17](#), $d_v = \delta_j$ for every $v \in L_j$, with the sum over neighbours splitting as a_j terms landing in L_{j-1} and b_j terms landing in L_{j+1} (no other layer is reached). Hence

$$H|\lambda_j\rangle = \delta_j|\lambda_j\rangle - \frac{1}{\sqrt{n_j}} \sum_{v \in L_j} \sum_{\substack{u \sim v \\ u \in L_{j-1}}} |u\rangle - \frac{1}{\sqrt{n_j}} \sum_{v \in L_j} \sum_{\substack{u \sim v \\ u \in L_{j+1}}} |u\rangle.$$

Consider the middle double sum. It ranges over all edges between L_j and L_{j-1} , each counted once for its L_j endpoint. By [Theorem 4.17](#) applied at layer $j - 1$, every $u \in L_{j-1}$ has exactly b_{j-1} neighbours in L_j ; so summing over all edges between the two layers and grouping terms by their L_{j-1} endpoint u instead of their L_j endpoint v , each $u \in L_{j-1}$ is counted exactly b_{j-1}

times. Hence

$$\sum_{v \in L_j} \sum_{\substack{u \sim v \\ u \in L_{j-1}}} |u\rangle = b_{j-1} \sum_{u \in L_{j-1}} |u\rangle = b_{j-1} \sqrt{n_{j-1}} |\lambda_{j-1}\rangle.$$

Symmetrically, grouping the last double sum by its L_{j+1} endpoint (each such endpoint has exactly a_{j+1} neighbours in L_j , by [Theorem 4.17](#) at layer $j+1$),

$$\sum_{v \in L_j} \sum_{\substack{u \sim v \\ u \in L_{j+1}}} |u\rangle = a_{j+1} \sqrt{n_{j+1}} |\lambda_{j+1}\rangle.$$

Substituting both back in and dividing by $\sqrt{n_j}$ gives [\(E.1\)](#). Since the right-hand side lies in $\mathcal{H}_{\text{layer}} = \text{span}\{|\lambda_0\rangle, \dots, |\lambda_{2d+1}\rangle\}$ for every j , this subspace is H -invariant, and the matrix of H restricted to it, in the orthonormal basis $\{|\lambda_j\rangle\}$, is read off directly from [\(E.1\)](#) as $\tilde{H}$. The equality of the two expressions given for $\tilde{H}_{j,j+1}$ follows from the edge-counting identity $n_j a_j = n_{j-1} b_{j-1}$ (both sides count the edges between L_{j-1} and L_j), rearranged with indices shifted by one: $n_{j+1} a_{j+1} = n_j b_j$, i.e. $a_{j+1} \sqrt{n_{j+1}/n_j} = a_{j+1} n_{j+1} / \sqrt{n_j n_{j+1}} = b_j n_j / \sqrt{n_j n_{j+1}} = b_j \sqrt{n_j/n_{j+1}}$. $\square$

$\leftarrow$ *Back to the main statement.*

E.3 Proof of [Theorem 4.17](#)

Restated. For each j , every $v \in L_j$ has exactly a_j neighbours in L_{j-1} and exactly b_j neighbours in L_{j+1} (and no neighbours in any other layer), with (a_j, b_j) as tabulated in [Theorem 4.17](#).

Proof. Immediate from [Theorems 4.1](#) and [4.15](#), case by case in j :

- $j = 0$: $L_0 = \{A\}$, the root of T_A , which has no parent and two children, both in L_1 . So $(a_0, b_0) = (0, 2)$.
- $1 \leq j \leq d-1$: $v \in L_j$ is a non-root, non-leaf vertex of T_A , with one parent (in L_{j-1}) and two children (in L_{j+1}). So $(a_j, b_j) = (1, 2)$.
- $j = d$: $v \in L_d$ is a leaf of T_A , with one parent (in L_{d-1}) and one matching-edge neighbour, a leaf of T_B (in L_{d+1} , by definition of L_{d+1} as the T_B -leaves). So $(a_d, b_d) = (1, 1)$.
- $j = d+1$: $v \in L_{d+1}$ is a leaf of T_B , with one matching-edge neighbour (a leaf of T_A , in L_d) and one parent — moving *towards* B , hence at T_B -depth one less, i.e. in L_{d+2} by definition of the T_B layers. So $(a_{d+1}, b_{d+1}) = (1, 1)$.
- $d+2 \leq j \leq 2d$: $v \in L_j$ is a non-root, non-leaf vertex of T_B , with two children (depth one more in T_B , i.e. in L_{j-1}) and one parent (in L_{j+1}). So $(a_j, b_j) = (2, 1)$.
- $j = 2d+1$: $L_{2d+1} = \{B\}$, the root of T_B , which has two children (in L_{2d}) and no parent. So $(a_{2d+1}, b_{2d+1}) = (2, 0)$.

This covers every $j = 0, \dots, 2d+1$ and matches the claimed table exactly. $\square$

$\leftarrow$ *Back to the main statement.*

E.4 Proof of Theorem 4.18

Restated. With δ_j, a_j, b_j as in Theorem 4.17 and n_j as in Theorem 4.16, $\tilde{H}_{jj} = 2$ for $j \in \{0, d, d+1, 2d+1\}$ and 3 otherwise; $\tilde{H}_{j,j+1} = -1$ for $j = d$ and $-\sqrt{2}$ for every other $0 \leq j \leq 2d$.

Proof. The diagonal entries $\tilde{H}_{jj} = \delta_j$ (Theorem E.2) are immediate from Theorem 4.1: A, B , and the two leaf layers have degree 2; every other layer has degree 3.

For the off-diagonal entries, use $\tilde{H}_{j,j+1} = -a_{j+1}\sqrt{n_{j+1}/n_j}$ (Theorem E.2) together with Theorems 4.16 and 4.17, in three ranges:

- $0 \leq j \leq d-1$ (within T_A , including the step into the leaf layer): $n_{j+1}/n_j = 2^{j+1}/2^j = 2$, and $a_{j+1} = 1$ (layer $j+1$ vertices — ordinary T_A vertices for $j+1 \leq d-1$, or T_A -leaves if $j+1 = d$ — each have exactly 1 neighbour in L_j , namely their parent, by Theorem 4.17). Hence $\tilde{H}_{j,j+1} = -1 \cdot \sqrt{2} = -\sqrt{2}$.
- $j = d$ (the matching edge, the one exceptional case): $n_{d+1}/n_d = 2^d/2^d = 1$, and $a_{d+1} = 1$ (each T_B -leaf has exactly 1 neighbour in L_d , its matching partner). Hence $\tilde{H}_{d,d+1} = -1 \cdot 1 = -1$.
- $d+1 \leq j \leq 2d$ (within T_B): $n_{j+1}/n_j = 2^{2d+1-(j+1)}/2^{2d+1-j} = \frac{1}{2}$, and $a_{j+1} = 2$ (layer $j+1$ vertices — ordinary T_B vertices once $j+1 \geq d+2$ — each have 2 neighbours in L_j , namely their two children). Hence $\tilde{H}_{j,j+1} = -2/\sqrt{2} = -\sqrt{2}$.

This accounts for every $j = 0, \dots, 2d$, completing the proof. $\square$

[← Back to the main statement.](#)

E.5 Derivation of Theorem 4.20

Proof. Write the discrete Schrodinger equation generated by (4.4) (cf. (3.5)) for the amplitude $\alpha_j(t) := \langle j | \psi(t) \rangle$:

$$i\dot{\alpha}_j(t) = 3\alpha_j(t) - \sqrt{2}\alpha_{j-1}(t) - \sqrt{2}\alpha_{j+1}(t), \quad j \in \mathbb{Z}.$$

Substitute the plane-wave ansatz $\alpha_j(t) = e^{i(kj-\omega t)}$:

$$\begin{aligned} i \cdot (-i\omega) e^{i(kj-\omega t)} &= 3e^{i(kj-\omega t)} - \sqrt{2}e^{i(k(j-1)-\omega t)} - \sqrt{2}e^{i(k(j+1)-\omega t)}, \\ \omega e^{i(kj-\omega t)} &= e^{i(kj-\omega t)} \left(3 - \sqrt{2}e^{-ik} - \sqrt{2}e^{ik} \right) = e^{i(kj-\omega t)} (3 - 2\sqrt{2}\cos k), \end{aligned}$$

using $e^{ik} + e^{-ik} = 2\cos k$. Cancelling the common nonzero factor $e^{i(kj-\omega t)}$ gives $\omega(k) = 3 - 2\sqrt{2}\cos k$ exactly, for every k , confirming the ansatz solves the equation for this choice of ω . Differentiating, $v_g(k) = \omega'(k) = 2\sqrt{2}\sin k$, which is maximised in absolute value at $k = \pm\pi/2$ (where $|\sin k| = 1$), giving $v_g^{\max} = 2\sqrt{2}$. $\square$

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Remark E.3 (Boundary effects, scoped honestly). The infinite-chain calculation above ignores: (a) the four special on-site energies (2 instead of 3 at $j \in \{0, d, d+1, 2d+1\}$), (b) the single weak bond of strength 1 at $j = d$ instead of $\sqrt{2}$, and (c) the fact that the chain is finite (length $2d+2$) rather than infinite, so a wavepacket eventually reflects off both ends. Each of these is a *localised* perturbation to an otherwise translation-invariant chain (in the sense of scattering theory: a finite-rank or boundary perturbation of an infinite hopping Hamiltonian), and the standard expectation — borne out by the detailed calculations of Childs, Farhi and Gutmann (2002), which we cite rather than reproduce — is that such perturbations affect only $O(1)$ factors in the transmission probability through the chain, not the $\Theta(d)$ ballistic timescale itself, which is set by the bulk group velocity $v_g^{\max} = 2\sqrt{2}$ of [Theorem 4.20](#) and is insensitive to $O(1)$ changes at $O(1)$ many sites.

E.6 The full DTQW oracle circuit

We now give the four-stage circuit summarised in [Section 4.4](#) in complete, register-by-register detail, verifying the 14-neighbour-query, 2-degree-query total of [Theorem 4.24](#). Throughout, the state entering the step is $|\Psi\rangle_0 = \sum_{v,i} \alpha_{v,i} |v\rangle_P |i\rangle_C |0\rangle_N |0\rangle_B |0\rangle_S |0\rangle_f$; linearity lets us track a single basis term $|v\rangle_P |i\rangle_C$ and drop the $\alpha_{v,i}$ and the sum throughout.

Stage 1: controlled coin (cost: 2 degree queries).

- (1.1) Query $O_{\deg}(v)$ into a 2-qubit scratch register: $|v\rangle_P |0\rangle_S \mapsto |v\rangle_P |d_v\rangle_S$.
- (1.2) Apply, controlled on the scratch register reading 2 or 3, the corresponding Grover coin D_2 or D_3 ([Theorem 3.36](#)) to register C . (If $d_v = 2$, only indices 1, 2 are meaningful in C ; D_2 is applied on that 2-dimensional subspace.) This step makes no oracle call.
- (1.3) Query $O_{\deg}(v)$ again: since $O_{\deg}$ XORs d_v into its target register and the target currently is d_v , this query returns the scratch register to $|0\rangle_S$.

After Stage 1: $|v\rangle_P (D_{d_v} |i\rangle)_C |0\rangle_N |0\rangle_B |0\rangle_S |0\rangle_f$.

Stage 2: neighbour lookup (cost: 1 neighbour query). Query $O_{\text{nbr}}(v, i)$ with target N :

$$|v\rangle_P |i\rangle_C |0\rangle_N \longmapsto |v\rangle_P |i\rangle_C |w\rangle_N, \quad w := u_i(v).$$

State: $|v\rangle_P |i\rangle_C |w\rangle_N |0\rangle_B |0\rangle_S |0\rangle_f$ (suppressing the coin superposition notation; this holds termwise for every i in the support of $D_{d_v}|\cdot\rangle$).

Stage 3: back-index computation (cost: 6 neighbour queries). For $x = 1, 2, 3$ in turn:

- (3.1) Query $O_{\text{nbr}}(w, x)$ with target S : $|w\rangle_N |0\rangle_S \mapsto |w\rangle_N |u_x(w)\rangle_S$. (1 query.)
- (3.2) Apply CNOT from each of the m qubits of P to the corresponding qubit of S : $|v\rangle_P |u_x(w)\rangle_S \mapsto |v\rangle_P |u_x(w) \oplus v\rangle_S$.

- (3.3) Apply X (bit-flip) to every qubit of S : S now reads $\overline{u_x(w) \oplus v}$ (bitwise complement), which is all-1s iff $u_x(w) = v$.
- (3.4) Apply an m -controlled Toffoli, controlled on all qubits of S reading 1, targeting f : f flips to 1 iff $u_x(w) = v$, i.e. iff x is exactly the index, at w , of the edge back to v .
- (3.5) Controlled on $f = 1$, copy the (classically known, fixed) value x into register B , via at most 2 Toffoli gates (one per bit of the 2-bit value x , each controlled on f). If $f = 0$ this is a no-op for this value of x ; across the three values $x = 1, 2, 3$, exactly one will have $f = 1$ (since v is a neighbour of w at a unique local index), so B ends this stage holding exactly that index.
- (3.6) Uncompute f : repeat the multi-controlled Toffoli of (3.4), which is a self-inverse gate, returning f to $|0\rangle$.
- (3.7) Uncompute the bit-flips of (3.3) and the CNOTs of (3.2), in reverse order, returning S to $|u_x(w)\rangle_S$.
- (3.8) Query $O_{\text{nbr}}(w, x)$ again with target S : since S currently holds $u_x(w)$, this XORs it away, returning S to $|0\rangle_S$. (1 query.)

Steps (3.1) and (3.8) are the only oracle calls in each pass through x ; over $x = 1, 2, 3$ this is $3 \times 2 = 6$ neighbour queries, matching the claimed cost.

State after Stage 3: $|v\rangle_P |i\rangle_C |w\rangle_N |j\rangle_B |0\rangle_S |0\rangle_f$, where j is the unique index with $u_j(w) = v$ — i.e. exactly the back-index $j = j_{w,v}$ of [Theorem 4.23](#).

Stage 4: shift and uncompute (cost: 7 neighbour queries).

- (4.1) Swap registers $P \leftrightarrow N$ and $C \leftrightarrow B$. State: $|w\rangle_P |j\rangle_C |v\rangle_N |i\rangle_B |0\rangle_S |0\rangle_f$. This implements S : the position has moved from v to w , and the coin register now correctly holds j , the local index of the edge just traversed *as seen from w* — exactly the convention needed for the next call to Stage 1 at the new position w .
- (4.2) *Clear B (currently = i).* Repeat the routine of Stage 3 verbatim, with the roles of P and N exchanged: i.e., for $x = 1, 2, 3$, query $O_{\text{nbr}}(v, x)$ into S (note: register N now holds v , so we query neighbours of v , not of w), compare S against the contents of P which is now w , flip f iff $u_x(v) = w$ (i.e. iff x is the index, at v , of the edge to w — this x equals the original coin value i , by construction), and XOR x into B (rather than copying into a fresh register) controlled on f , cancelling the value i already present; uncompute f and S as before. Cost: 6 neighbour queries. State: $|w\rangle_P |j\rangle_C |v\rangle_N |0\rangle_B |0\rangle_S |0\rangle_f$.
- (4.3) *Clear N (currently = v).* Query $O_{\text{nbr}}(w, j)$ with target N : since j is exactly the back-index satisfying $u_j(w) = v$ (established in Stage 3, and untouched since), and N currently holds v , this query XORs v away: $|w\rangle_P |j\rangle_C |v\rangle_N \mapsto |w\rangle_P |j\rangle_C |v \oplus v\rangle_N = |w\rangle_P |j\rangle_C |0\rangle_N$. Cost: 1 neighbour query.

Final state: $|w\rangle_P |j\rangle_C |0\rangle_N |0\rangle_B |0\rangle_S |0\rangle_f$ — position and coin correctly updated, all scratch registers restored to $|0\rangle$, ready for the next step's Stage 1. Stage 4's total cost is $6 + 1 = 7$ neighbour queries, confirming [Section 4.4.1](#)'s claim that B must be cleared (using the still-available value of N) strictly before N is cleared.

Grand total. 1 (Stage 2) + 6 (Stage 3) + 7 (Stage 4) = 14 neighbour queries, and 2 (Stage 1) degree queries, per step — exactly [Theorem 4.24](#). The gate count beyond oracle calls is $O(m) = O(\log n)$ per step: $O(1)$ calls to a fixed 2- or 3-dimensional unitary D_{d_v} , plus $O(m)$ CNOT, X , and (multi-controlled) Toffoli gates in Stages 3 and 4 (each pass through x uses $O(m)$ gates for the comparison and conditional copy, and there are $O(1)$ such passes per step).

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Appendix F

Proofs Deferred from Chapter 6 (Search Problems)

F.1 Proof of the Gray-code Hamiltonian cycle (Lemma 5.12)

Restatement. The reflected binary Gray code $\gamma_n : \{0, \dots, 2^n - 1\} \rightarrow \{0, 1\}^n$ of [Theorem 5.12](#) satisfies: (i) γ_n is a bijection; (ii) consecutive values $\gamma_n(k), \gamma_n(k+1)$ ($0 \leq k < 2^n - 1$) differ in exactly one bit; (iii) $\gamma_n(2^n - 1)$ and $\gamma_n(0)$ also differ in exactly one bit (so the path closes into a cycle).

Proof. We argue by induction on n . For $n = 1$, $\gamma_1(0) = 0$, $\gamma_1(1) = 1$: this is trivially a bijection $\{0, 1\} \rightarrow \{0, 1\}^1$, the unique consecutive pair $(\gamma_1(0), \gamma_1(1)) = (0, 1)$ differs in its one bit, and the wrap-around pair $(\gamma_1(1), \gamma_1(0)) = (1, 0)$ also differs in that one bit. All three claims hold.

Assume the claims hold for $n - 1$ ($n \geq 2$). By definition,

$$\gamma_n(k) = \begin{cases} 0 \gamma_{n-1}(k), & 0 \leq k \leq 2^{n-1} - 1, \\ 1 \gamma_{n-1}(2^n - 1 - k), & 2^{n-1} \leq k \leq 2^n - 1. \end{cases}$$

Bijectivity. On $k \in \{0, \dots, 2^{n-1} - 1\}$, $\gamma_n(k)$ ranges, by the inductive hypothesis applied to γ_{n-1} , exactly over $\{0\} \times \{0, 1\}^{n-1}$ as k ranges over its domain, bijectively. On $k \in \{2^{n-1}, \dots, 2^n - 1\}$, the substitution $k' := 2^n - 1 - k$ ranges bijectively (it is a decreasing bijection) over $\{0, \dots, 2^{n-1} - 1\}$ as k ranges over $\{2^{n-1}, \dots, 2^n - 1\}$, so $\gamma_n(k) = 1 \gamma_{n-1}(k')$ ranges, again by the inductive hypothesis, exactly over $\{1\} \times \{0, 1\}^{n-1}$, bijectively. The two ranges $\{0\} \times \{0, 1\}^{n-1}$ and $\{1\} \times \{0, 1\}^{n-1}$ partition $\{0, 1\}^n$, so γ_n is a bijection $\{0, \dots, 2^n - 1\} \rightarrow \{0, 1\}^n$.

Consecutive pairs within a half. For $0 \leq k \leq 2^{n-1} - 2$, both $\gamma_n(k) = 0 \gamma_{n-1}(k)$ and $\gamma_n(k+1) = 0 \gamma_{n-1}(k+1)$ share the prepended bit 0, so they differ in exactly one bit iff $\gamma_{n-1}(k)$ and $\gamma_{n-1}(k+1)$ do — true by the inductive hypothesis (ii). Symmetrically, for $2^{n-1} \leq k \leq 2^n - 2$, write $k' = 2^n - 1 - k$, so $\gamma_n(k) = 1 \gamma_{n-1}(k')$ and $\gamma_n(k+1) = 1 \gamma_{n-1}(k' - 1)$ (since incrementing k by 1 decrements $k' = 2^n - 1 - k$ by 1); these differ in exactly one bit iff $\gamma_{n-1}(k' - 1)$ and $\gamma_{n-1}(k')$ do, i.e. iff $\gamma_{n-1}(k' - 1), \gamma_{n-1}((k' - 1) + 1)$ do — again true by (ii) of the inductive hypothesis, applied at index $k' - 1 \in \{0, \dots, 2^{n-1} - 2\}$.

The boundary pair $k = 2^{n-1} - 1$. Here $\gamma_n(2^{n-1} - 1) = 0 \gamma_{n-1}(2^{n-1} - 1)$ and $\gamma_n(2^n - 1) =$

$1\gamma_{n-1}(2^n - 1 - 2^{n-1}) = 1\gamma_{n-1}(2^{n-1} - 1)$: both have the *same* suffix $\gamma_{n-1}(2^{n-1} - 1)$, differing only in the prepended bit (0 vs. 1). Hence they differ in exactly one bit.

The wrap-around pair $k = 2^n - 1$ to $k = 0$. $\gamma_n(2^n - 1) = 1\gamma_{n-1}(2^n - 1 - (2^n - 1)) = 1\gamma_{n-1}(0)$ and $\gamma_n(0) = 0\gamma_{n-1}(0)$: again the same suffix $\gamma_{n-1}(0)$, differing only in the prepended bit. Hence they too differ in exactly one bit.

All cases of (ii) and (iii) are covered, completing the induction. $\square$

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Theorem 5.10 run along $\gamma_n(0), \dots, \gamma_n(2^n - 1)$ is therefore a valid Hamiltonian-path traversal of Q_n : by bijectivity every vertex is visited exactly once, and by (ii) every consecutive pair in the traversal is an edge of Q_n (we do not even need the wrap-around property (iii) for **Theorem 5.11**, since the traversal of **Theorem 5.10** need not close into a cycle; (iii) is recorded because it comes for free from the same induction and is occasionally convenient elsewhere, e.g. if one wants to begin the search from an arbitrary, rather than a fixed, starting vertex).

F.2 Proof of the boustrophedon Hamiltonian path (Lemma 5.13)

Restatement. The d -dimensional grid $\{0, \dots, L - 1\}^d$ (vertices differing by ± 1 in exactly one coordinate are adjacent) admits a Hamiltonian path.

Proof. By induction on d . For $d = 1$, the path $0, 1, 2, \dots, L - 1$ visits every vertex of $\{0, \dots, L - 1\}$ exactly once, and consecutive vertices differ by 1 in the (unique) coordinate, hence are adjacent: a Hamiltonian path.

Assume a Hamiltonian path $u_1, u_2, \dots, u_{L^{d-1}}$ exists on $\{0, \dots, L - 1\}^{d-1}$ (the inductive hypothesis, $d \geq 2$). Build a path on $\{0, \dots, L - 1\}^d$ as follows: for $j = 1, \dots, L^{d-1}$, append the L vertices $(u_j, 0), (u_j, 1), \dots, (u_j, L - 1)$ to the path *if j is odd*, or the L vertices $(u_j, L - 1), (u_j, L - 2), \dots, (u_j, 0)$ *if j is even* (the last coordinate is swept forward on every odd-indexed “row” u_j and backward on every even-indexed one — the boustrophedon, or “ox-plough”, pattern), concatenating these L^{d-1} blocks of length L in order $j = 1, 2, \dots, L^{d-1}$.

This visits every vertex of $\{0, \dots, L - 1\}^d$ exactly once: the blocks are indexed by the L^{d-1} distinct values $u_1, \dots, u_{L^{d-1}}$ (a Hamiltonian path on the first $d - 1$ coordinates visits each value of $\{0, \dots, L - 1\}^{d-1}$ exactly once, by the inductive hypothesis), and within each block all L values of the last coordinate appear exactly once (the block is a permutation, forward or backward, of $\{0, \dots, L - 1\}$).

Consecutive vertices within a block differ by ± 1 in the last coordinate only (by construction, either the increasing or the decreasing enumeration of $\{0, \dots, L - 1\}$), hence are adjacent. It remains to check the $L^{d-1} - 1$ junctions between consecutive blocks j and $j + 1$: the last vertex of block j is $(u_j, L - 1)$ if j is odd or $(u_j, 0)$ if j is even, and the first vertex of block $j + 1$ is $(u_{j+1}, 0)$ if $j + 1$ is odd (i.e. j even) or $(u_{j+1}, L - 1)$ if $j + 1$ is even (i.e. j odd). In either parity case, the last vertex of block j and the first vertex of block $j + 1$ agree in their last coordinate ($L - 1$ when j is odd, 0 when j is even) and differ only in the first $d - 1$ coordinates, where they equal u_j and u_{j+1} respectively: these are adjacent in $\{0, \dots, L - 1\}^{d-1}$ since $u_1, \dots, u_{L^{d-1}}$ is a Hamiltonian *path* (consecutive entries adjacent, by the inductive hypothesis). Hence the

junction vertices, too, are adjacent in $\{0, \dots, L-1\}^d$ (they differ in exactly one of the first $d-1$ coordinates, by exactly ± 1 , and agree in the last coordinate).

All $L^d - 1$ consecutive pairs of the constructed path are therefore edges of $\{0, \dots, L-1\}^d$, and all L^d vertices appear exactly once: a Hamiltonian path, completing the induction. $\square$

[\$\leftarrow\$ Back to the main statement.](#)